GOOGLE WORKSPACE TRAINING

Google Sheets Foundation and Intermediate

One-day course review draft

Build a reliable fictional task register. Labs 1–5 include 123 captured stages for navigation, formats, formulas, cleaning, validation and focused views. Actual rejection, clearing limits, source integrity and hidden-row totals are checked. Earlier import/paste limitations, later labs, independent challenges and learner timing remain pending.

Delivered by Asraf · Microsoft Certified Trainer

Google Sheets Fundamental and Intermediate

One-day course review draft · Review draft, version 0.4

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Resources: Use the practice files supplied with this review package.

Google, Google Workspace, Gmail, Google Drive, Google Docs, Google Sheets, Google Slides, Google Sites, Gemini and NotebookLM are names associated with Google. This independent training guide is not published or endorsed by Google.

Screenshots show the product at the time of writing. Google updates its apps often, so labels, availability and layouts in your account may differ. This review draft still requires full lab validation.

About the author

Asraf is a Microsoft Certified Trainer and HRD Corp-certified IT trainer based in Malaysia. He has been teaching working professionals since 2008, helping teams at banks, pharmaceutical companies, construction firms, and retailers get real work done with Microsoft 365, Power Platform, Excel, Google Workspace, and, more recently, AI tools like Copilot.

He holds a Bachelor of Information Technology (Multimedia) from the University of Malaya, along with Microsoft certifications in Power BI (PL-300) and AI for business (AB-730). Outside the classroom he writes his own automation tools in Python and Node.js, mostly to save himself from doing the same task twice.

He runs his training practice solo, which means he has personally felt every inefficiency this book tries to fix.

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INTRODUCTION

Before you begin

This review book accompanies Google Sheets Fundamental and Intermediate. Work through its chapters in order and keep the original practice inputs unchanged. The same notes and prompts also appear in the local course website.

Current Status

All chapters in this book are written drafts. Most have not yet been executed end to end in the application. The masterclass's reference brief chapter has a tested core path with a recorded recovery prompt; that does not validate the other exercises. Treat VERIFY notes and draft notices as production work still to complete.

What You Need

Use a desktop browser and the Google account approved for the class. Confirm app and AI access beforehand. Edition, administrator controls and language can affect availability.

Download the supplied fictional practice pack. No real client or personnel record is needed. Dataset week fields are relative practice weeks, not an agreed pilot start date.

How to Use the Exercises

Follow the numbered steps, attempt the output and then open the expected-result box. AI wording can vary; source facts and approval conditions must remain correct.

Values and prompts follow the notes. Replace source placeholders before submitting. Inspect proposed edits before applying them. Drafting does not authorise sending, inviting people or publishing.

Course Timing

This course is planned for 1 day, with 390 teaching minutes per day, excluding breaks and lunch. Estimates include practice, checks and discussion and need a classroom pilot.

Each course supplies independent starting inputs. Use a stated recovery file if an earlier output is missing.

Expanded Casework

Revision 0.4 adds casework, independent challenges and evidence gates to every chapter. The original short guided path alone is insufficient. Timings are targets, not measured learner results. Read each case input and keep baseline and scenario files separate.

01

CHAPTER 1

Create and Navigate a Workbook

In this chapter you will create a native Sheet, rename tabs and enter a small set of records.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 40 minutes: 10 explanation/demonstration, 10 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 30 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A named native workbook with an intact source and a separate practice tab.

> **Status:** author reference with 26 current-stage application captures through 1.4. Native conversion, navigation, separate practice tabs, address map, reconstruction and Drive reopen checked. The instructor completed the XLSX upload manually after the browser extension blocked it; automated upload, independent challenge and learner timing remain unverified.

What You Will Learn

- Create and name a native Sheet.
- Navigate tabs, cells and ranges.
- Keep the input records intact.

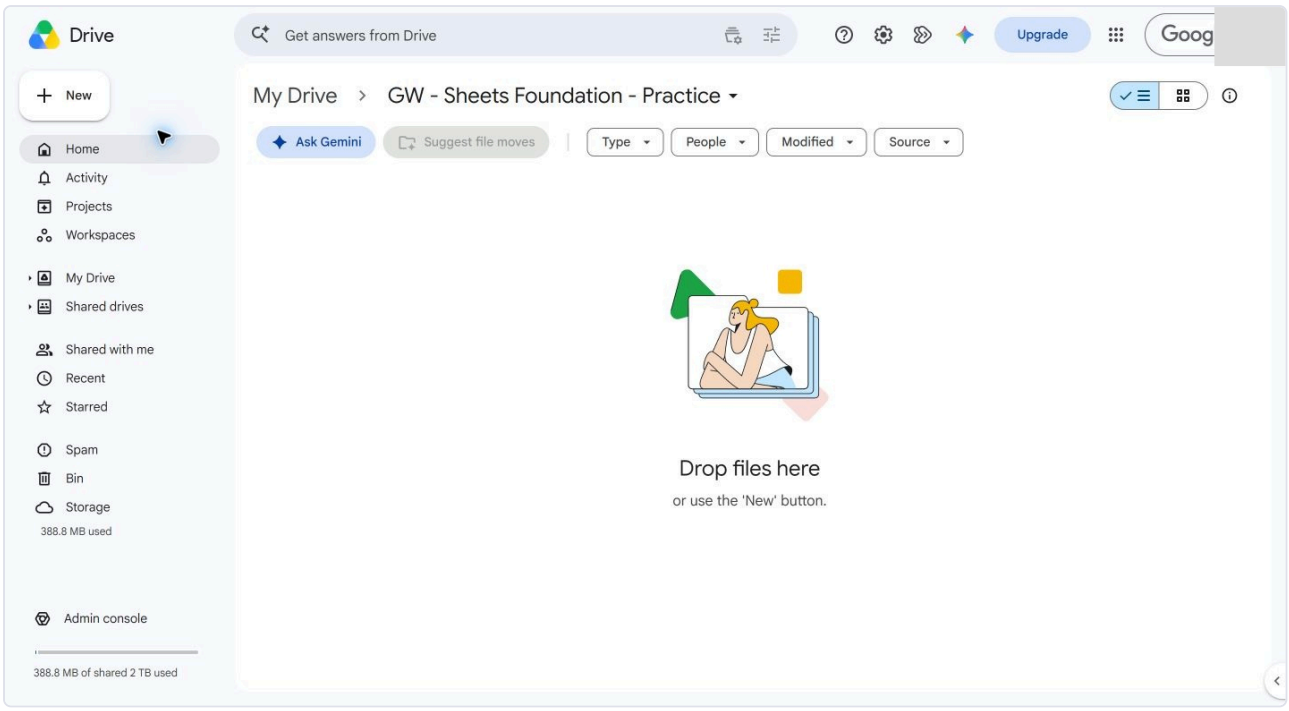
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

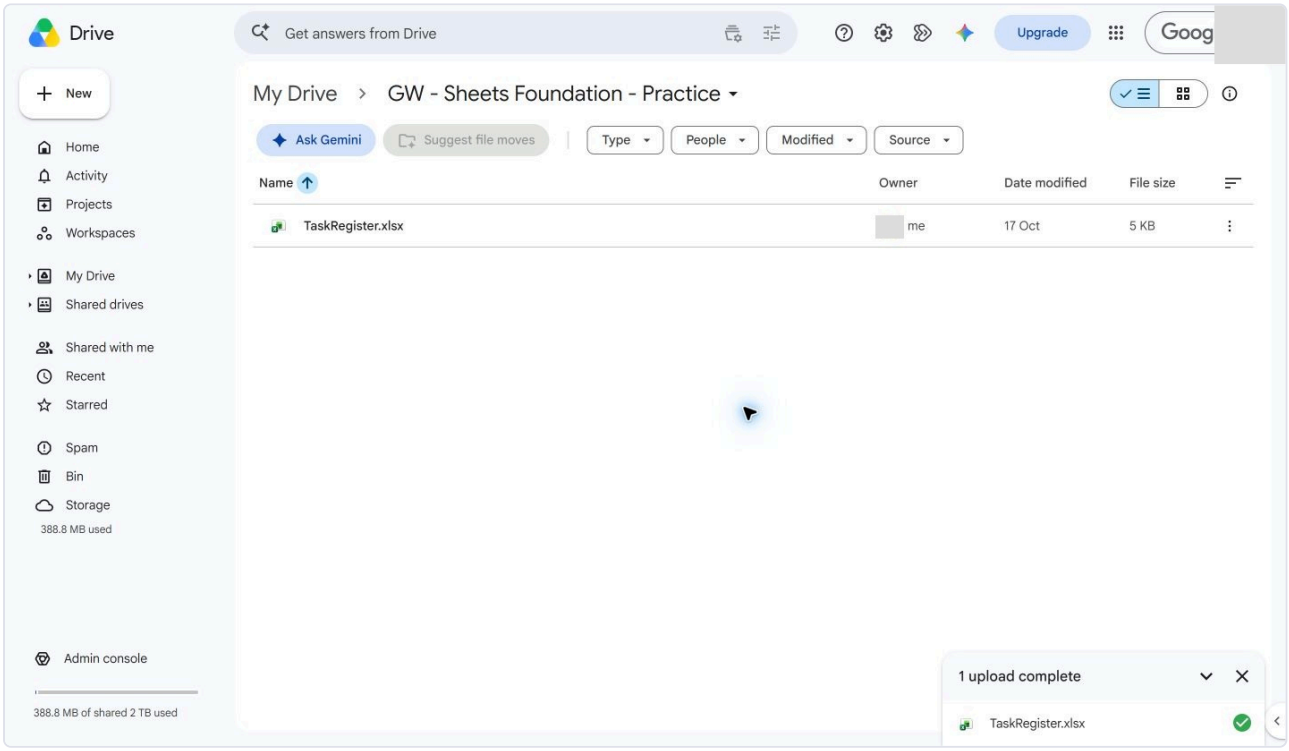
Download TaskRegister.xlsx. This workbook starts at row 1 so cell references in the course match exactly.

1.1 Open a Native Copy

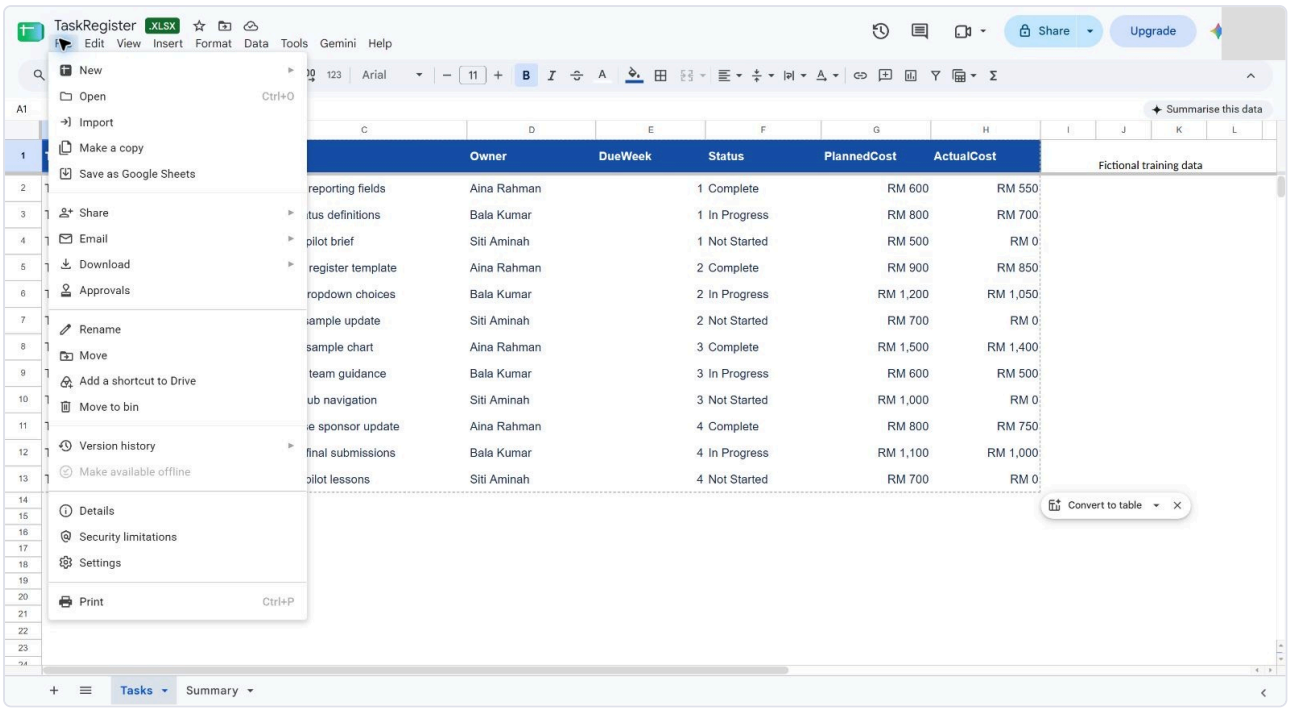
1. In the training account, create a Drive folder named `GW - Sheets Foundation - Practice`. Keep the starter file separate from later native practice.



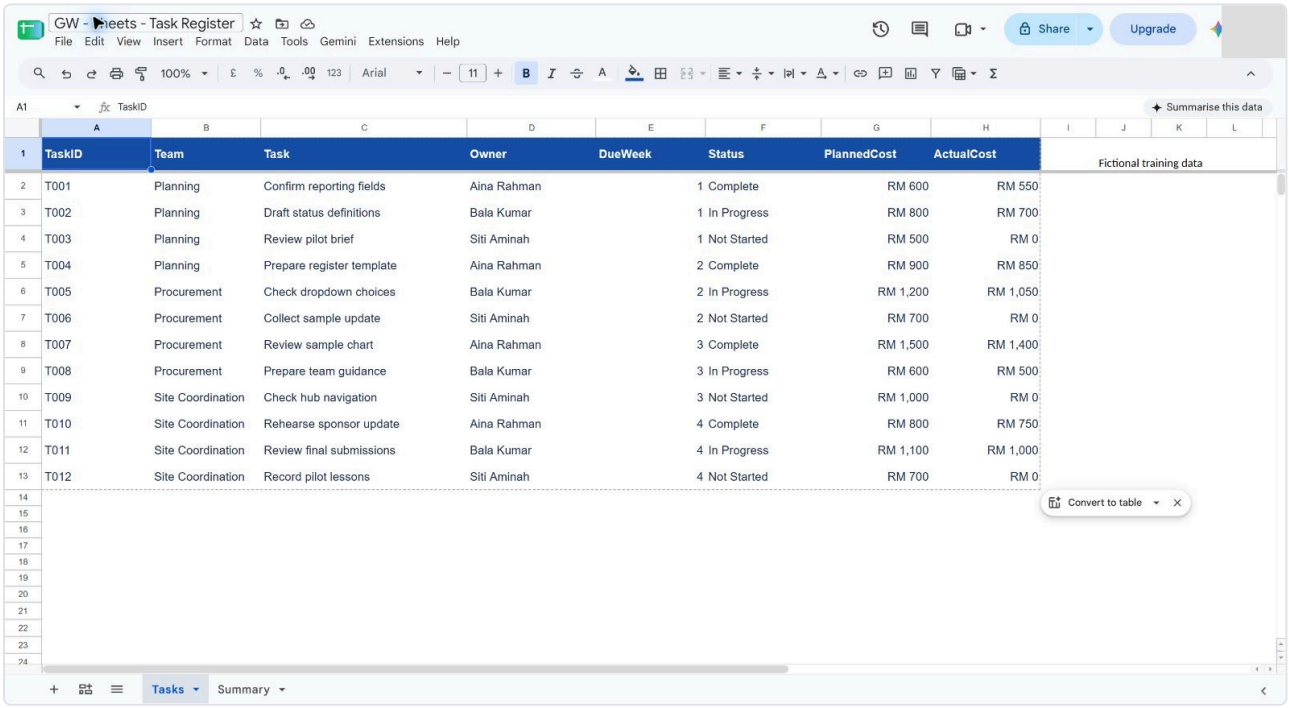
2. Upload **TaskRegister.xlsx** into this folder and confirm its **Microsoft Excel** type. In this author run the instructor uploaded the file manually: the extension still rejected file upload after its file-URL setting and browser restart. The captured uploaded file is real; the automated upload step is not claimed as passed.



3. Open the uploaded file in Sheets. While its **.XLSX / Microsoft Excel format** badge is visible, choose **File > Save as Google Sheets**. This creates a separate native file; the original Excel file remains in the folder.



4. Name the new native file `GW - Sheets - Task Register`. Confirm the XLSX badge is absent and **Tasks** and **Summary** are present. Do not rename the original XLSX instead.

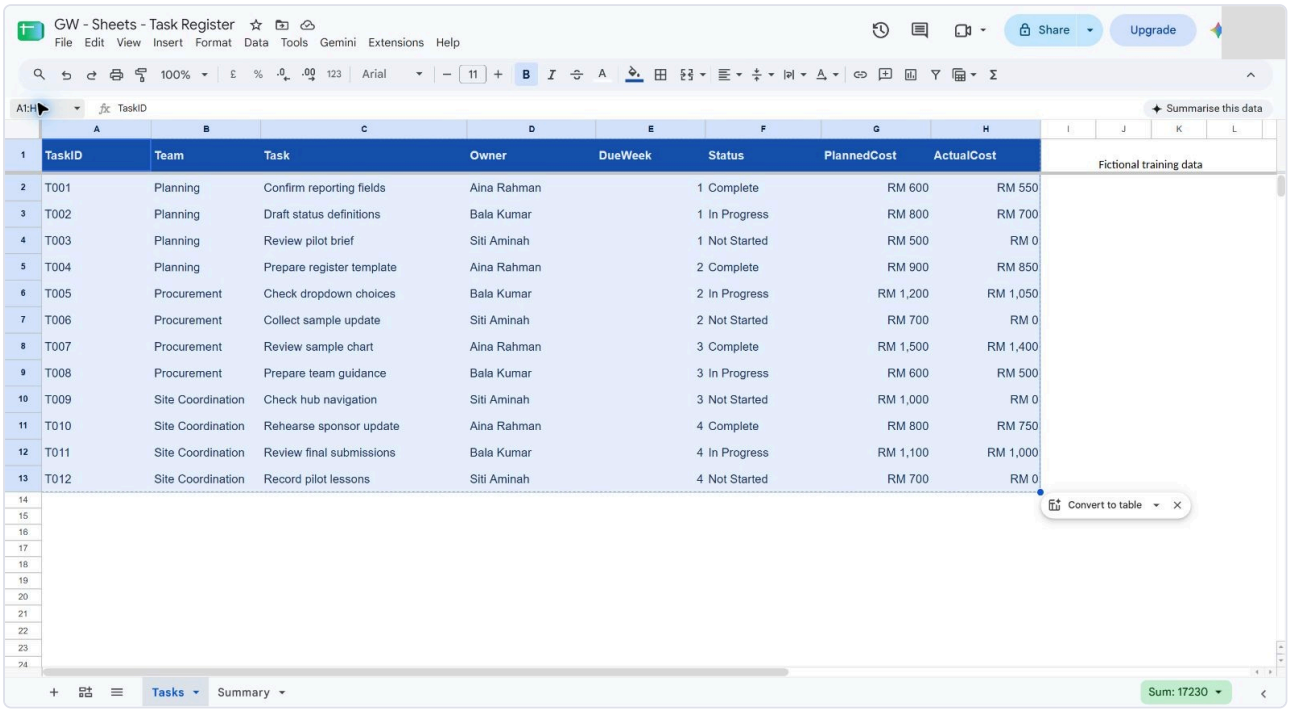


What should you see?

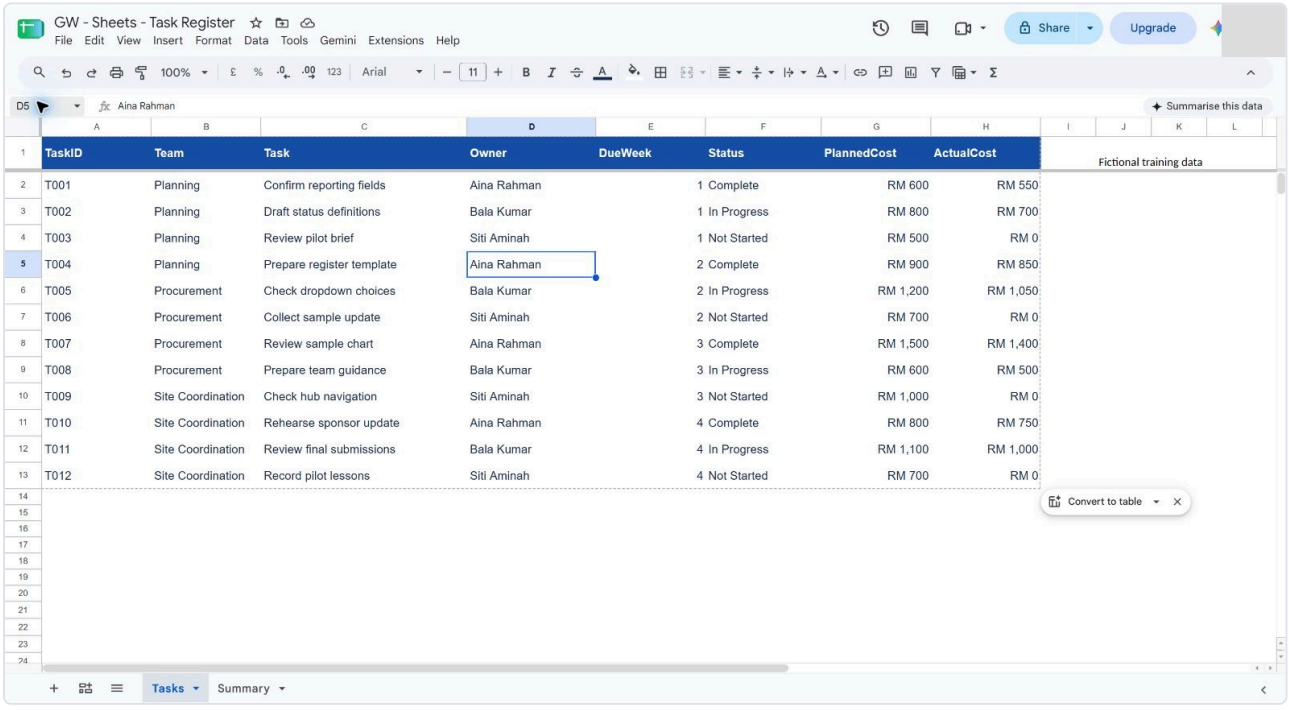
The native workbook has Tasks and Summary tabs, with 12 task records.

1.2 Navigate the Register

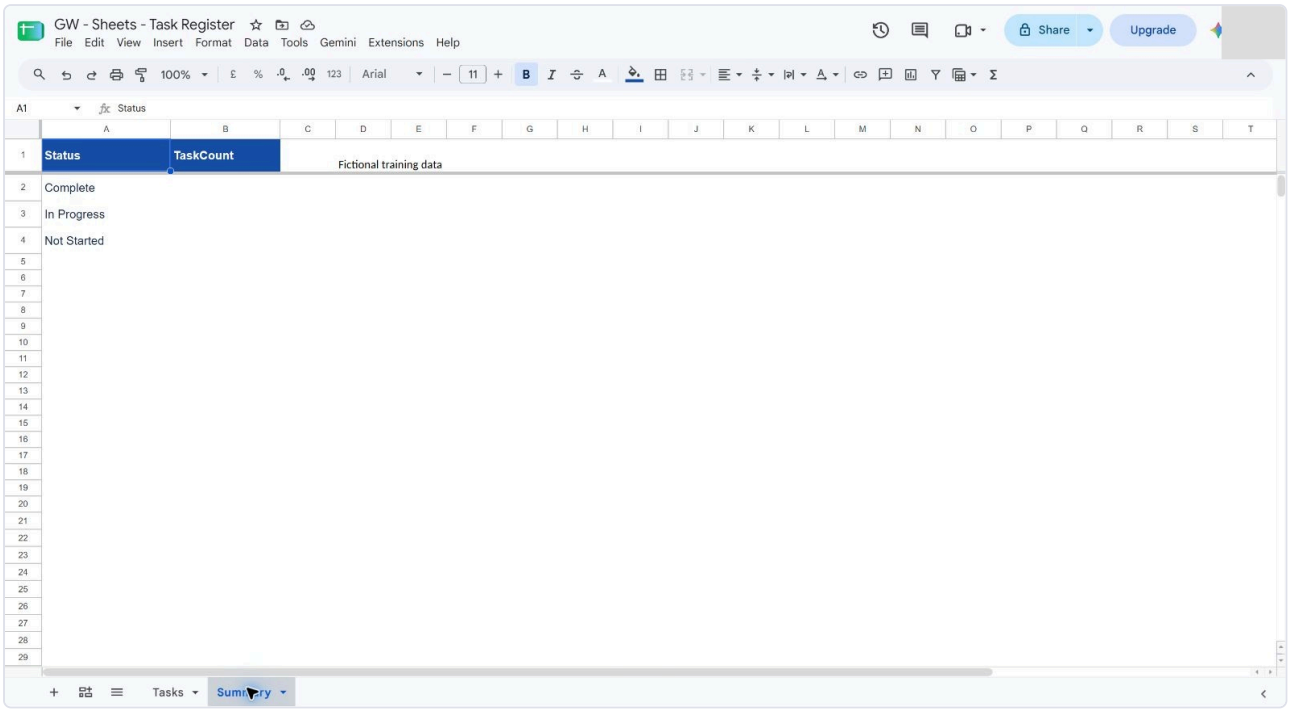
1. Open **Tasks**. Click the name box, enter `A1:H13` and press **Enter** to select all eight headers and twelve records.



2. Enter **D5** in the name box and press **Enter**. Read **Aina Rahman** from the formula bar; the corresponding row is **T004** in column A. A cell address locates the value now; TaskID identifies the record if rows later move.



3. Open **Summary** and return to **Tasks**. Summary contains the supplied starter headings/categories; do not build later formulas or charts yet.

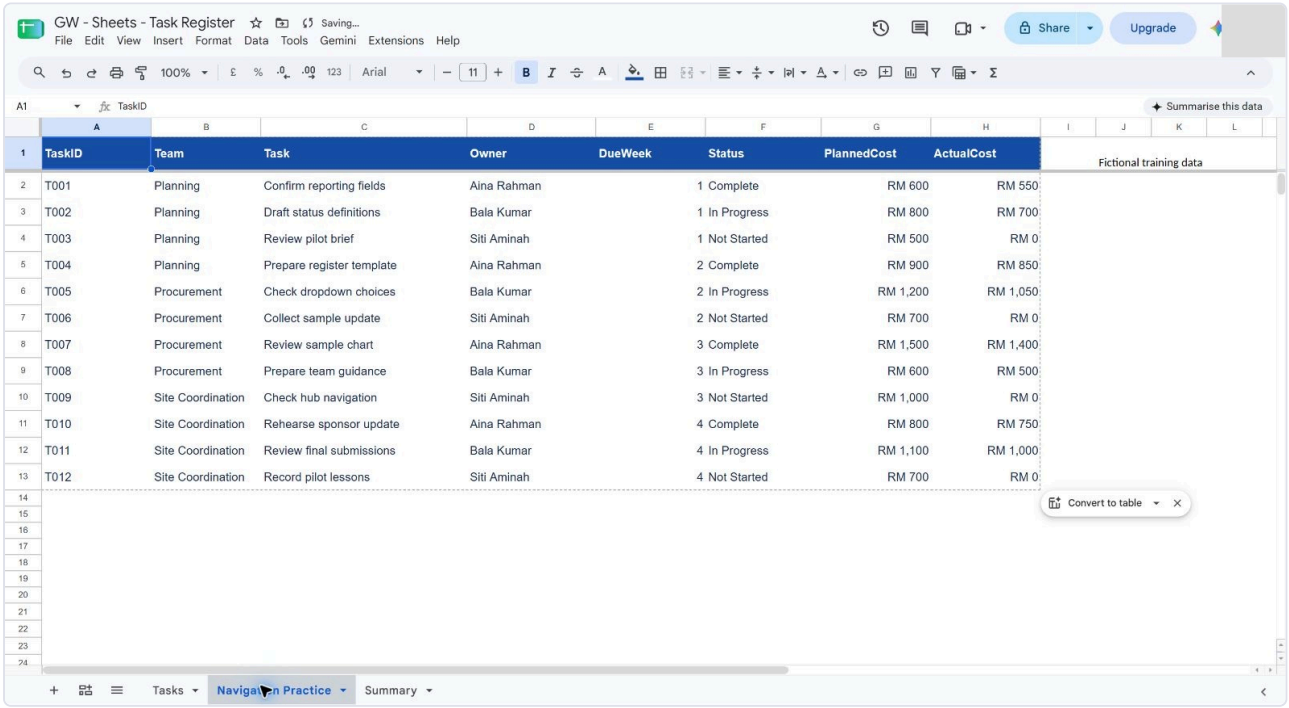


What should you see?

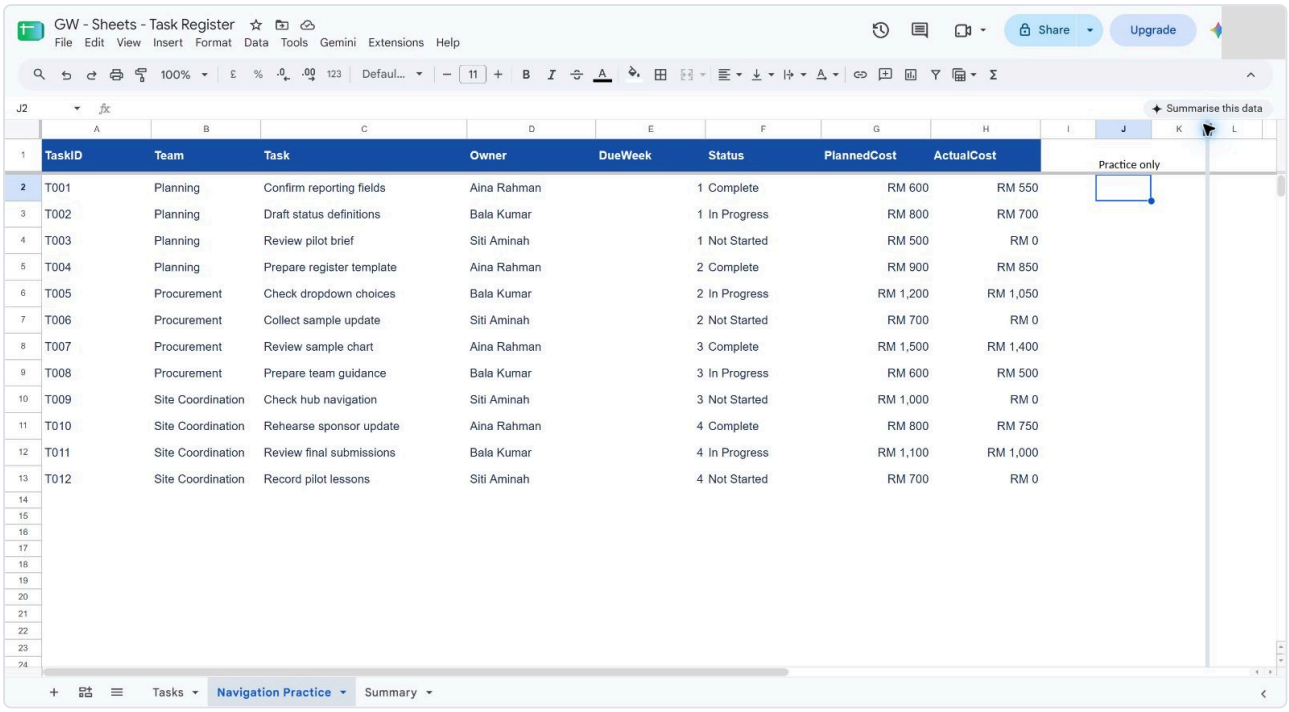
You can identify a cell address, a rectangular range and a worksheet tab.

1.3 Manage a Practice Tab

1. Open the **Tasks** tab menu and choose **Duplicate**. Rename **Copy of Tasks** to `Navigation Practice` using its tab label or **Rename** command.

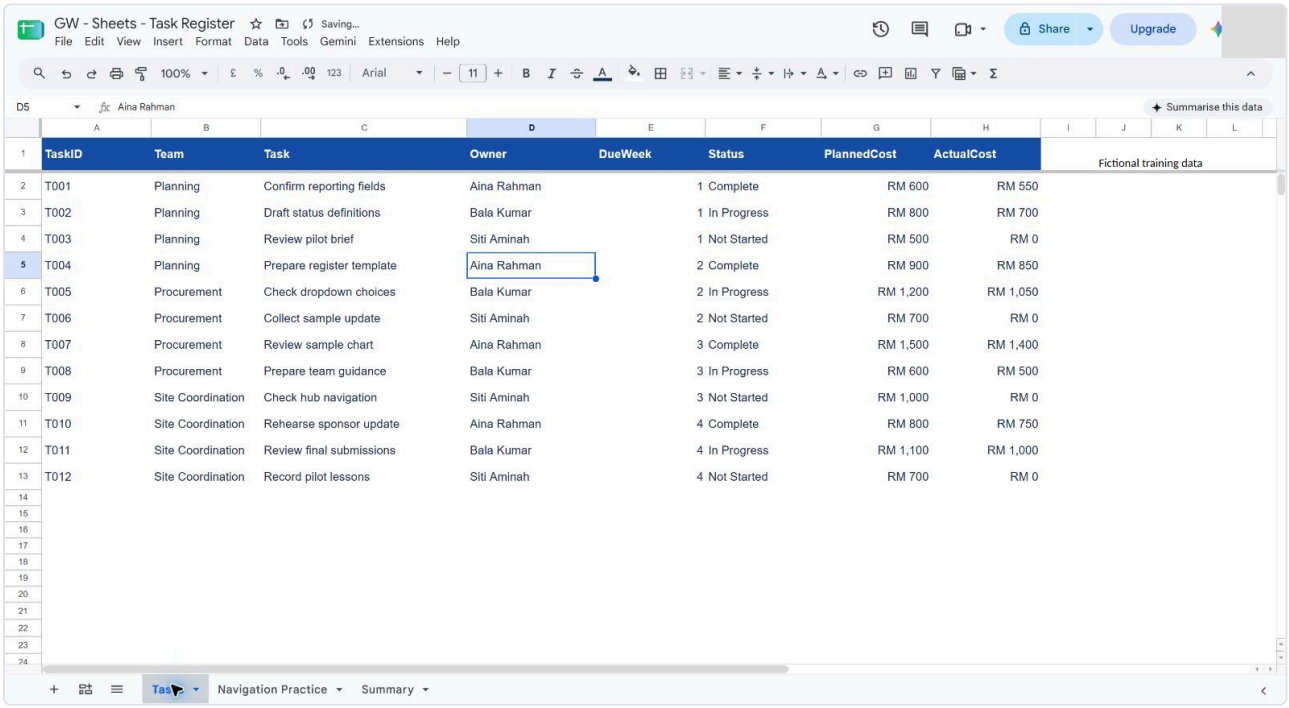


2. On **Navigation Practice**, enter `Practice only` in **J1**, commit the entry, and widen column J using its header boundary. Keep this experiment on the duplicate.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM 600	RM 550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM 800	RM 700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM 500	RM 0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM 900	RM 850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM 1,200	RM 1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM 700	RM 0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM 1,500	RM 1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM 600	RM 500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM 1,000	RM 0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM 800	RM 750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM 1,100	RM 1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM 700	RM 0

3. Return to **Tasks** and inspect **A1:H13**. The reference source was compared with its initial native cell data and remained unchanged, including all twelve records.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
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T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM 800	RM 700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM 500	RM 0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM 900	RM 850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM 1,200	RM 1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM 700	RM 0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM 1,500	RM 1,400
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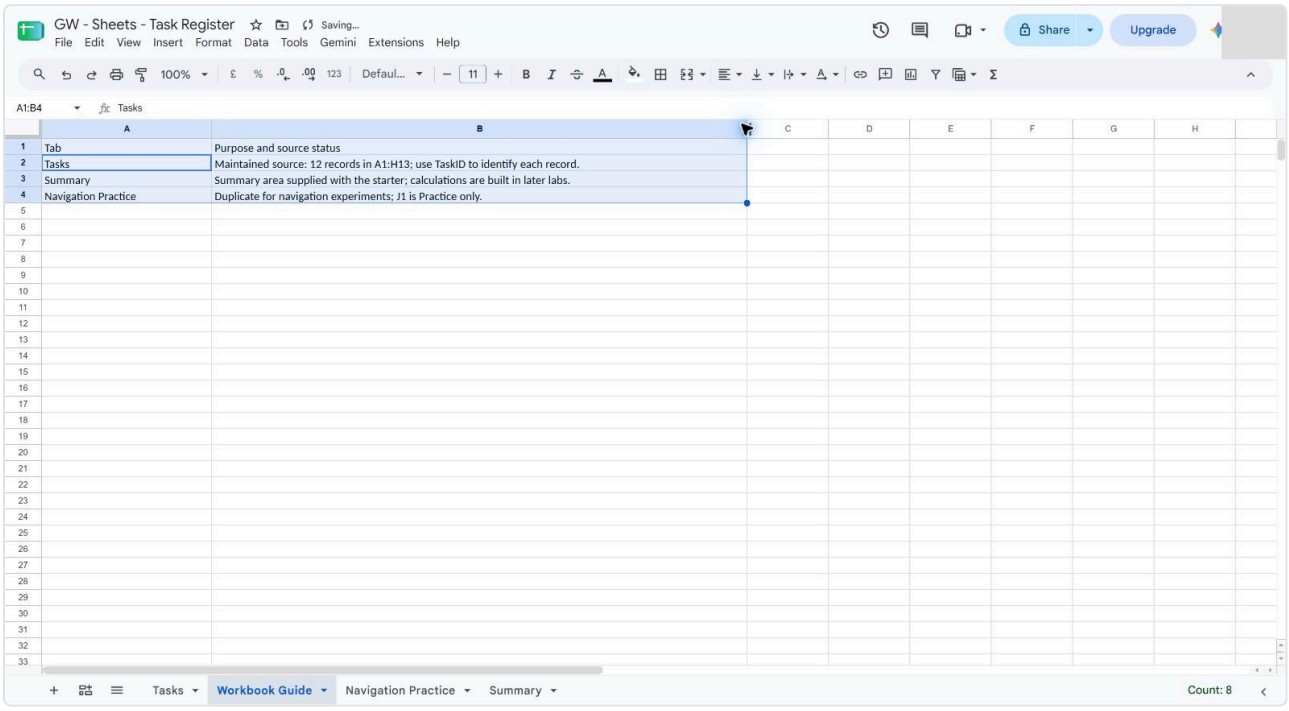
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM 1,100	RM 1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM 700	RM 0

What should you see?

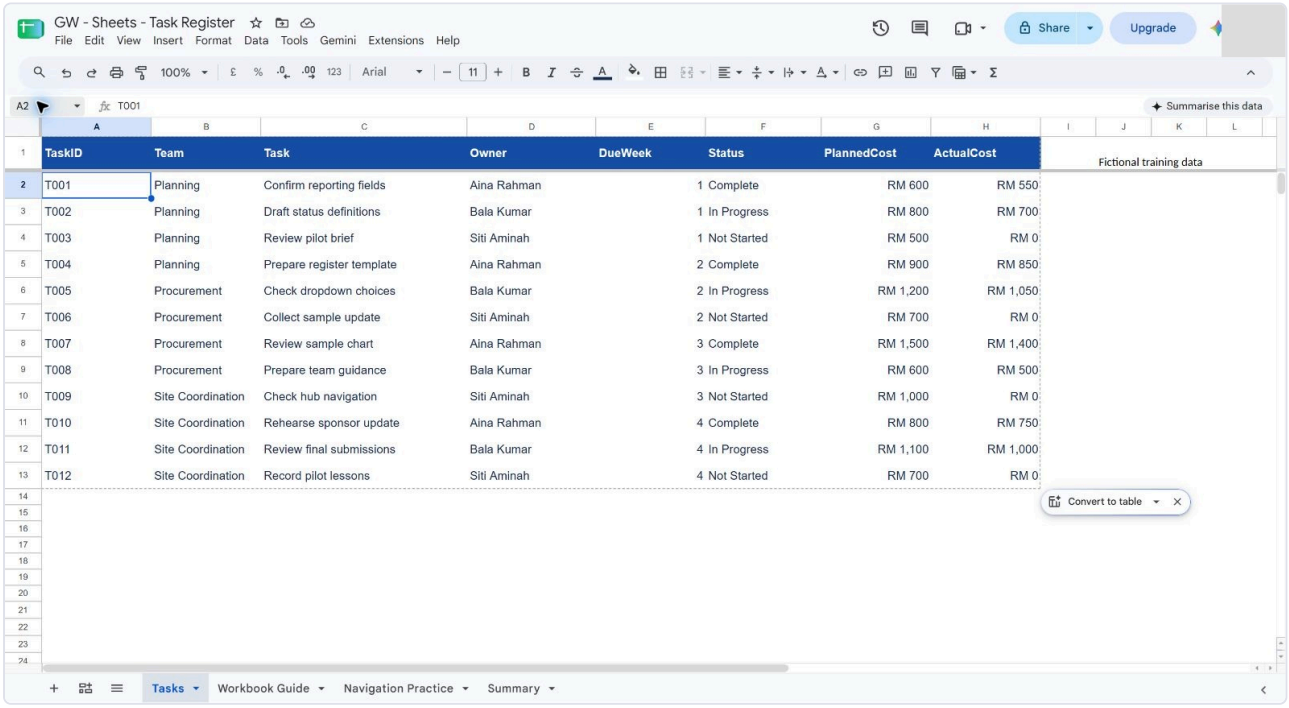
Experiments are on a clearly named copy; the baseline input is preserved.

1.4 Build a Workbook Map and Reconstruct a Small Register

1. Use **Add Sheet**, rename it **Workbook Guide**, and record the purpose of Tasks, Summary and Navigation Practice. Identify **Tasks!A1:H13** as the maintained twelve-record source. Widen the authored guide columns so the map is readable.



2. On **Tasks**, use the name box to visit **A2, D5, F8 and H13**. Inspect the value and the corresponding TaskID each time before recording the check in Workbook Guide. The reference values are below; zero actual cost is a value, not a blank.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM 600	RM 550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM 800	RM 700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM 500	RM 0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM 900	RM 850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM 1,200	RM 1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM 700	RM 0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM 1,500	RM 1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM 600	RM 500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM 1,000	RM 0
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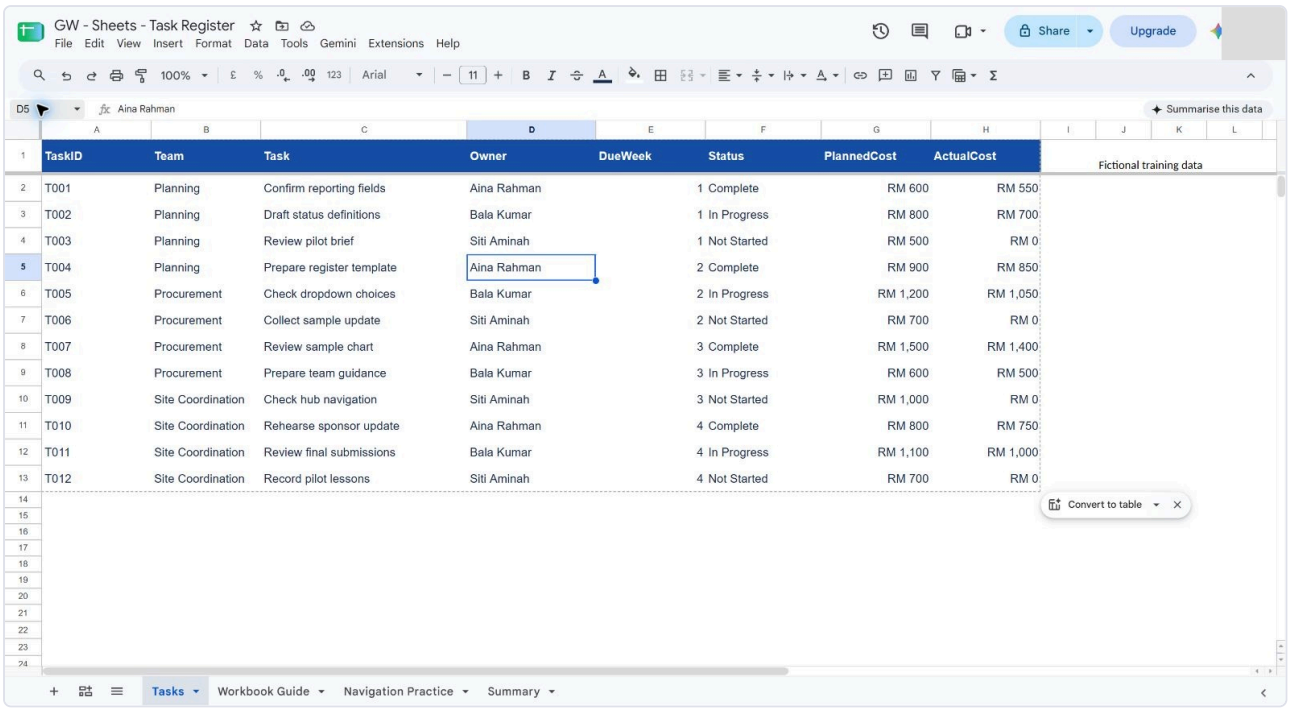
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM 700	RM 0



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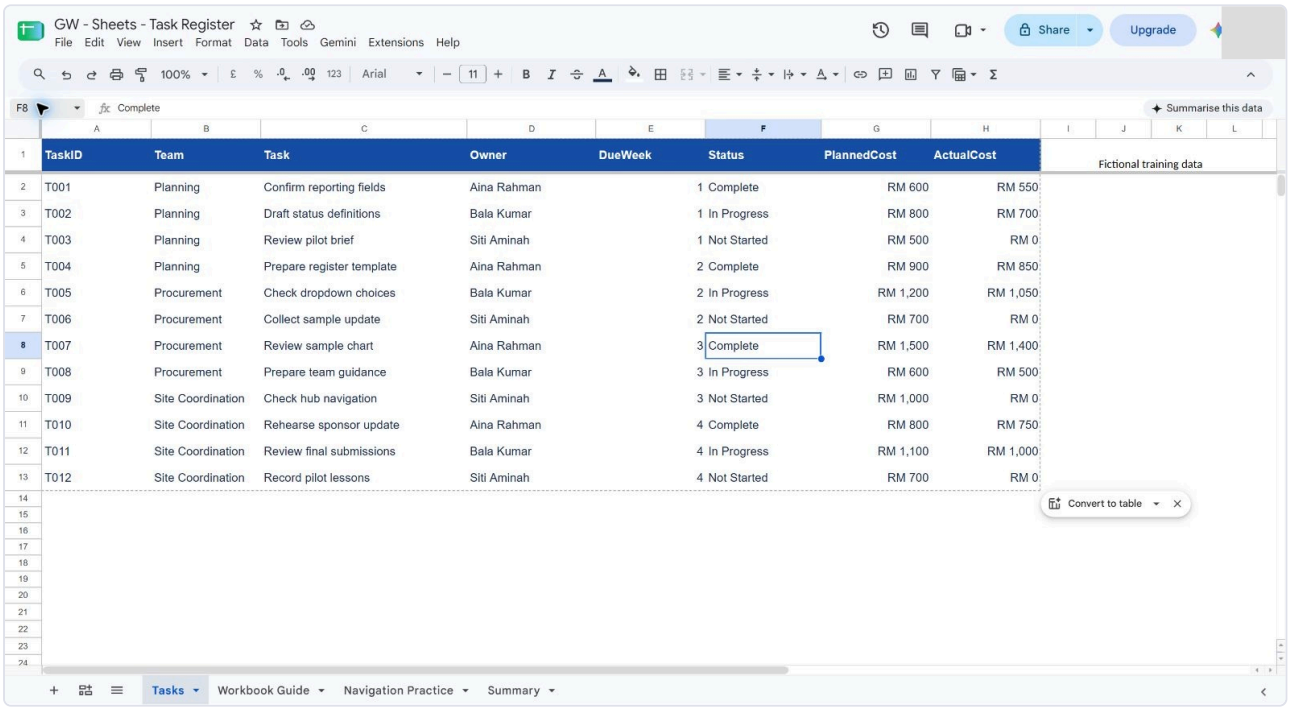

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T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM 700	RM 0
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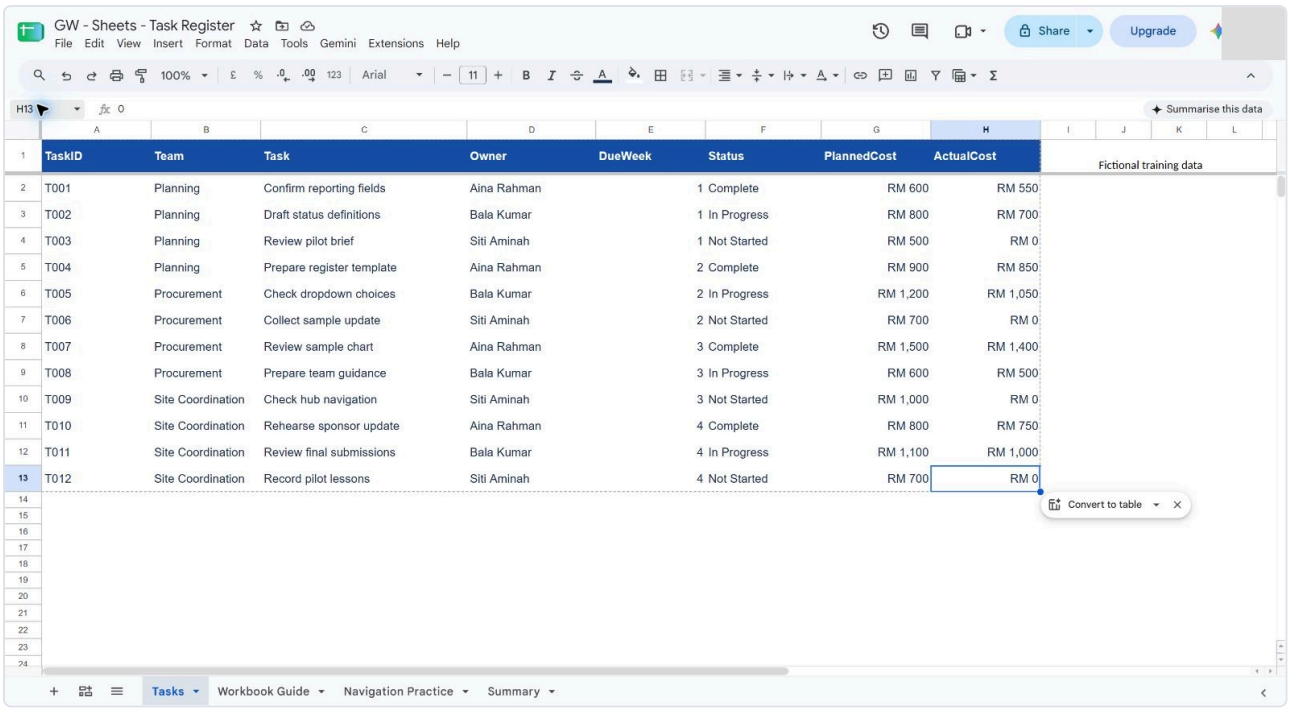

TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
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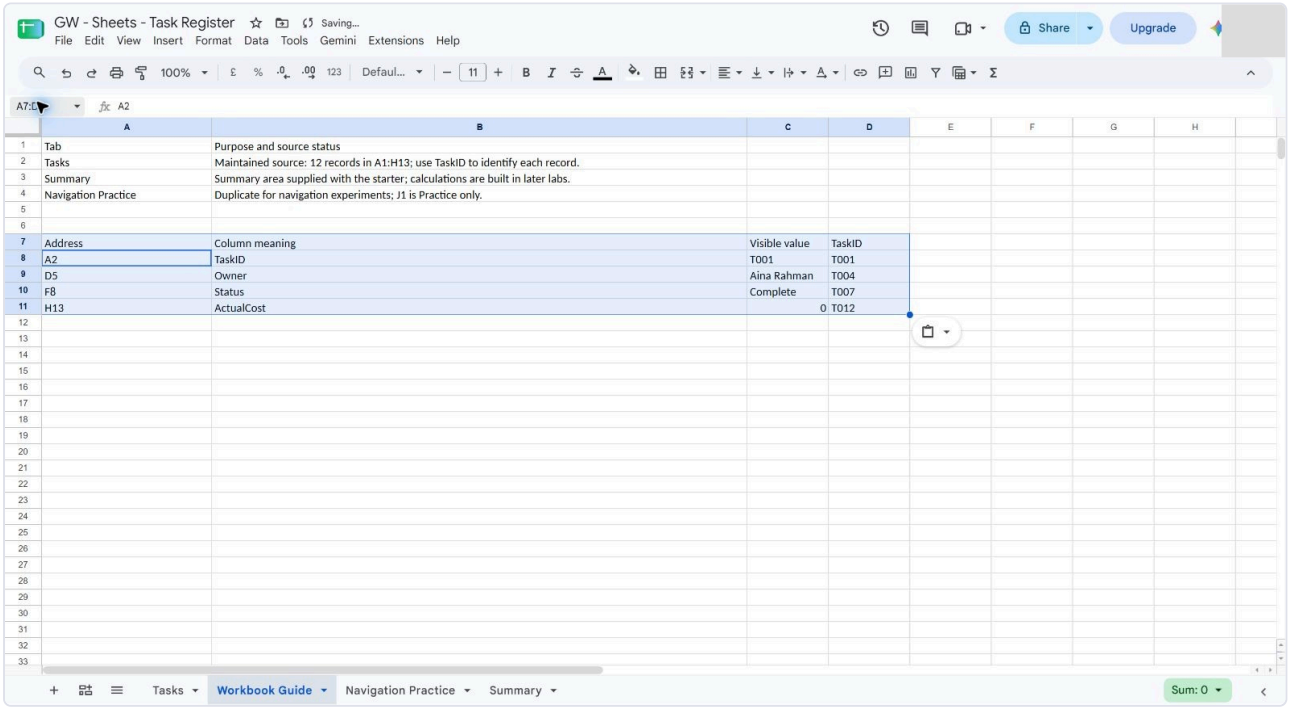
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM 700	RM 0

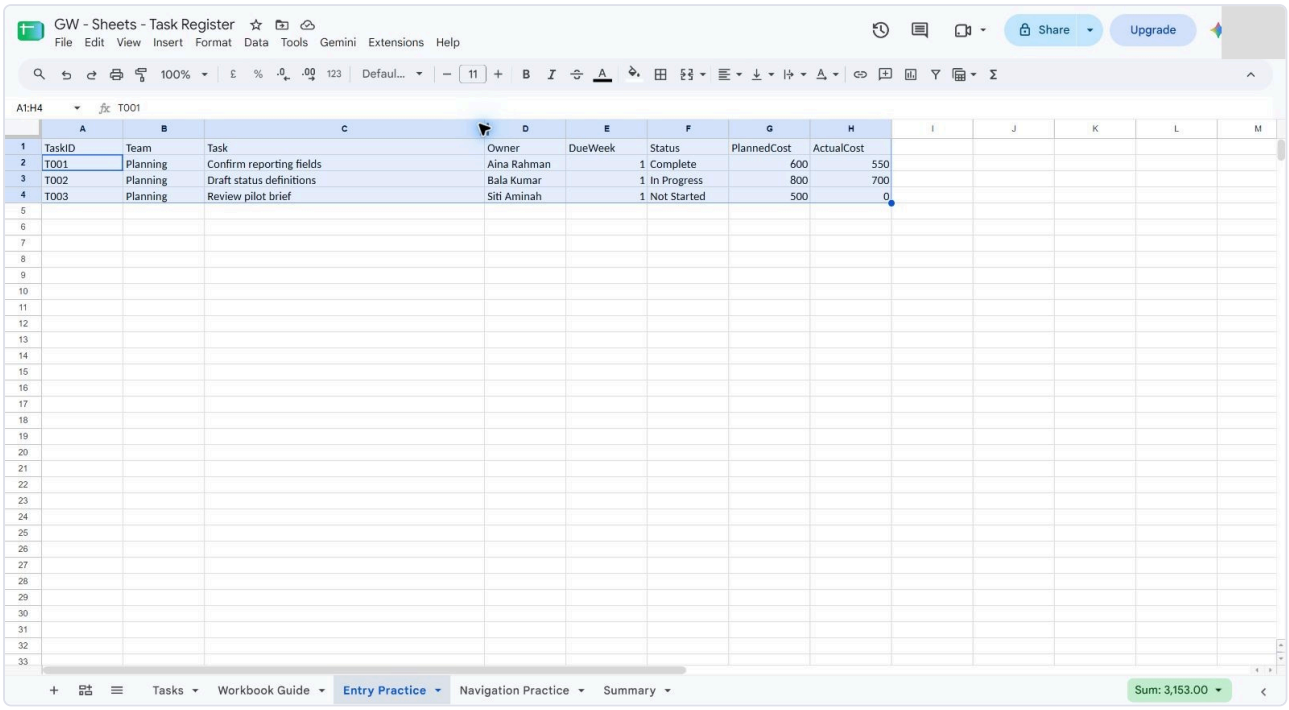
Address	Column meaning	Visible value	TaskID
A2	TaskID	T001	T001
D5	Owner	Aina Rahman	T004
F8	Status	Complete	T007
H13	ActualCost	0 (displayed RM 0 in Tasks)	T012



Tab	Purpose and source status
Tasks	Maintained source: 12 records in A1:H13; use TaskID to identify each record.
Summary	Summary area supplied with the starter; calculations are built in later labs.
Navigation Practice	Duplicate for navigation experiments; J1 is Practice only.

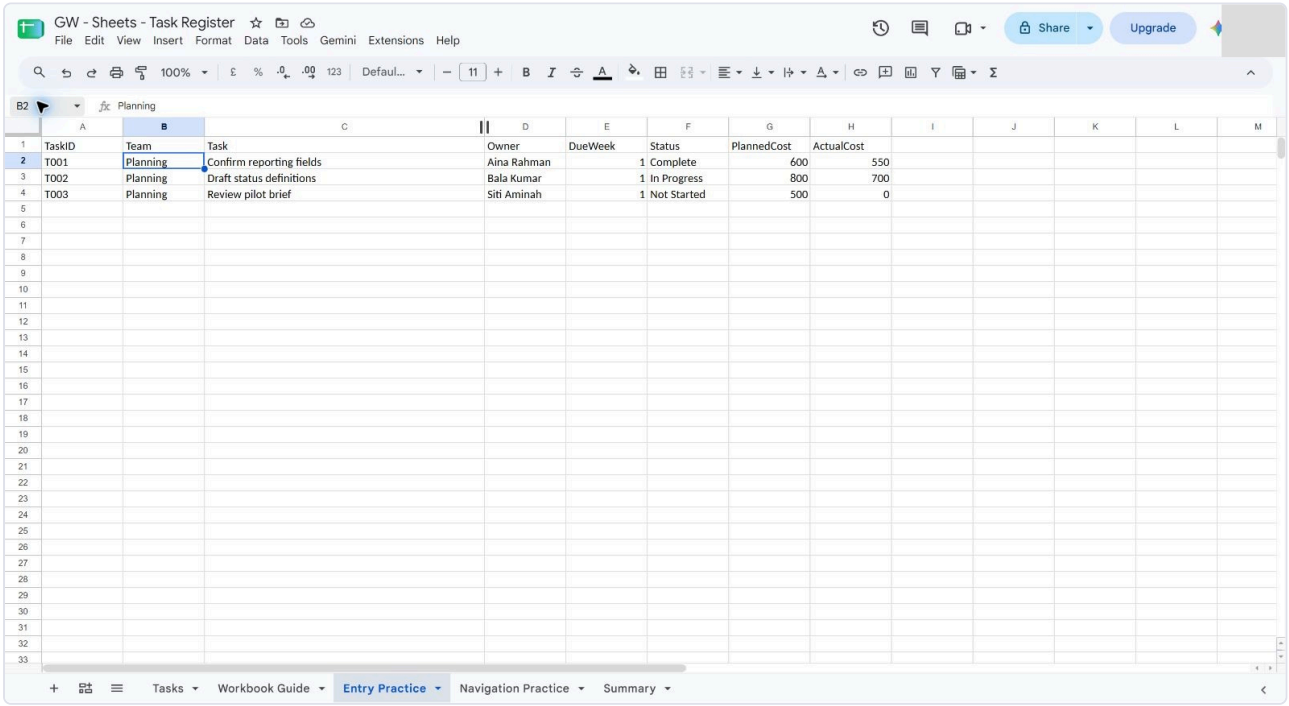
Address	Column meaning	Visible value	TaskID
A2	TaskID	T001	T001
D5	Owner	Aina Rahman	T004
F8	Status	Complete	T007
H13	ActualCost	0	T012

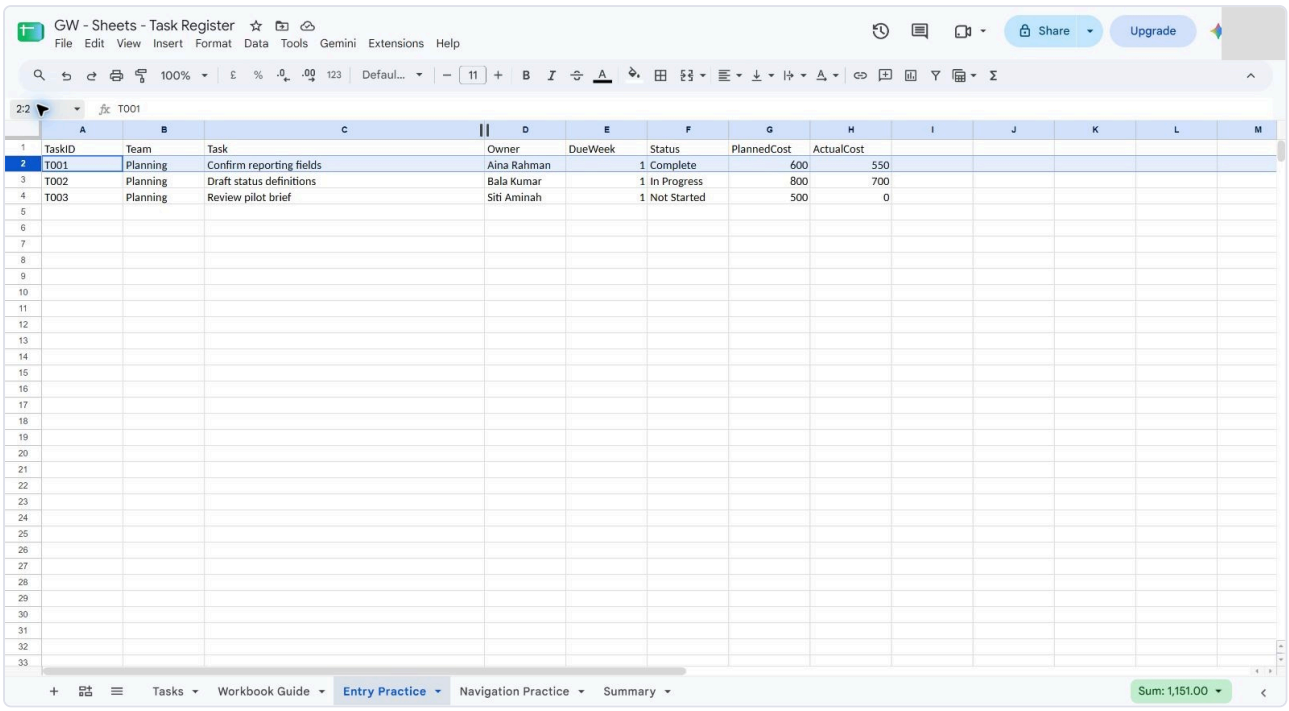
3. Add **Entry Practice**. Enter the eight source headers in **A1:H1**, then recreate **T001, T002 and T003** as values in **A2:H4**, preserving field positions. Widen the task-text column if needed; do not copy the entire Tasks tab for this reconstruction exercise.



4. Compare all eight fields by **TaskID** against **Tasks**. The reference three rows matched, so no copying error was invented or introduced. Record that result. The observed readability issue was clipped task text; widening the Entry Practice task column repaired the display without changing values.

5. Practise **B2** (one cell), **2:2** (whole row), **C:C** (whole column) and **A1:H4** (small table) through the name box. Record that **A1:H1** targets the header only, while a complete record must keep all its fields together. Selection alone does not move data.





TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
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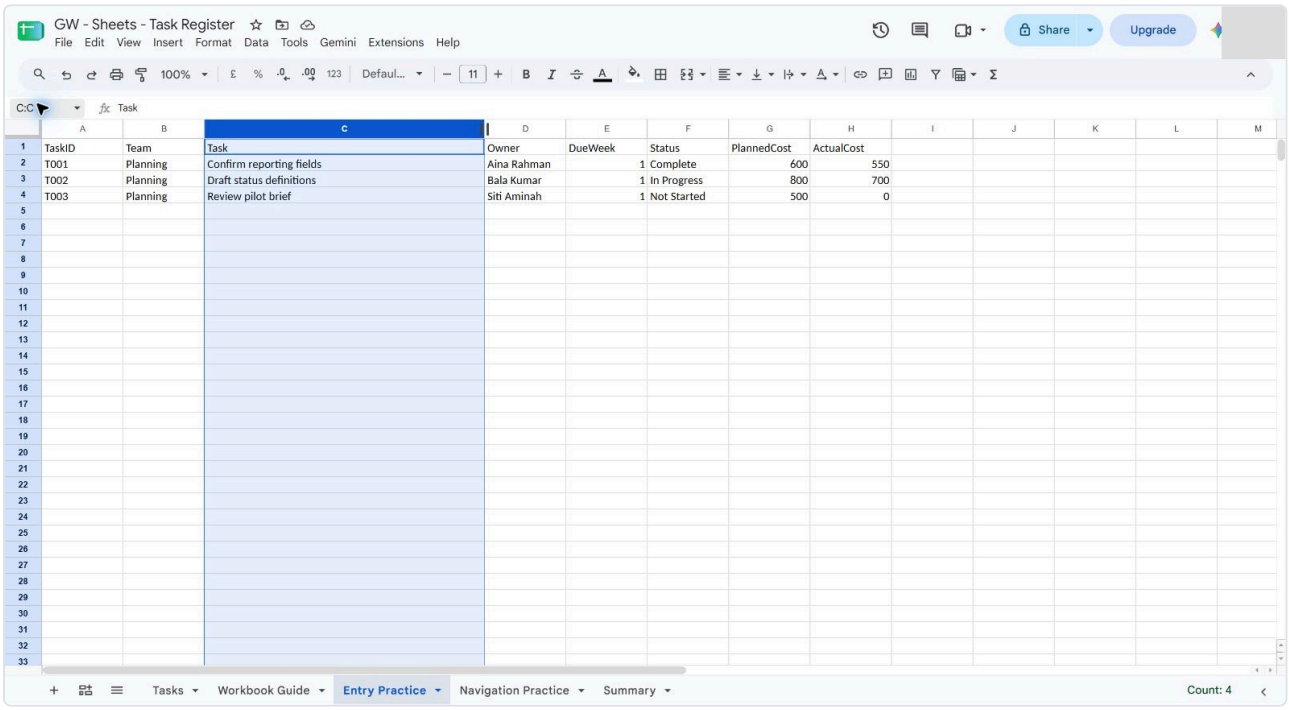
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	500	0



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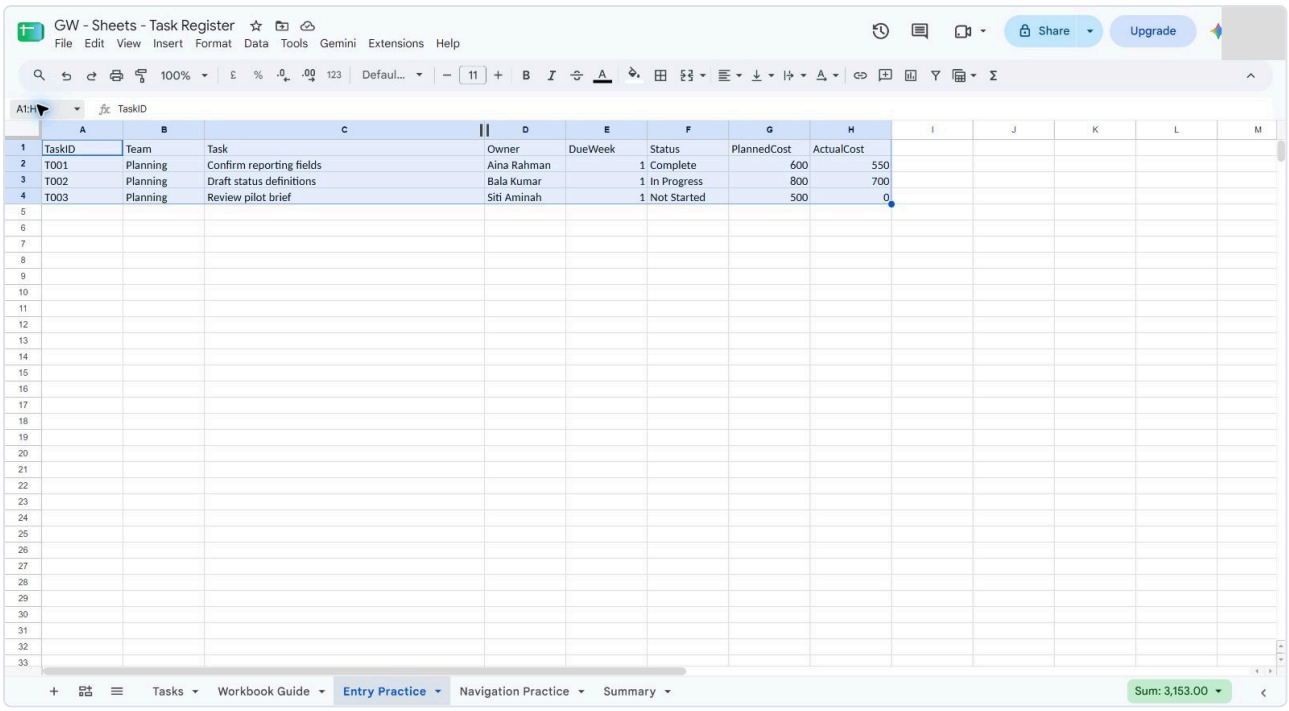
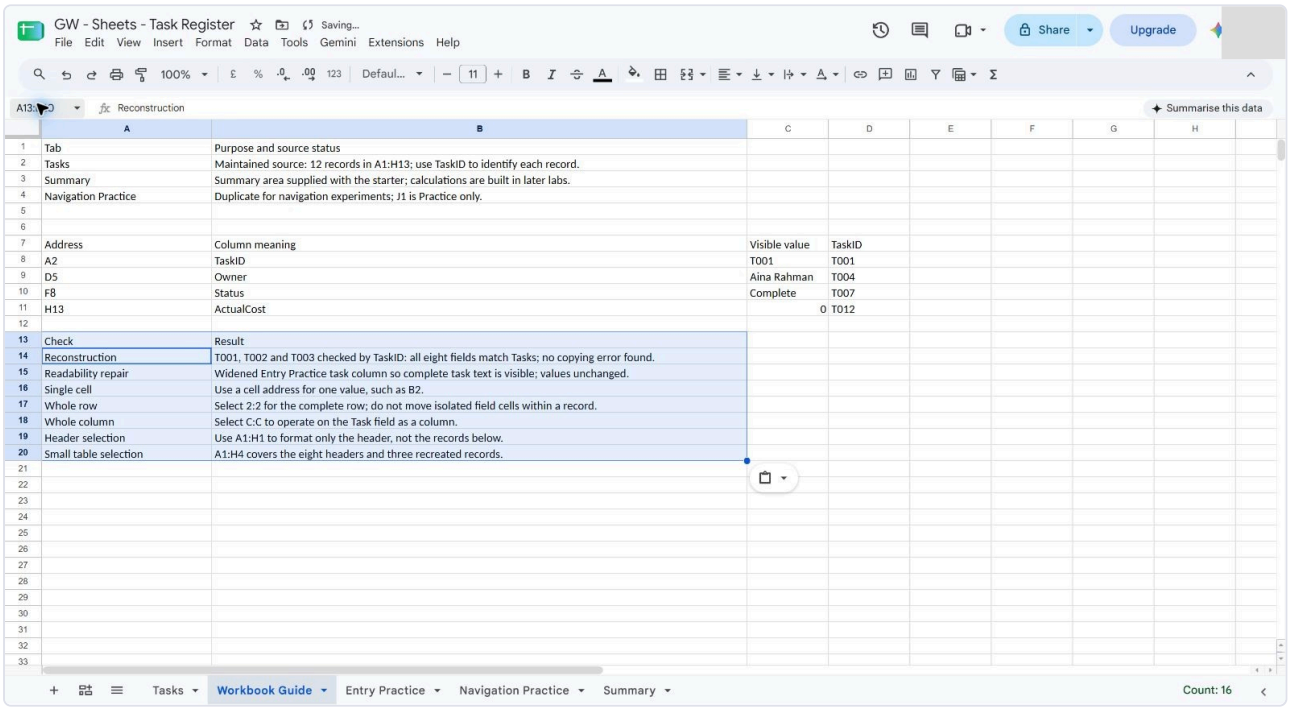
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	800	700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	500	0

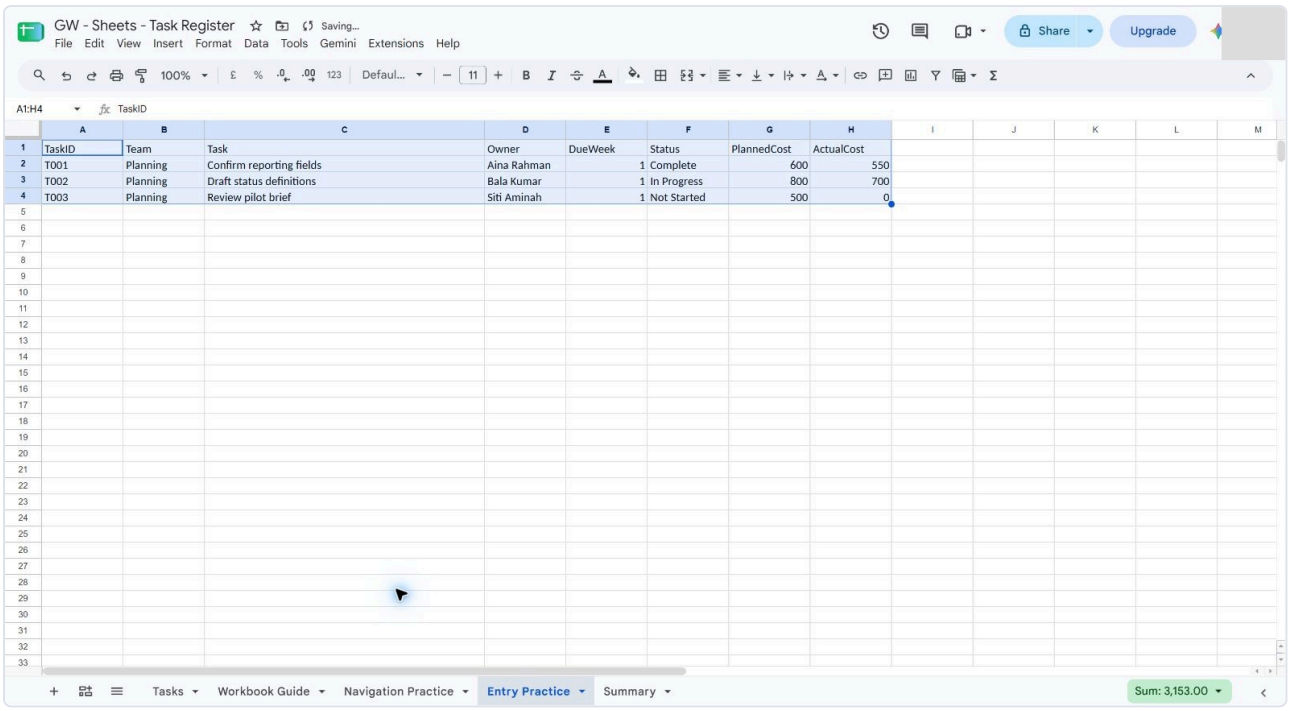


	A	B	C	D	E	F	G	H
1	TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
2	T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	600	550
3	T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	800	700
4	T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	500	0

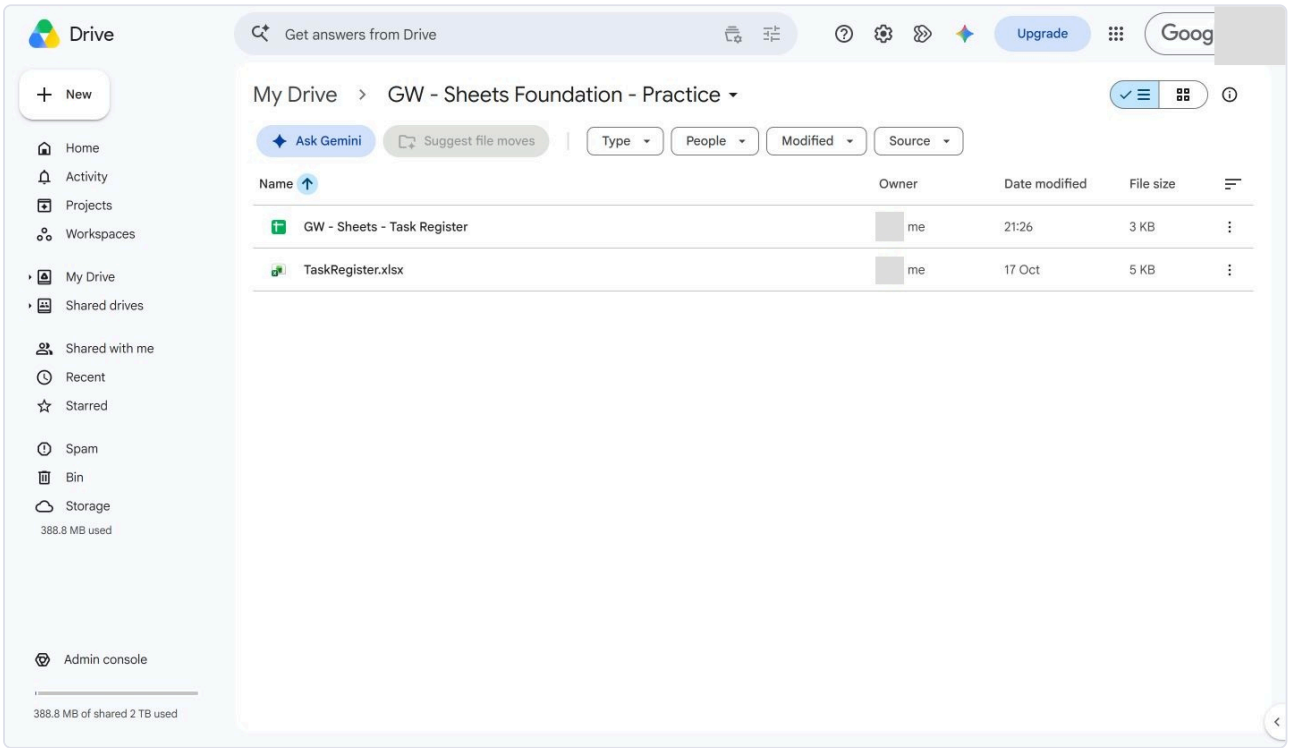


	A	B	C	D
1	Tab	Purpose and source status		
2	Tasks	Maintained source: 12 records in A1:H13; use TaskID to identify each record.		
3	Summary	Summary area supplied with the starter; calculations are built in later labs.		
4	Navigation Practice	Duplicate for navigation experiments; J1 is Practice only.		
5				
6				
7	Address	Column meaning	Visible value	TaskID
8	A2	TaskID	T001	T001
9	D5	Owner	Aina Rahman	T004
10	F8	Status	Complete	T007
11	H13	ActualCost	0	T012
12				
13	Check	Result		
14	Reconstruction	T001, T002 and T003 checked by TaskID: all eight fields match Tasks; no copying error found.		
15	Readability repair	Widened Entry Practice task column so complete task text is visible; values unchanged.		
16	Single cell	Use a cell address for one value, such as B2.		
17	Whole row	Select 2:2 for the complete row; do not move isolated field cells within a record.		
18	Whole column	Select C:C to operate on the Task field as a column.		
19	Header selection	Use A1:H1 to format only the header, not the records below.		
20	Small table selection	A1:H4 covers the eight headers and three recreated records.		

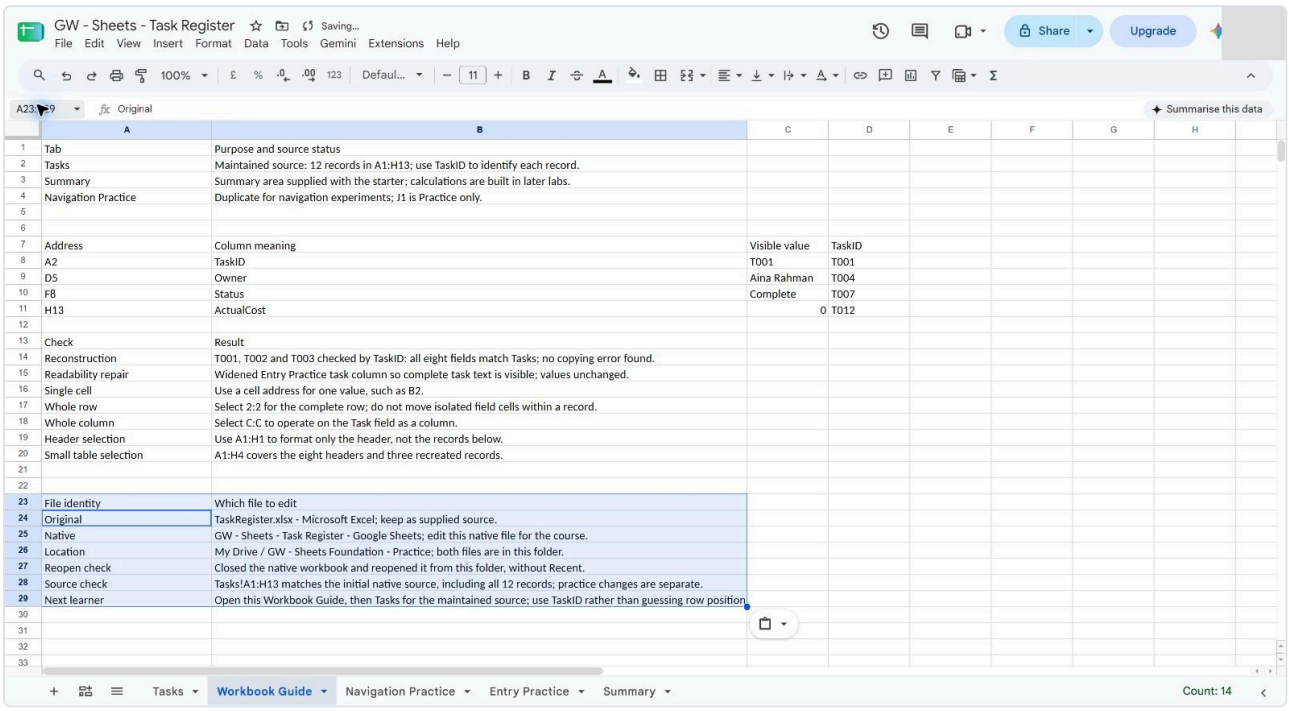
6. Use the Entry Practice tab menu **Move right** to place it after **Navigation Practice**. Close the native workbook, return to the practice folder in Drive and reopen the file by its name and type. This tests retrieval from the folder rather than using Recent.

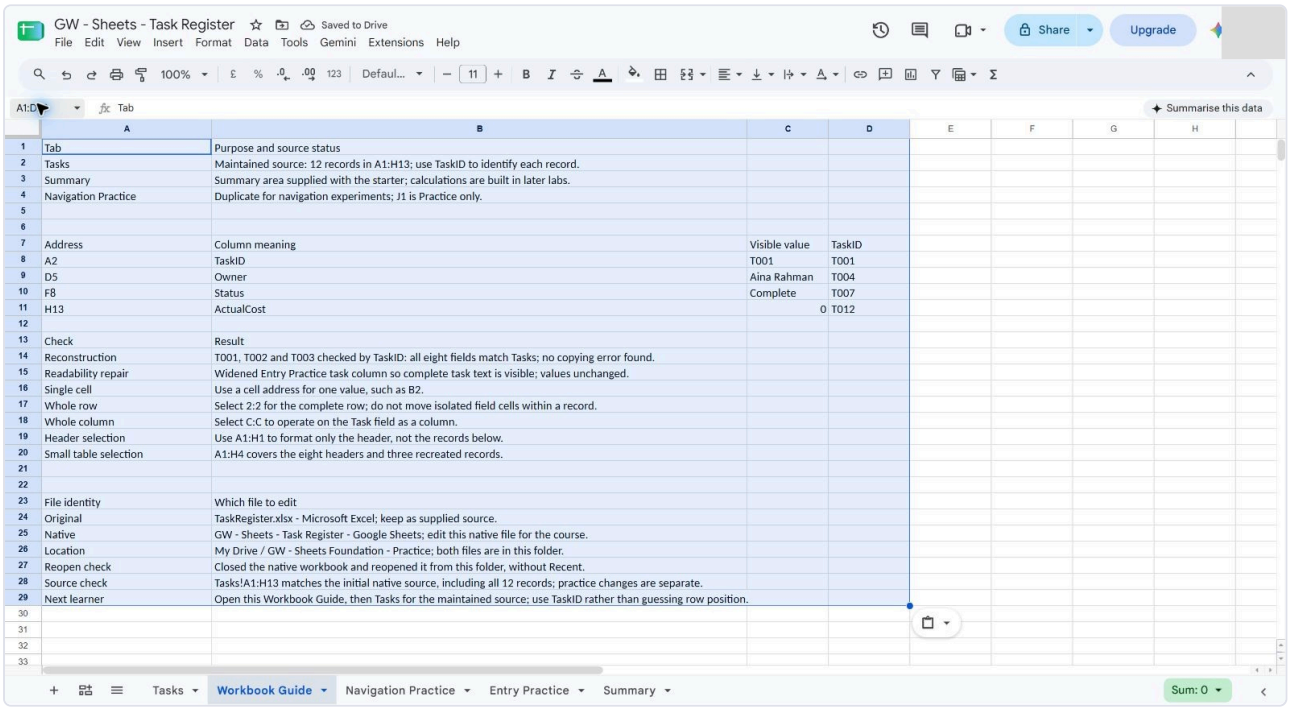


TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	600	550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	800	700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	500	0



7. In **Workbook Guide**, record both identities: **TaskRegister.xlsx — Microsoft Excel**, retained as supplied source; **GW - Sheets - Task Register — Google Sheets**, the course working file. Both are in **My Drive / GW - Sheets Foundation - Practice**. Tell the next learner to open Workbook Guide, then Tasks, and identify records by TaskID. Confirm the map and checks survived reopening.





What should you see?

Workbook Guide with four address-to-record checks. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Without using Recent, reopen the native file and locate the owner and actual cost for T008 using its record identity.
2. Create a two-row **New Entry Form** practice table with blank inputs and clear field labels. Do not append unfinished rows to Tasks.
3. Explain and demonstrate a cell address, a range, a tab and the source file identity. Have a partner use the guide or test it yourself after closing the workbook.

Evidence to Hand Over

- Workbook Guide with four address-to-record checks.
- Recreated three-row entry table and a corrected/verified copy record.
- Independent source retrieval and a labelled new-entry practice area.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
An address is mistaken for a permanent task identity	Use TaskID to identify a record because sorting can move it to another row.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

You can navigate and manage a native workbook. Next, make its data readable without changing its meaning.

Check yourself: What is the difference between a spreadsheet file and a worksheet tab?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
GW - Sheets Foundation - Practice
```

```
GW - Sheets - Task Register
```

```
A1:H13
```

```
Navigation Practice
```

```
Practice only
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Beginner navigation brief: You inherit an XLSX, a native copy and several tabs. Another learner must be able to identify the maintained source and find T008's owner and actual cost without guessing from row position.
```

Workbook Guide Reference

Address	Meaning	Value	TaskID
A2	TaskID	T001	T001
D5	Owner	Aina Rahman	T004
F8	Status	Complete	T007
H13	ActualCost	0	T012

Tab purposes: **Tasks** is the maintained twelve-record source; **Summary** is the supplied summary area; **Navigation Practice** is the duplicate for experiments; **Entry Practice** is a three-row reconstruction; **Workbook Guide** records the map and checks. Final reference order: Tasks, Workbook Guide, Navigation Practice, Entry Practice, Summary.

Entry Practice Values

Enter these eight fields without moving their positions. Compare each row with Tasks by TaskID.

```
TaskID  Team    Task    Owner   DueWeek Status  PlannedCost     ActualCost
T001    Planning        Confirm reporting fields        Aina Rahman     1       Complete
600     550
T002    Planning        Draft status definitions        Bala Kumar      1       In Progress
800     700
T003    Planning        Review pilot brief      Siti Aminah     1       Not Started     500
0
```

Verification record: all eight fields for T001, T002 and T003 matched; no copying error found. Task text was clipped on the new tab and its column was widened; values remained unchanged. B2 selects one cell, 2:2 a whole row, C:C a whole column, A1:H4 the small table and A1:H1 only its header.

Edit **GW - Sheets - Task Register (Google Sheets)** in **My Drive / GW - Sheets Foundation - Practice**. Retain **TaskRegister.xlsx (Microsoft Excel)** as the supplied source. The author reopened the native file from this folder and confirmed Tasks!A1:H13 unchanged. The instructor performed the upload manually after an extension permission block; automated browser upload, independent challenge and classroom timing remain unverified.

02

CHAPTER 2

Enter and Format Reliable Data

In this chapter you will use dates, numbers, currency, wrapping and sensible column widths.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 45 minutes: 10 explanation/demonstration, 15 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 35 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A readable register with numeric currency values and sensible dimensions.

> **Status:** partial author reference with 18 staged application captures through 2.4. Currency/week formats, wrapping/row fit, numeric versus text tests, SUM/COUNT and format-only behaviour checked. Values-only target-format preservation did not pass in the controlled session and needs manual verification. Source values remain unchanged; independent challenge and learner timing are pending.

What You Will Learn

- Apply formats to typed numbers.
- Use wrapping and fitted dimensions.
- Distinguish appearance from the stored value.

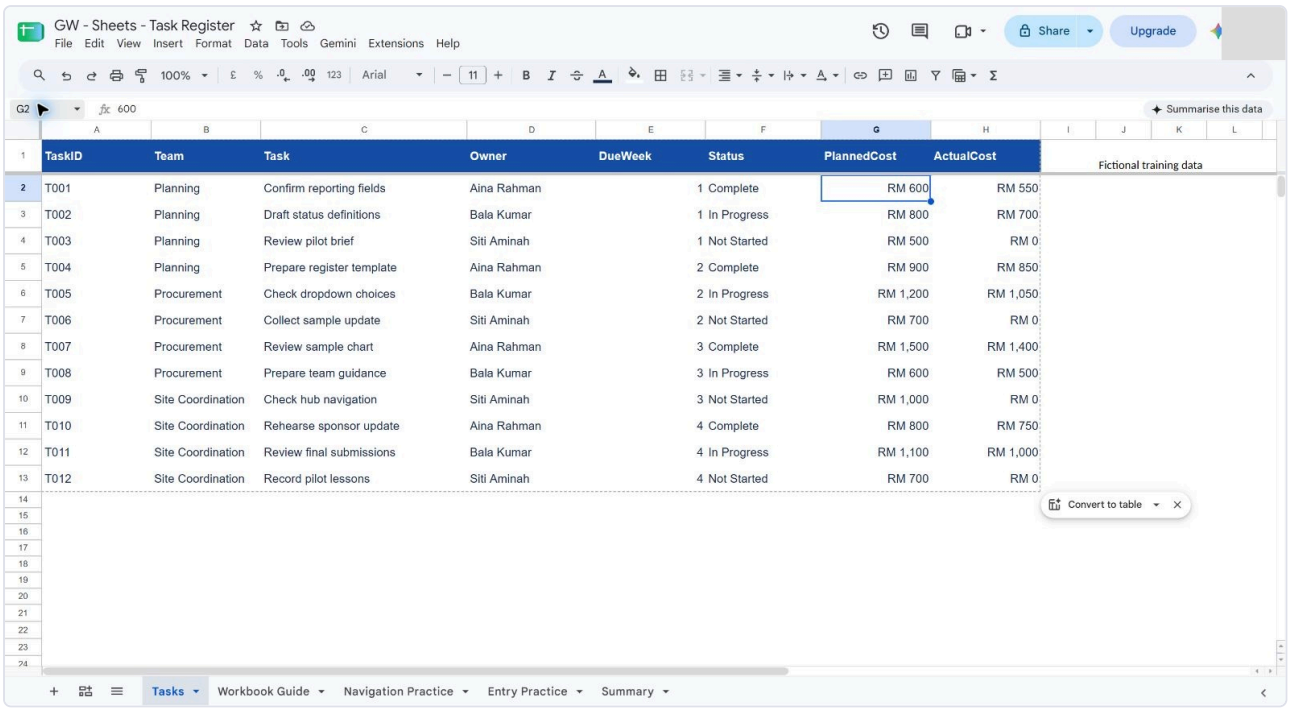
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use your original Tasks tab. If needed, reopen a fresh native copy of TaskRegister.xlsx.

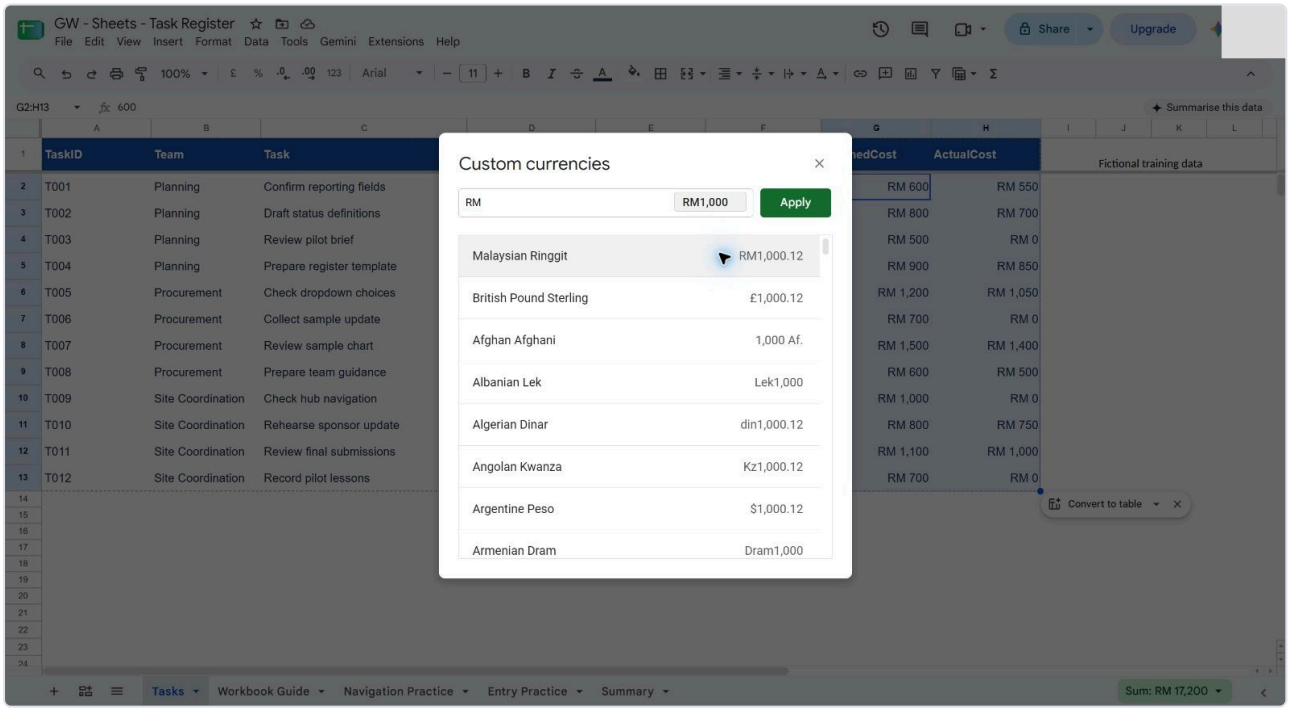
2.1 Inspect Values and Types

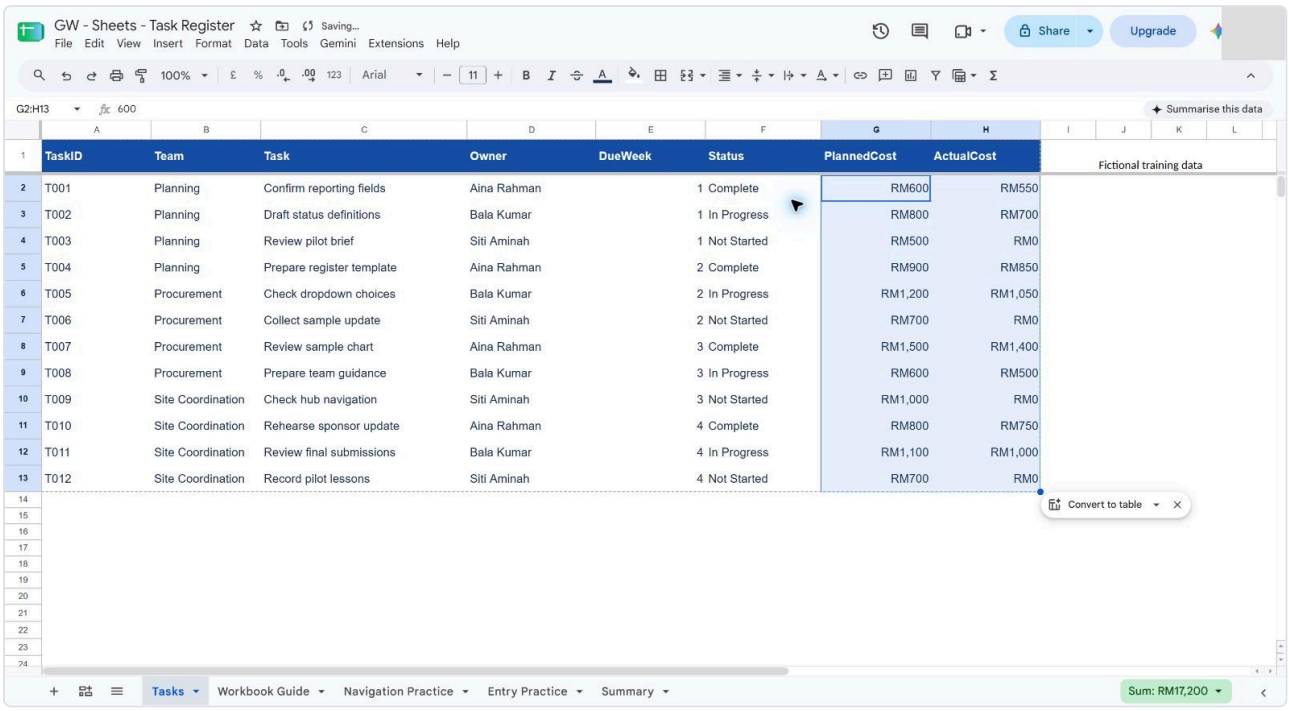
1. On **Tasks**, select **G2** through the name box. The formula bar shows numeric **600**, while the cell displays its currency. Do not type the RM label into the source number.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM 600	RM 550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM 800	RM 700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM 500	RM 0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM 900	RM 850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM 1,200	RM 1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM 700	RM 0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM 1,500	RM 1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM 600	RM 500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM 1,000	RM 0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM 800	RM 750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM 1,100	RM 1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM 700	RM 0

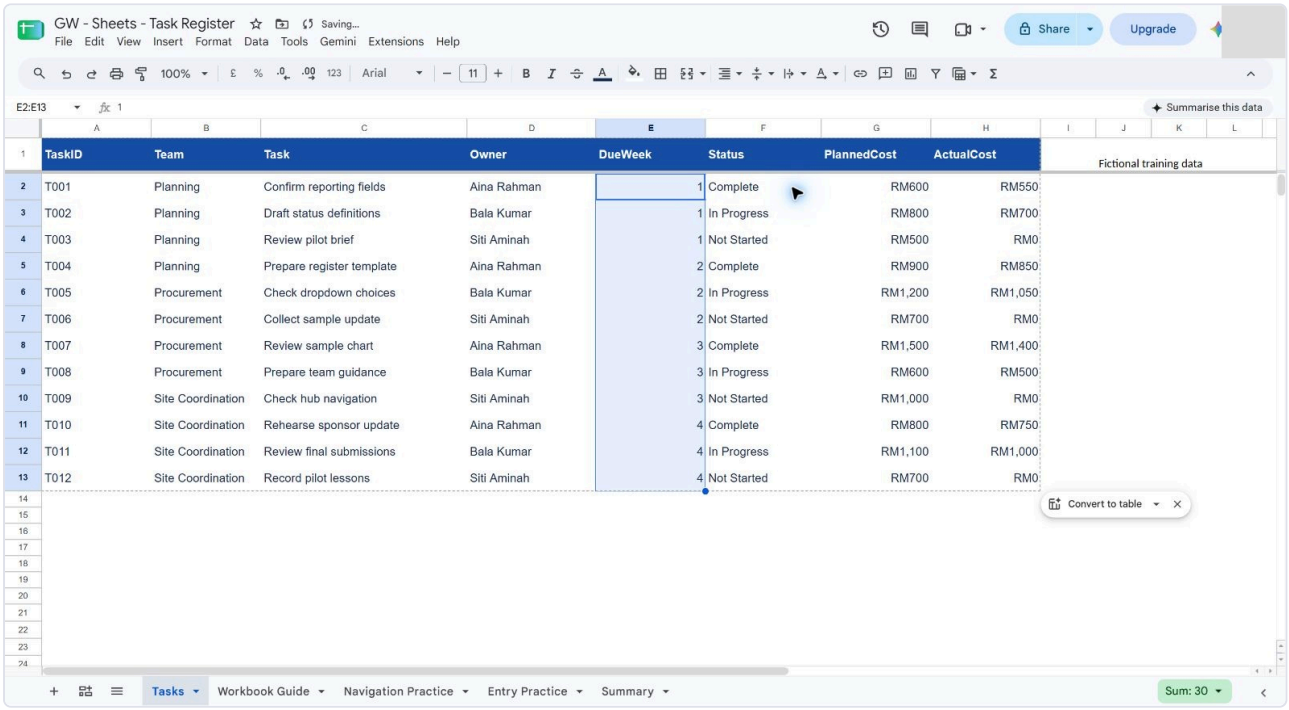
2. Select **G2:H13**. Open **More formats > Custom currency**, choose **Malaysian Ringgit**, select **RM1,000** without decimals, and **Apply**. The toolbar currency shortcut follows the workbook locale, so it may offer pounds; select RM explicitly.





TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM600	RM550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM800	RM700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM500	RM0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM900	RM850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM1,200	RM1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM700	RM0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM1,500	RM1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM600	RM500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM1,000	RM0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM800	RM750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM1,100	RM1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM700	RM0

3. Select **E2:E13**, use **More formats > Custom number format**, enter **0** and **Apply**. DueWeek remains an index from 1 to 4, not a calendar date.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM600	RM550
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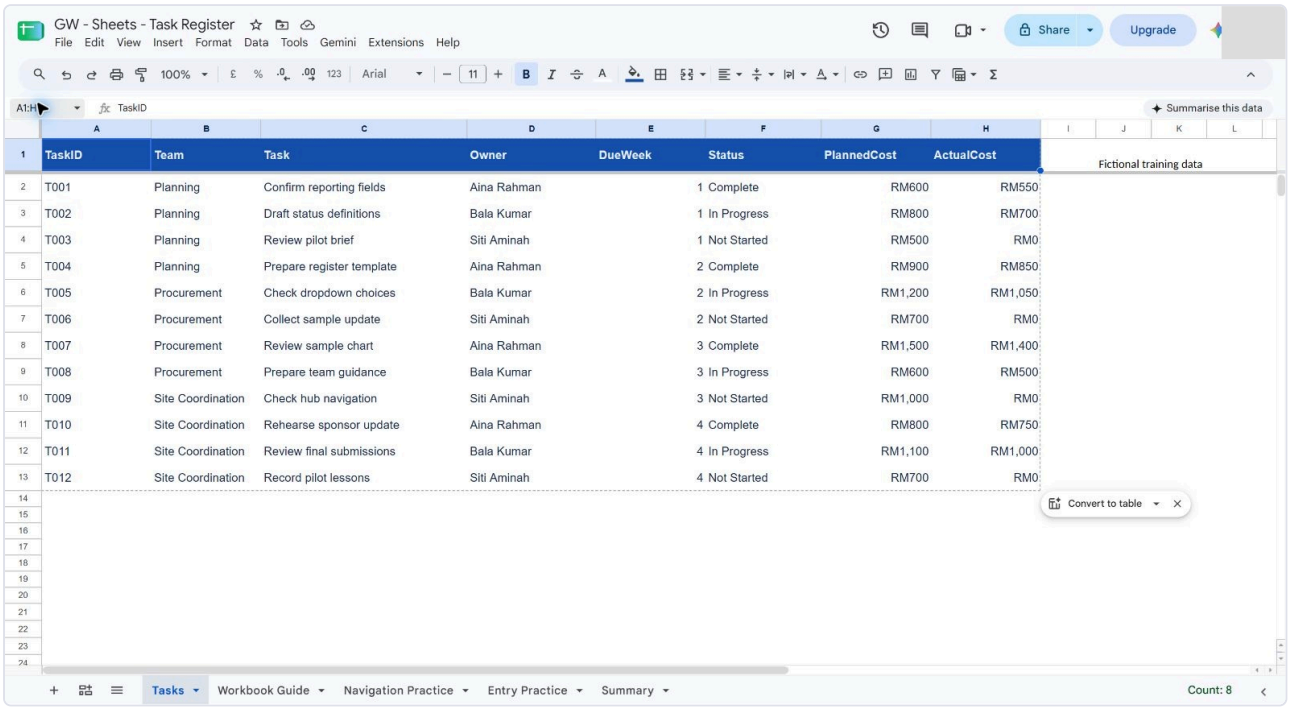
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM500	RM0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM900	RM850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM1,200	RM1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM700	RM0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM1,500	RM1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM600	RM500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM1,000	RM0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM800	RM750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM1,100	RM1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM700	RM0

What should you see?

Costs show their currency while remaining numeric. Week values still range from 1 to 4.

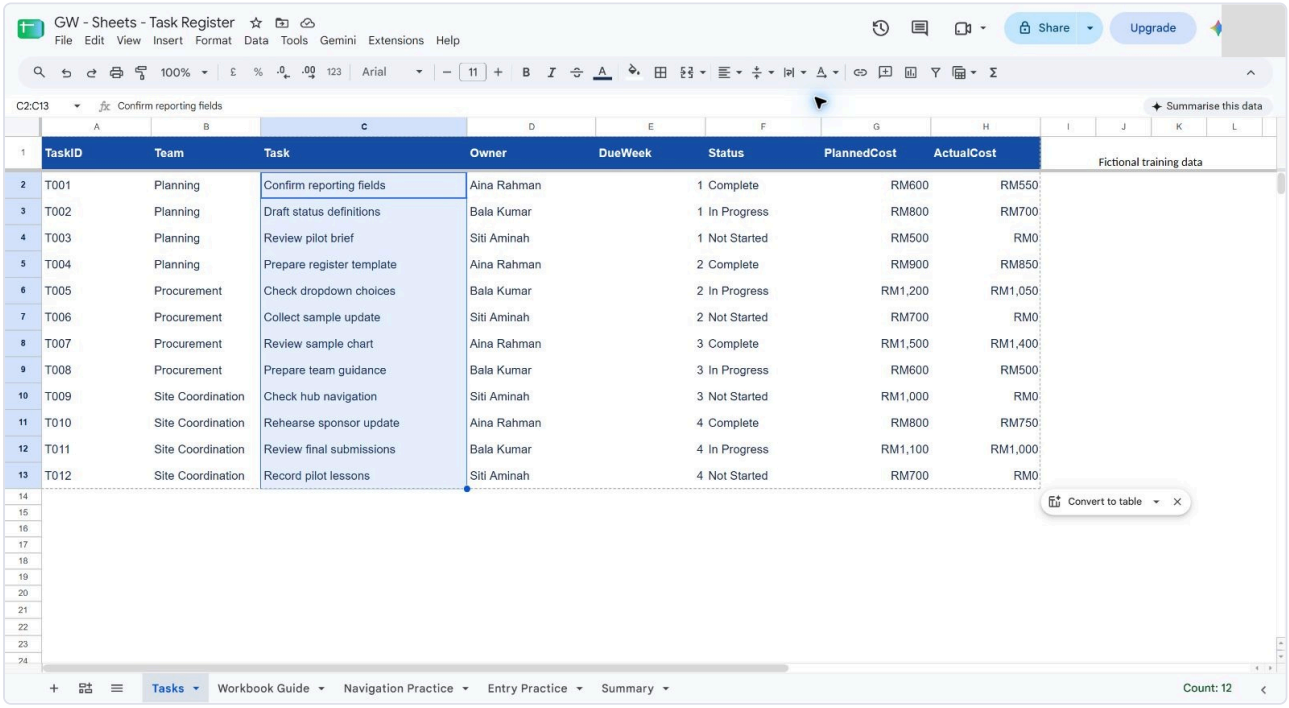
2.2 Improve Readability

1. Select **A1:H1** and inspect its contrast. The starter already has bold white labels on a dark blue fill, so the reference retains that suitable style. A reasoned no-change decision is preferable to changing a readable header just to demonstrate a button.



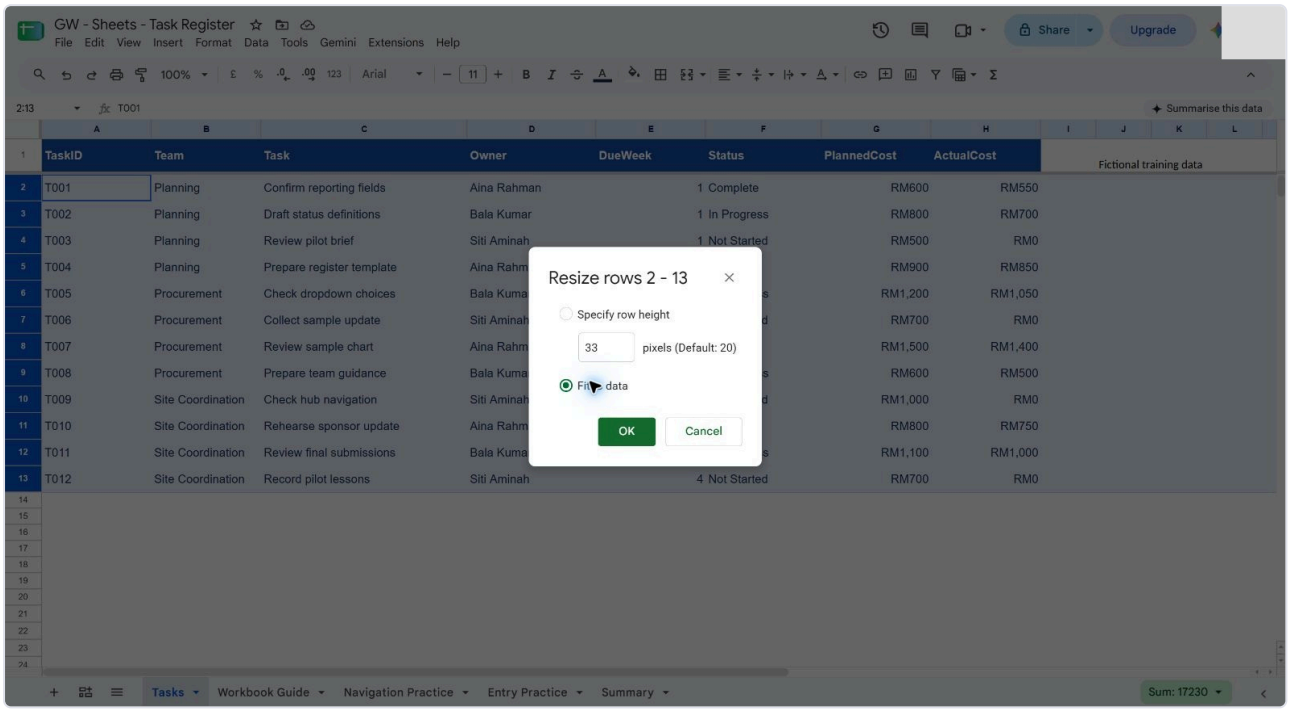
TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM600	RM550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM800	RM700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM500	RM0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM900	RM850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM1,200	RM1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM700	RM0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM1,500	RM1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM600	RM500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM1,000	RM0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM800	RM750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM1,100	RM1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM700	RM0

2. Inspect Task and Owner widths and widen only if text is cut off. The supplied source widths already show each task and owner. Select **C2:C13**, choose **Text wrapping > Wrap**, and inspect complete descriptions.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM600	RM550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM800	RM700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM500	RM0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM900	RM850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM1,200	RM1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM700	RM0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM1,500	RM1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM600	RM500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM1,000	RM0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM800	RM750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM1,100	RM1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM700	RM0

3. Select rows **2:13**, open the row-header menu, choose **Resize rows 2–13 > Fit to data > OK**. Keep each data column separate; do not merge cells in the register.

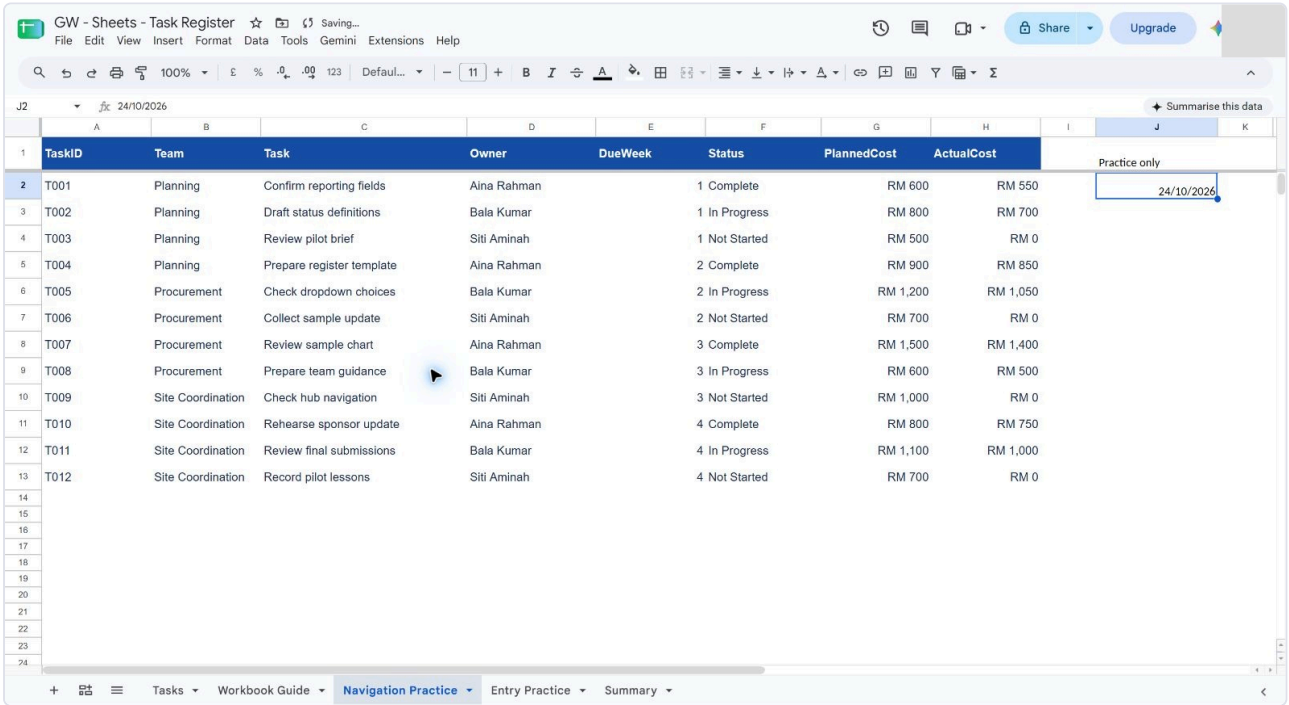


What should you see?

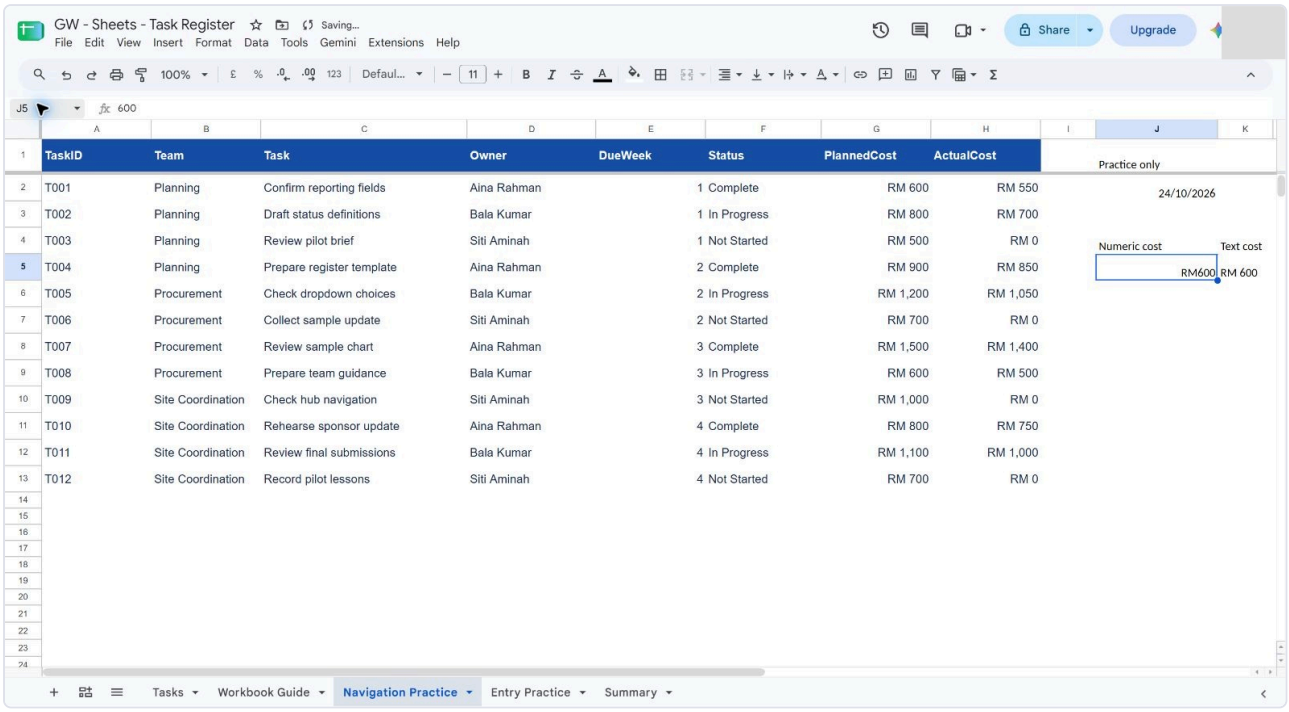
Headers, names and cost figures are readable without cutting off text.

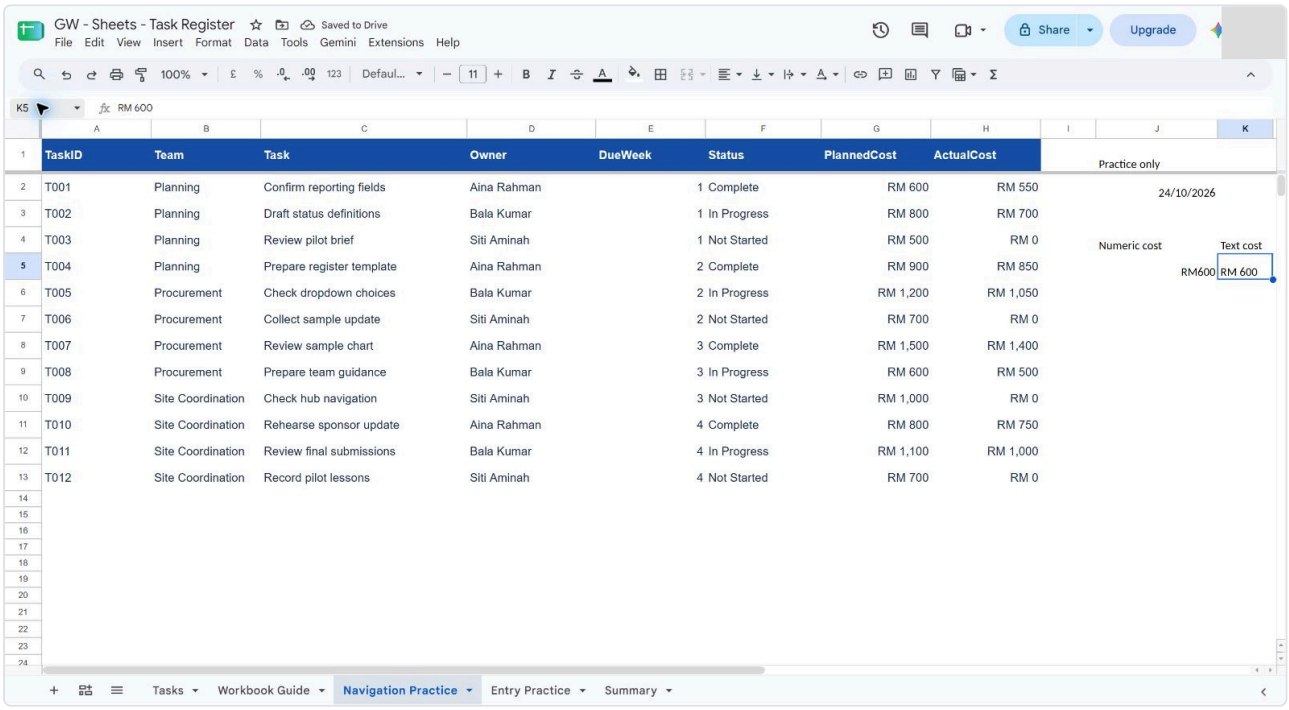
2.3 Inspect Formatting Effects

1. On **Navigation Practice**, enter **2026-10-24** in **J2**, commit it, then use **More formats > Date**. In the reference locale it displays **24/10/2026**. This is a practice example only, not an agreed pilot start.



2. Enter **Numeric cost / Text cost** in **J4:K4**, then numeric **600** in **J5** and text **RM 600** in **K5**. Apply RM currency to J5. Select each cell and compare its formula-bar value.





TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM 600	RM 550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM 800	RM 700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM 500	RM 0
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM 900	RM 850
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM 1,200	RM 1,050
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM 700	RM 0
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM 1,500	RM 1,400
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM 600	RM 500
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM 1,000	RM 0
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM 800	RM 750
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM 1,100	RM 1,000
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM 700	RM 0

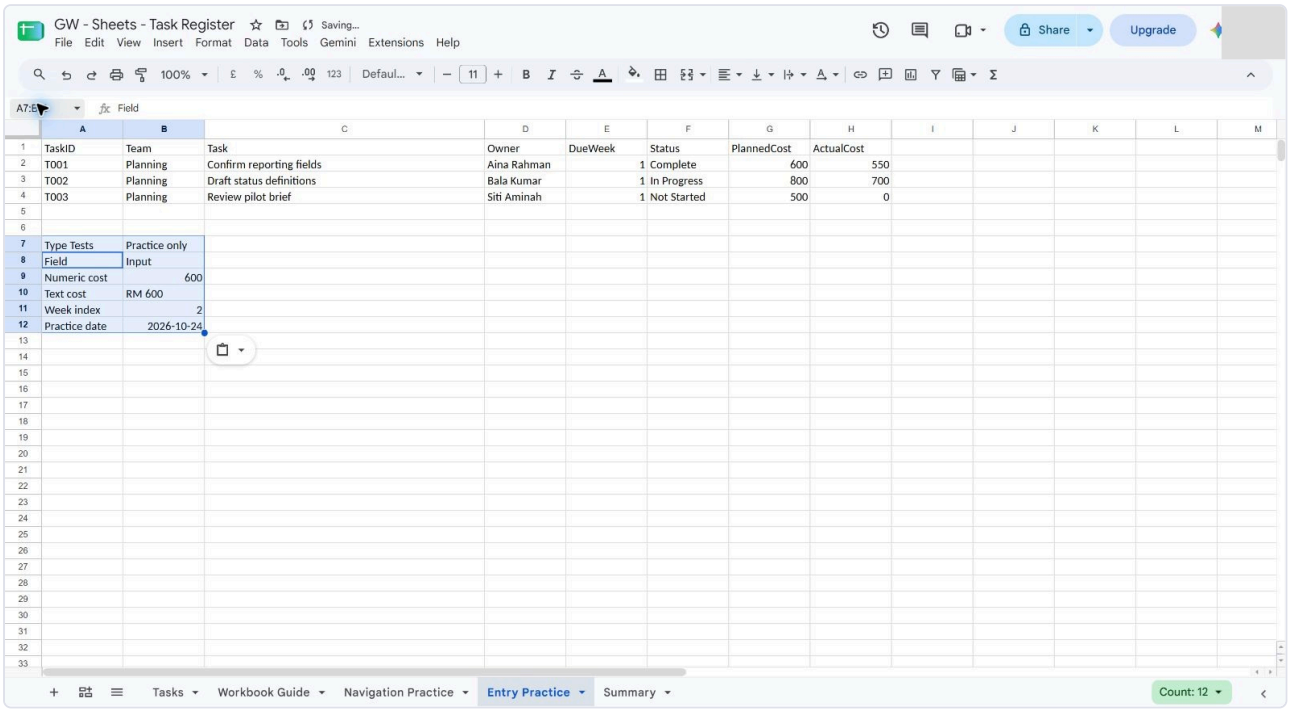
3. Keep these clearly labelled practice entries as evidence, or undo them when no longer useful. The reference retains them outside the baseline. Compare the original source values before and after formatting; display changes must not alter the twelve records.

What should you see?

You can explain why number formatting does not convert arbitrary text into a number.

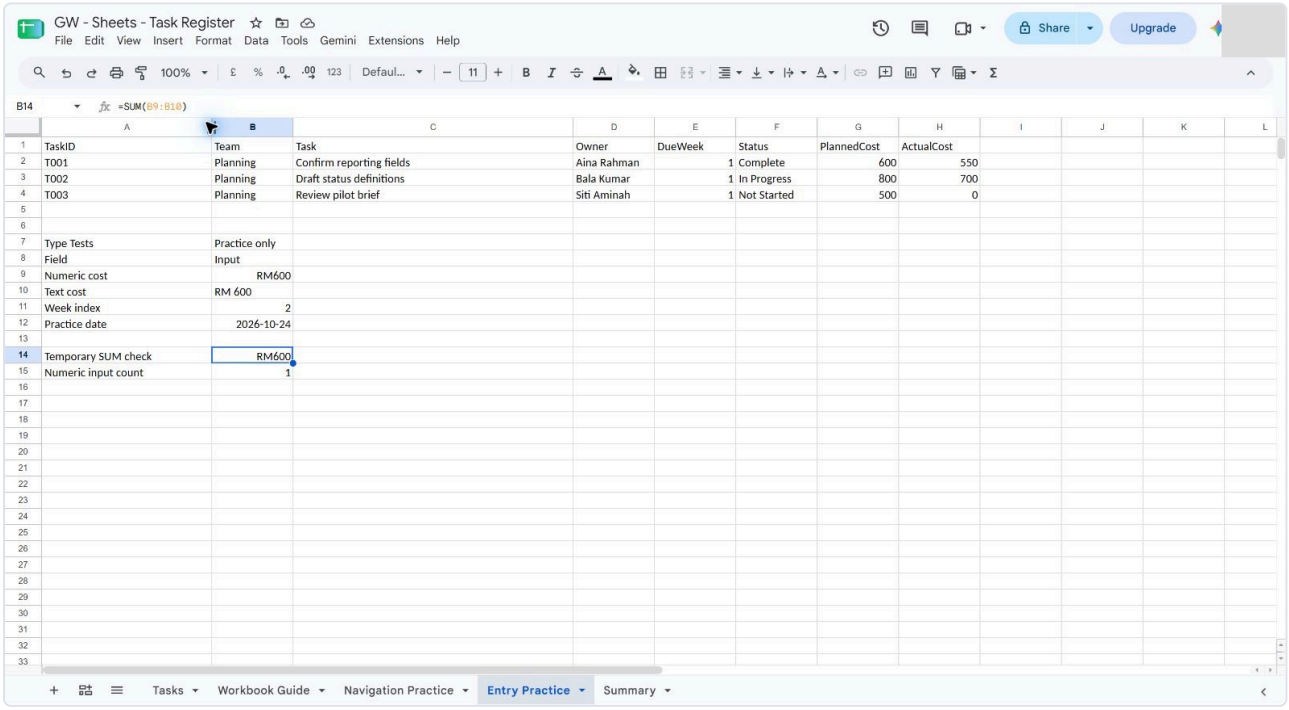
2.4 Diagnose Display, Type and Readability Problems

1. On **Entry Practice**, keep **A1:H4** as the copied three-record table. Create **Type Tests** in **A7:B12**: labels in A, numeric **600** in B9, text **RM 600** in B10, week index **2** in B11 and practice date **2026-10-24** in B12.

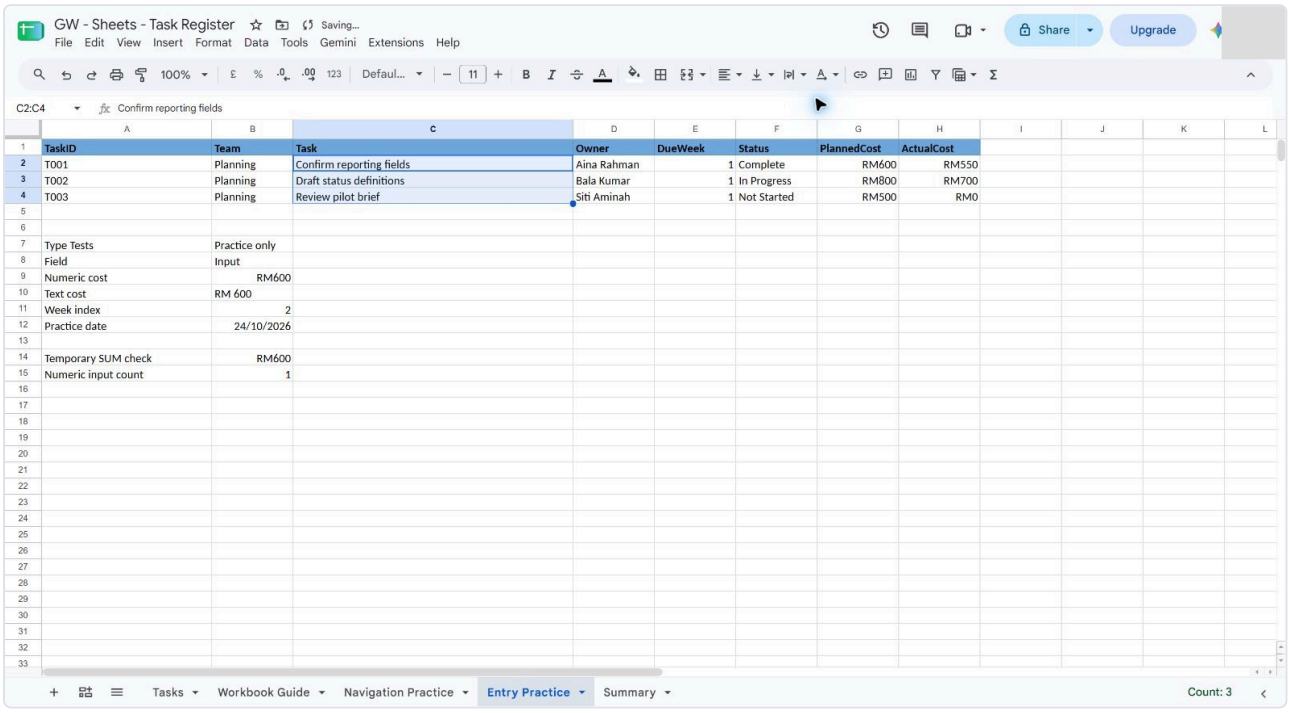


2. Apply RM currency to **B9:B10**, then inspect both values. B9 remains a number and B10 remains text. Give B12 a date display; keep the practice date outside the source week column.

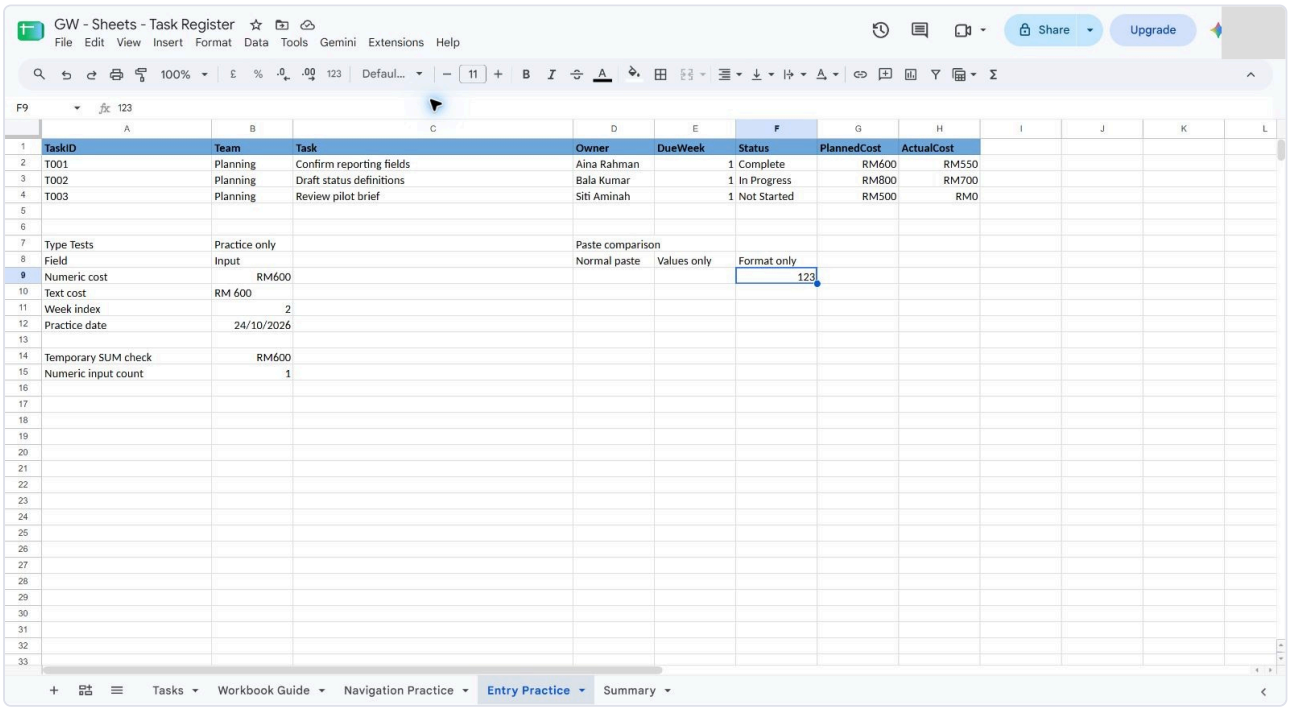
3. Add **=SUM(B9:B10)** in B14 and **=COUNT(B9:B10)** in B15. The reference results are **600** and **1**: numeric 600 contributes, while text RM 600 is ignored by this range SUM. Widen the label column if needed. These checks remain on the practice tab.



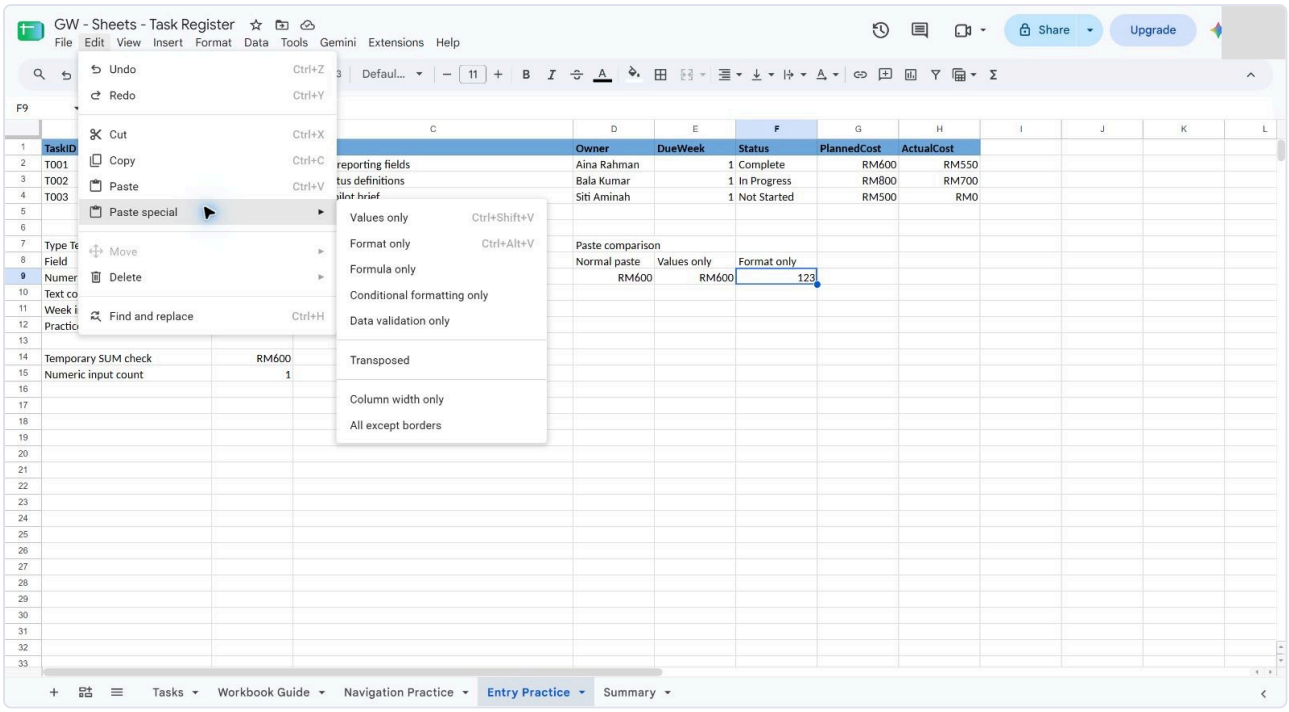
4. Give **G2:H4** the same custom RM currency display and **E2:E4** custom number format **0**. B11 remains a whole-number practice index. Dates stay in B12 and Navigation Practice J2; they do not supply missing source schedule evidence.
5. Apply a bold, light-blue header with readable dark text to **A1:H1**, wrap **C2:C4**, and retain the widened task column. Make each complete description readable. No source cells are merged.

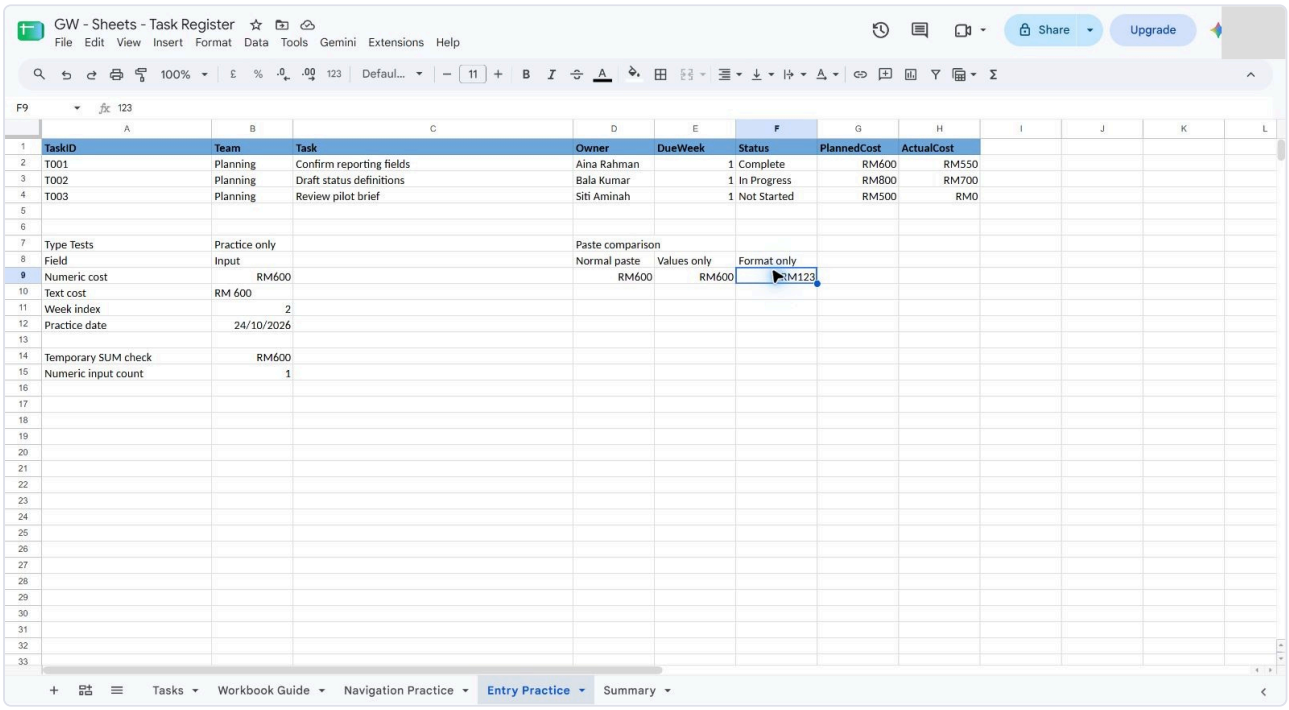


6. Prepare **D8:F9** as a separate paste comparison: headers **Normal paste / Values only / Format only**, with **123** already in F9. Use B9 (numeric 600 with RM display) as the source. Select destination cells without pasting a cell address over your copied clipboard content.



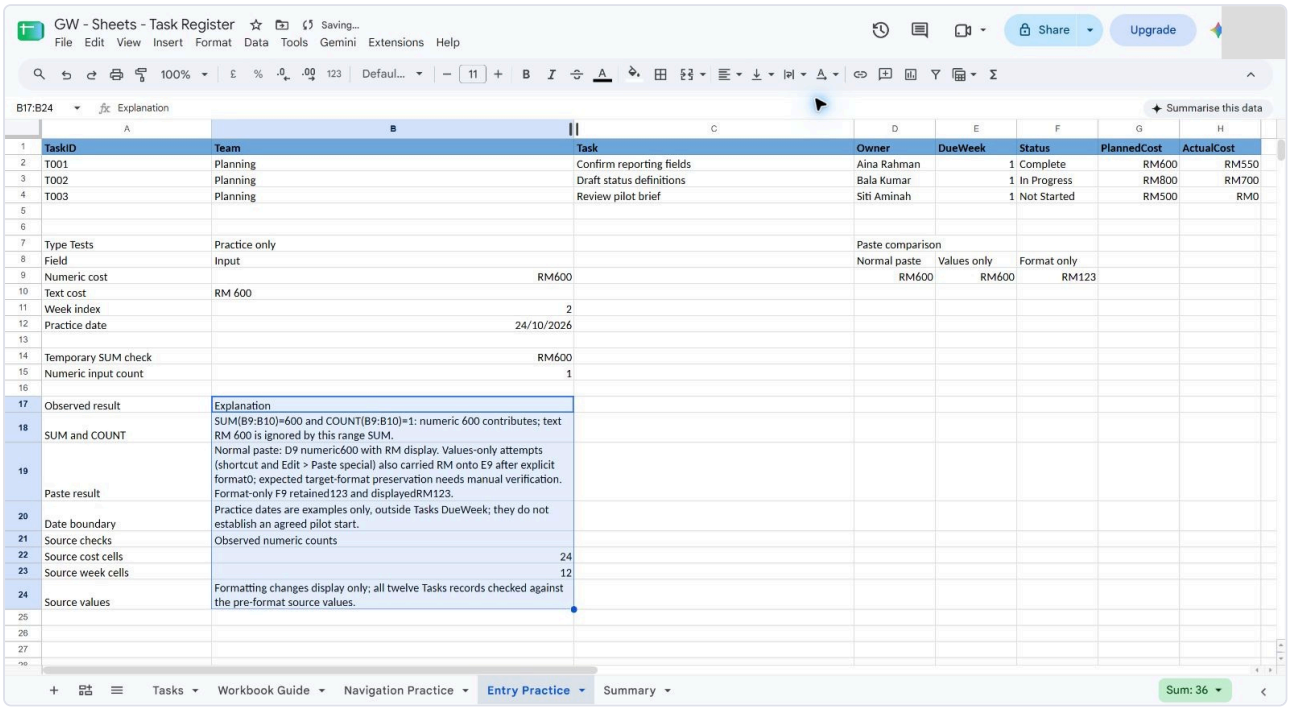
Use **Edit > Paste special** to test the two special transfers separately from normal paste. Normal paste D9 was verified as numeric 600 with RM display. Format-only F9 retained numeric **123** and displayed **RM123**.



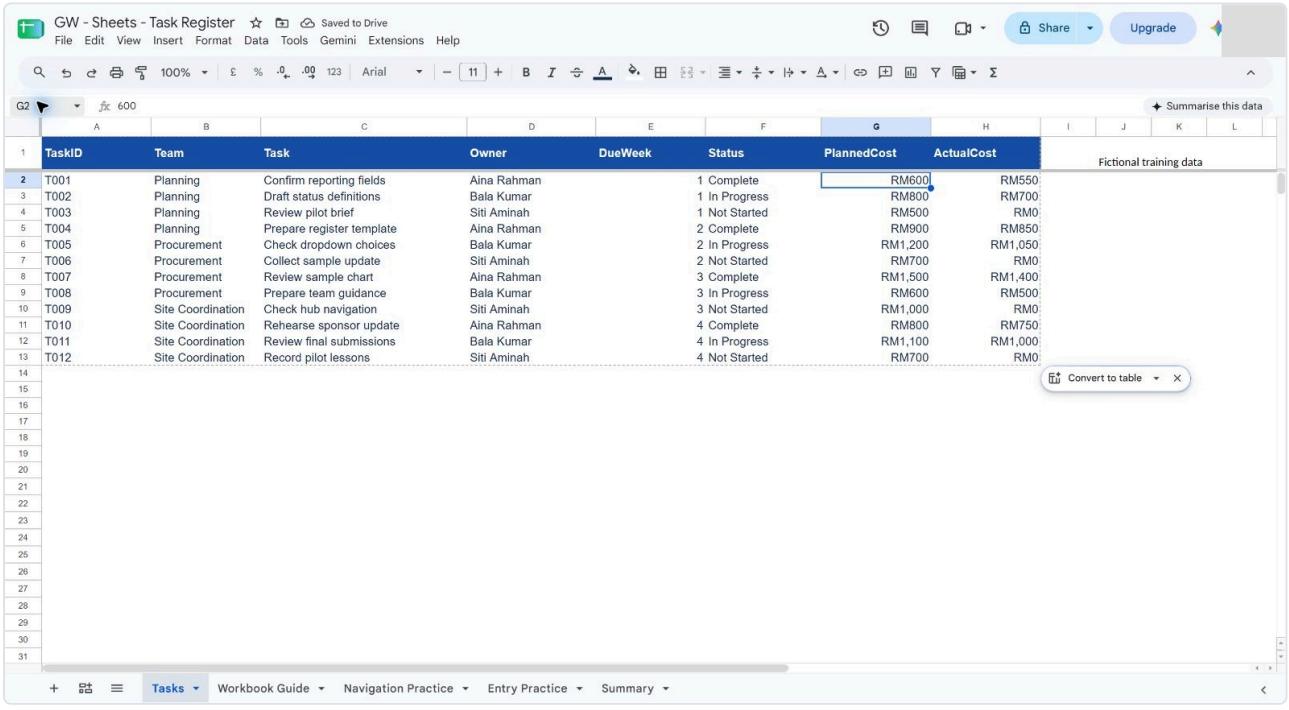


> **Verification limitation:** the controlled session's values-only shortcut and direct menu both carried the RM format onto E9 even after its target was explicitly set to number format 0. The number remained 600, but expected preservation of the target format was **not verified**. Repeat this step manually, compare E9's number format before and after, and record the outcome. Do not treat visually similar cells or an attempted command as proof that the property transfer passed.

7. Record the results and the unresolved paste check on the practice tab. **=COUNT(Tasks!G2:H13)** returns **24** numeric cost cells; **=COUNT(Tasks!E2:E13)** returns **12** numeric week cells. A bounded comparison confirmed that every original **Tasks!A1:H13** underlying value remained unchanged. Wrap the observation text so it is readable.



Return to **Tasks > G2** and inspect **600** again. Formatting did not create a confirmed calendar date or change the twelve records.



What should you see?

Type-test block with observed calculation behaviour. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Repair the card's three proposed entries in a separate intake table. Preserve raw input and record the clean value or clarification needed.
2. Prepare a readable three-record handout view with consistent currency and whole-week display. Prove the underlying numeric values remain numeric.
3. Give a partner two visually similar cost cells and ask them which one can be calculated reliably. Use formula-bar and SUM evidence in your explanation.

Evidence to Hand Over

- Type-test block with observed calculation behaviour.
- Raw-to-clean intake table and readable entry-practice layout.
- Independent numeric evidence and unchanged 12-row source.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
Currency looks correct but totals are wrong	Inspect numeric type and formula-bar value before changing more display formats.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Typed values and readable formatting prepare the register for calculation.

Check yourself: Why should you avoid merging cells inside a dataset?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Use the supplied source and the explicit practice values below. Keep dates and type tests outside the original register.

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Formatting intake: PlannedCost “RM 600” supplied as text; DueWeek “next Monday” with no agreed calendar start; a long Task description clipped in a narrow column. Treat type, missing date evidence and presentation as three different issues.
```

Exact Practice Values

- Source costs: Tasks!G2:H13, custom Malaysian Ringgit **RM1,000** without decimals.
- Source week indices: Tasks!E2:E13, custom number format **0**.
- Practice date: **2026-10-24**, displayed **24/10/2026** in the reference locale; not a confirmed project start.
- Navigation Practice: J2 date; J4:K4 Numeric cost / Text cost; J5 numeric600, K5 text RM 600.

Entry Practice type block: A7:B12, labelled numeric600 / text RM 600 / week2 / practice date. B14 **=SUM(B9:B10)** gives600; B15 **=COUNT(B9:B10)** gives1. Both formulas reference the cells as a range; the observed result does not establish behaviour for every possible way of passing text to a function.

Paste comparison: D9 normal paste from B9 gives numeric600 with currency format; F9 format-only retains123 and displaysRM123. **Values-only target-format preservation remains unverified**: both shortcut and menu carried the RM format in this controlled session after setting E9 to number format0. Repeat manually and compare formats as well as values. Do not claim this property-transfer check passed.

Source controls: **=COUNT(Tasks!G2:H13)** gives24; **=COUNT(Tasks!E2:E13)** gives12. All twelve source records retain their original underlying values. The source header already had readable contrast and its widths were retained; wrapping and Fit to data were applied. Entry Practice header/widths/wrapping were improved. Independent intake repair, partner test and classroom timing remain pending.

03

CHAPTER 3

Calculate and Copy Formulas

In this chapter you will use arithmetic, SUM, AVERAGE and COUNT; explain relative and absolute references.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 65 minutes: 15 explanation/demonstration, 22 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 50 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: Copied variance formulas, reconciled totals and a tested fixed reference.

> **Status:** author reference with 23 staged application captures through guided sections 3.1–3.3 and casework 3.4. Row formulas, summaries, fixed-reference change/restoration, zero/blank/text checks, deliberate reference repair and reconciliation verified. Independent estimate/omitted-range challenge and learner timing remain pending.

What You Will Learn

- Write arithmetic and aggregation formulas.
- Copy relative references correctly.
- Anchor a shared assumption.

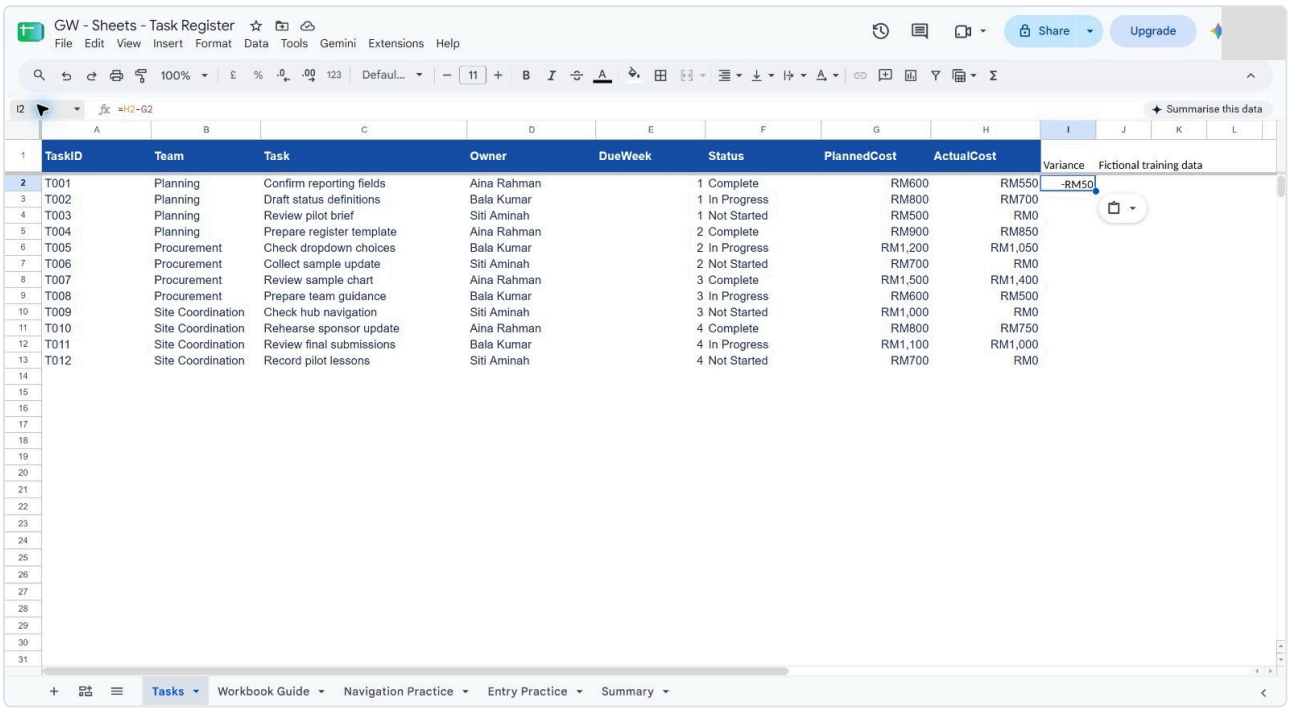
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Tasks A1:H13. The workbook has no prefilled answer formulas.

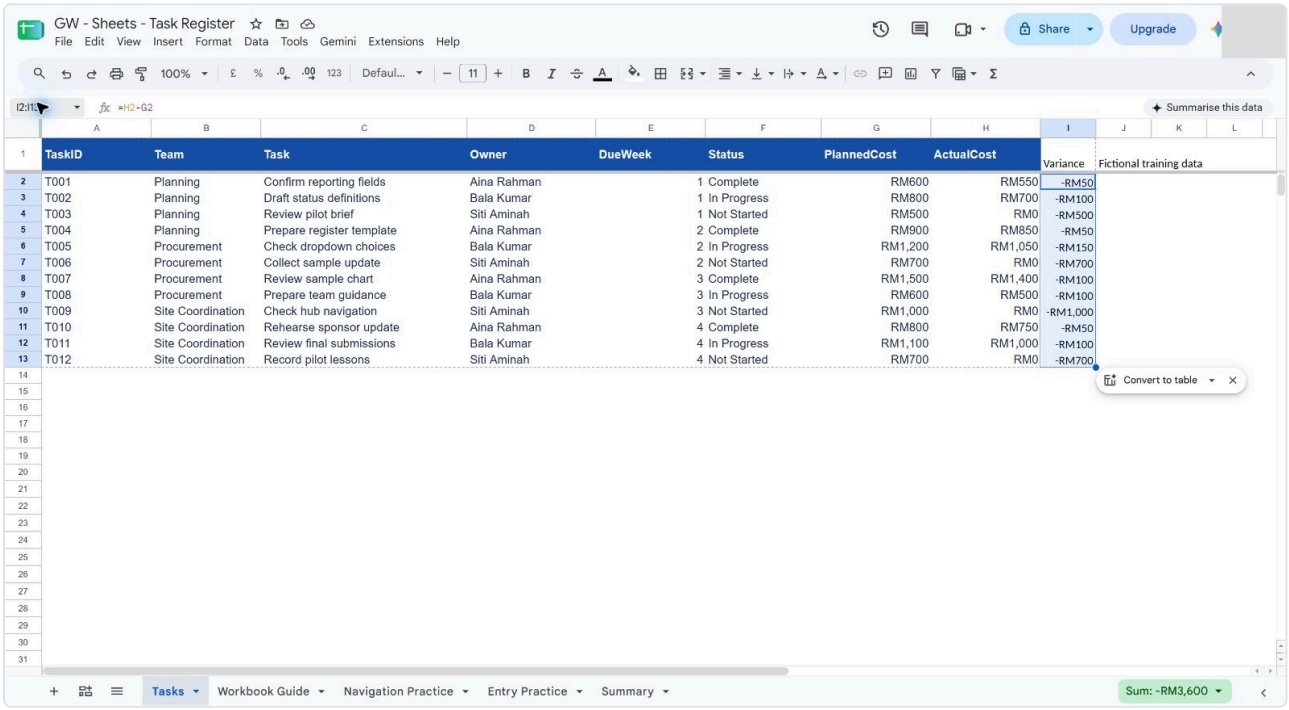
3.1 Calculate Variance

1. On **Tasks**, use the name box to select **I1** and enter **Variance**. In **I2** enter **=H2-G2** and commit it. T001 displays **-RM50**; the formula bar shows the underlying calculation.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost	Variance	Fictional training data
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM600	RM550	-RM50	
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM800	RM700		
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM500	RM0		
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM900	RM850		
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM1,200	RM1,050		
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM700	RM0		
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM1,500	RM1,400		
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM600	RM500		
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM1,000	RM0		
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM800	RM750		
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM1,100	RM1,000		
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM700	RM0		

2. Select **I2:I13** with the name box and press **Ctrl+D** to fill down from the first cell. The fill handle or normal copy/paste is an alternative. The range sum is **-RM3,600**.



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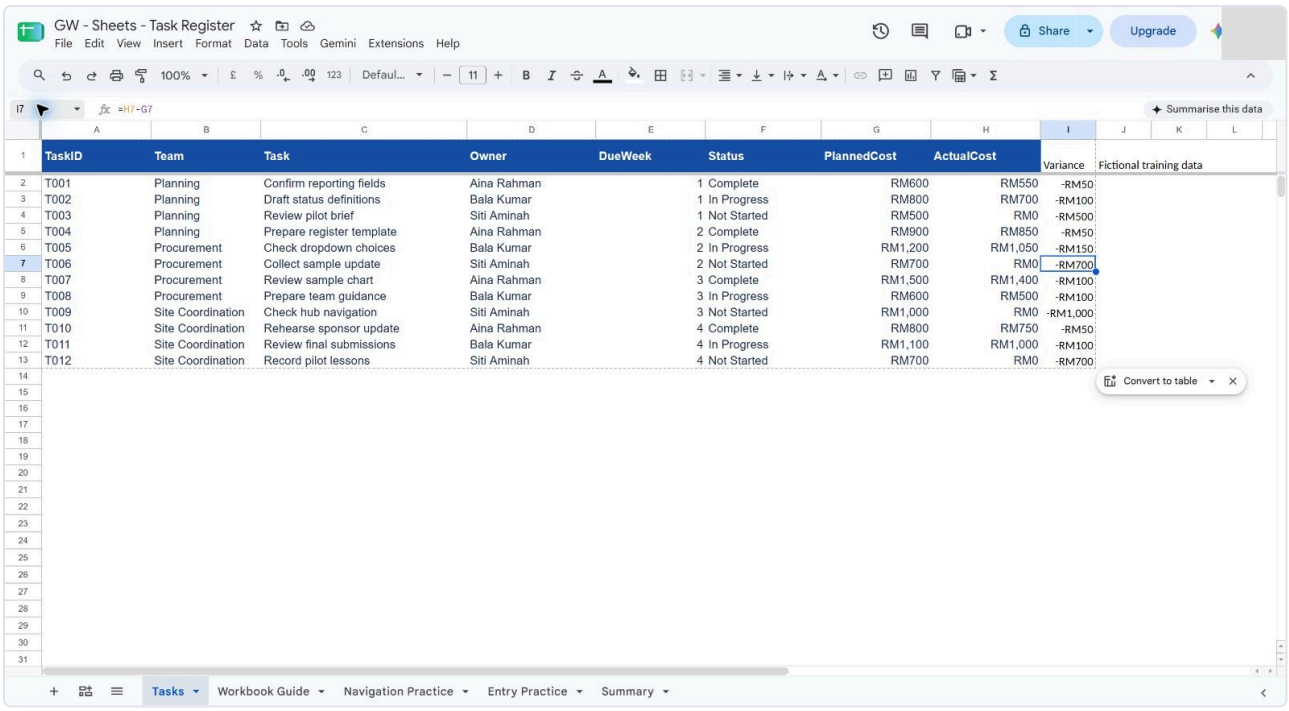
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM500	RM0	-RM500	
T004	Planning	Prepare register template	Aina Rahman	2	Complete	RM900	RM850	-RM50	
T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM1,200	RM1,050	-RM150	
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM700	RM0	-RM700	
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM1,500	RM1,400	-RM100	
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM600	RM500	-RM100	
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM1,000	RM0	-RM1,000	
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM800	RM750	-RM50	
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM1,100	RM1,000	-RM100	
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM700	RM0	-RM700	

3. Inspect **I2**, **I7** and **I13** in the formula bar. Each refers to its own row: **=H2-G2**, **=H7-G7** and **=H13-G13**. The final two checks display **-RM700**. A negative variance means actual cost is below planned; it does not by itself establish a final saving or project completion.



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T005	Procurement	Check dropdown choices	Bala Kumar	2	In Progress	RM1,200	RM1,050	-RM150	
T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM700	RM0	-RM700	
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM1,500	RM1,400	-RM100	
T008	Procurement	Prepare team guidance	Bala Kumar	3	In Progress	RM600	RM500	-RM100	
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM1,000	RM0	-RM1,000	
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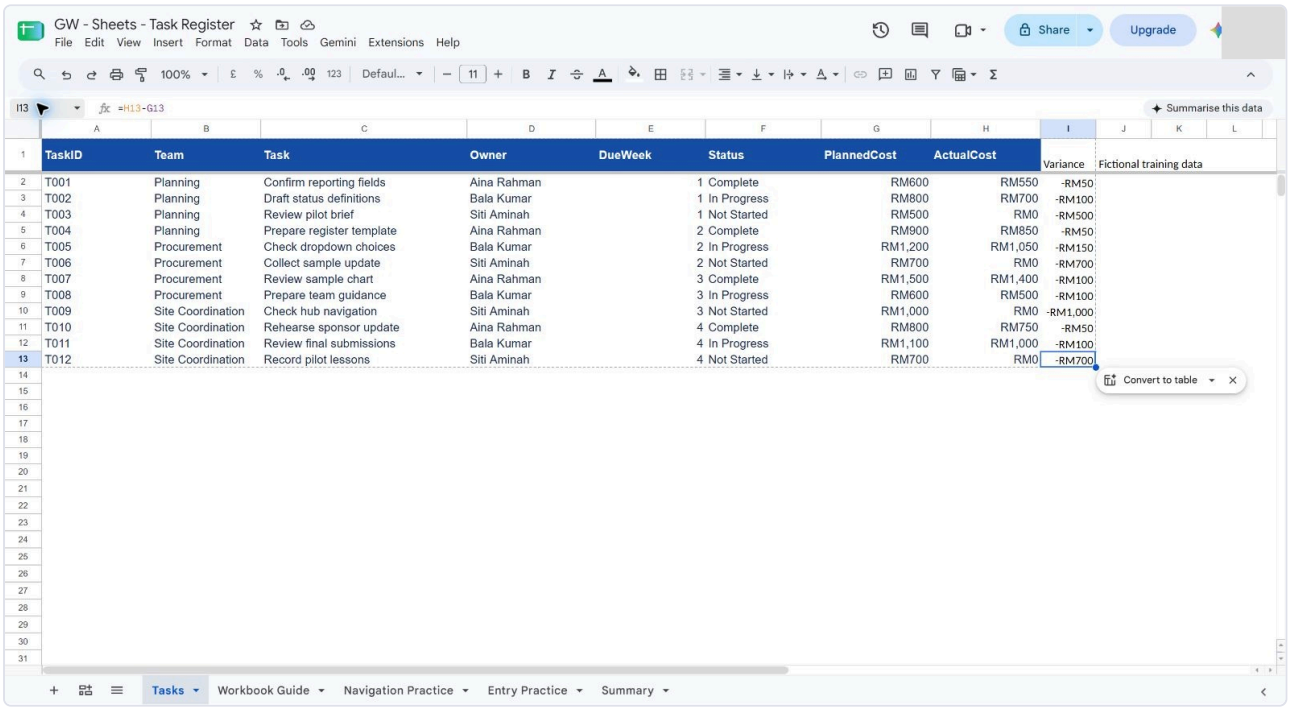
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM1,100	RM1,000	-RM100	
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM700	RM0	-RM700	



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost	Variance	Fictional training data
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T006	Procurement	Collect sample update	Siti Aminah	2	Not Started	RM700	RM0	-RM700	
T007	Procurement	Review sample chart	Aina Rahman	3	Complete	RM1,500	RM1,400	-RM100	
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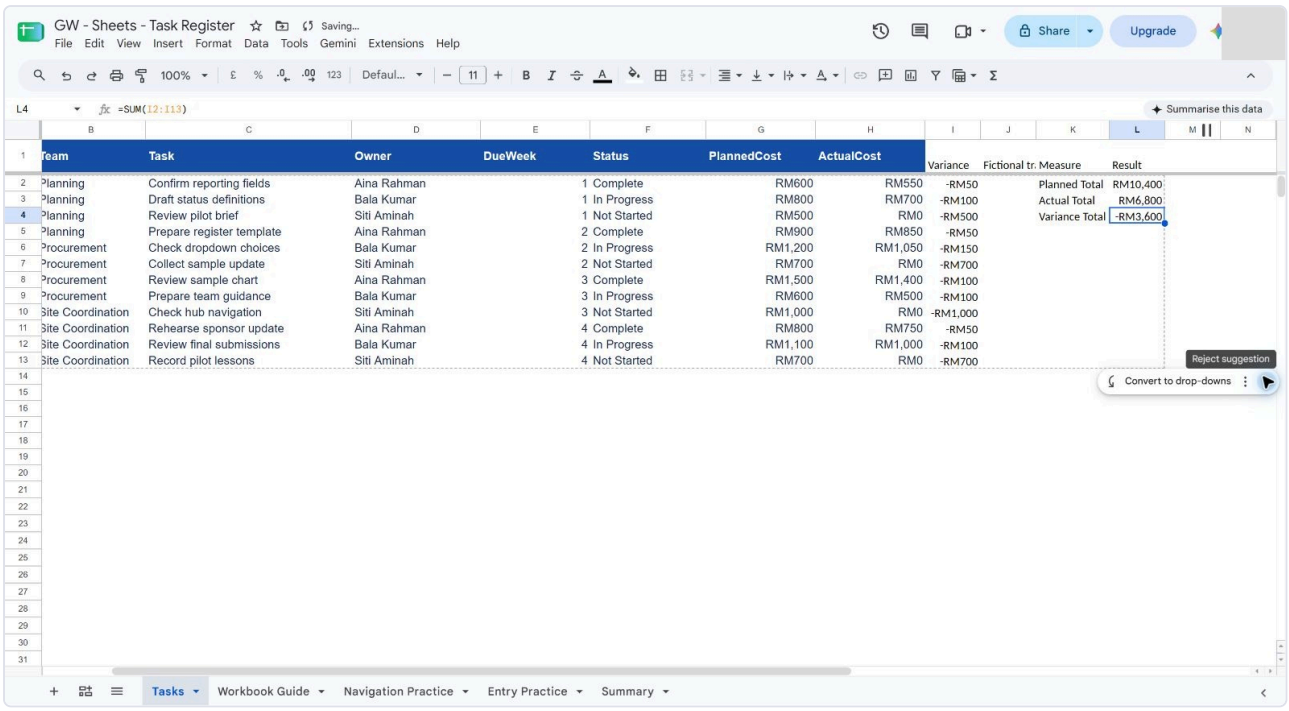
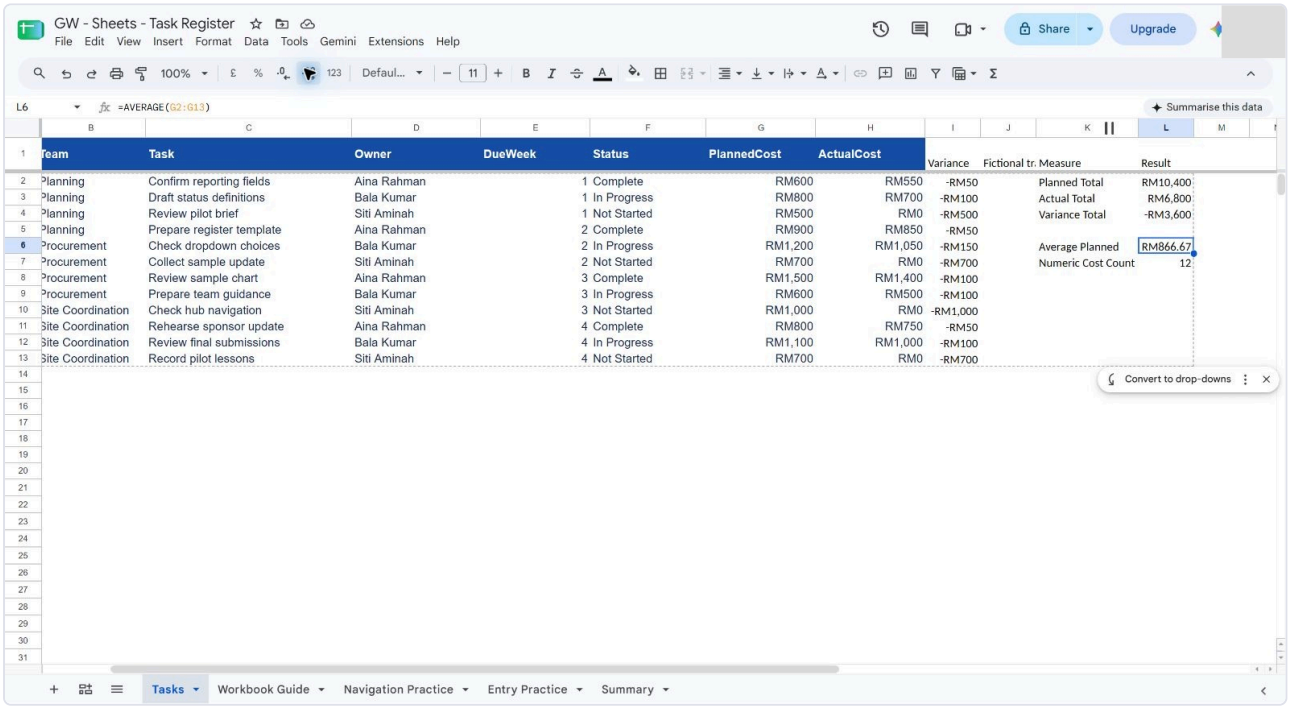
T009	Site Coordination	Check hub navigation	Siti Aminah	3	Not Started	RM1,000	RM0	-RM1,000	
T010	Site Coordination	Rehearse sponsor update	Aina Rahman	4	Complete	RM800	RM750	-RM50	
T011	Site Coordination	Review final submissions	Bala Kumar	4	In Progress	RM1,100	RM1,000	-RM100	
T012	Site Coordination	Record pilot lessons	Siti Aminah	4	Not Started	RM700	RM0	-RM700	

3.2 Summarise the Columns

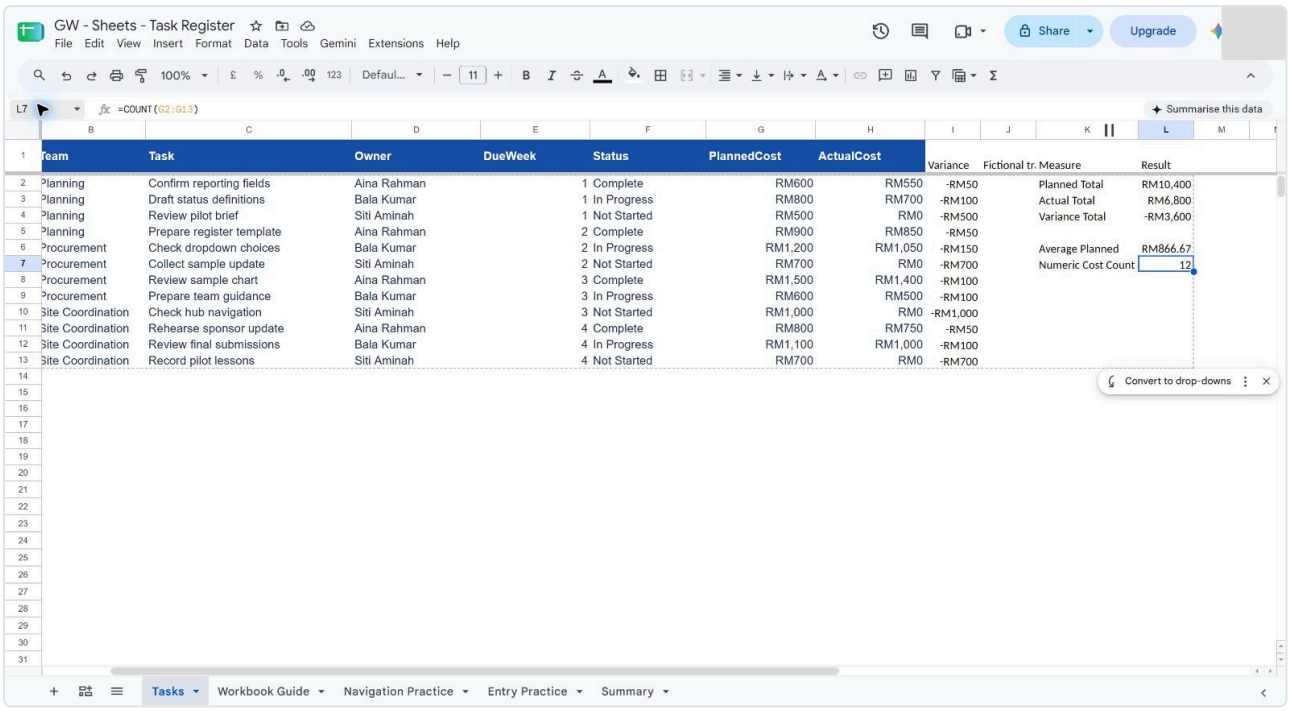
1. In **K1:L1** enter **Measure / Result**. Label **K2:K4 Planned Total / Actual Total / Variance Total**. In **L2:L4** enter **=SUM(G2:G13)**, **=SUM(H2:H13)** and **=SUM(I2:I13)**. Double-click the right edge of the K column header to fit its labels. Scroll horizontally when needed to show the complete summary.



2. Add **Average Planned** in K6 with **=AVERAGE(G2:G13)** in L6, and **Numeric Cost Count** in K7 with **=COUNT(G2:G13)** in L7. Fit K again after adding the longer labels. Select L6 and increase decimal places twice if it inherited the source's whole-ringgit display. The displayed average is **RM866.67**; its underlying calculation remains 10,400 divided by 12.

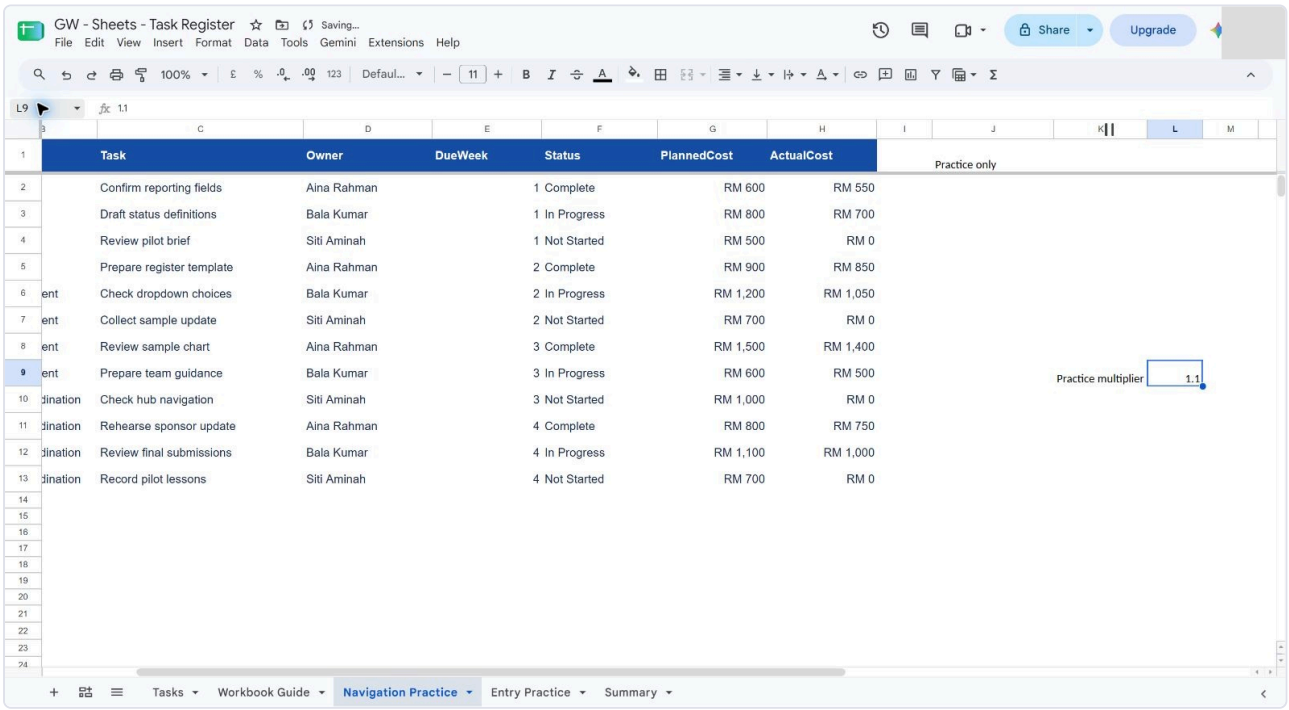


3. Inspect L7's formula and **12** result. The totals are **RM10,400 planned**, **RM6,800 actual** and **-RM3,600 variance**. Do not replace the formulas with those expected numbers.

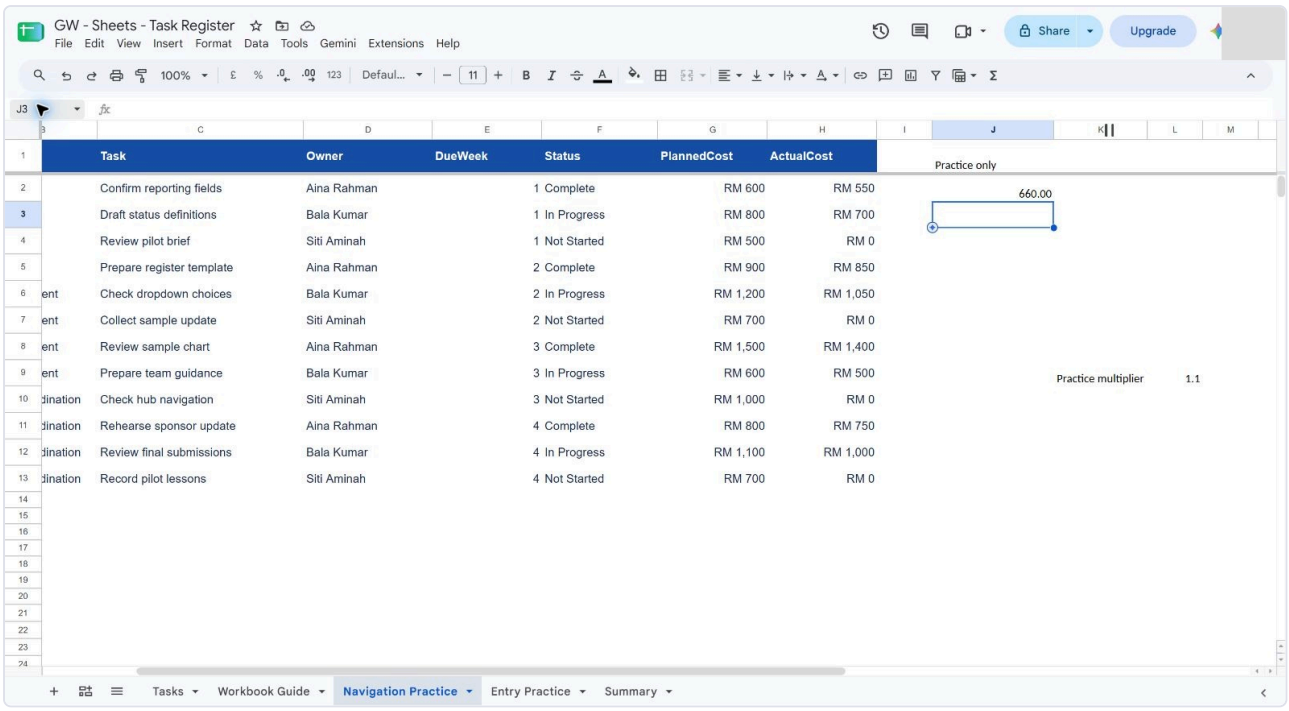


3.3 Anchor an Assumption

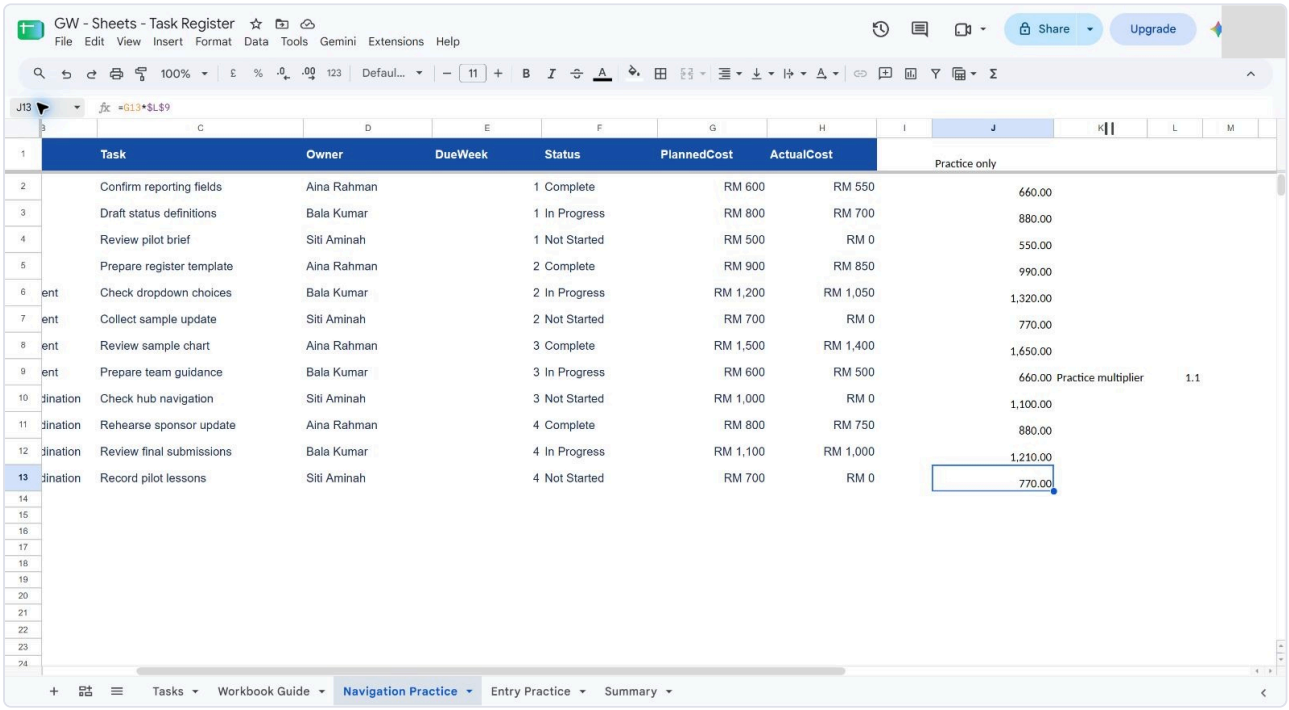
1. On **Navigation Practice**, reuse the clearly labelled practice area: clear the earlier Lab 2 date/cost examples in **J2:K6**. Their earlier screenshots remain in Lab 2. Set **J2:J13** to **More formats > Number**, so the old J2 date format cannot disguise a numeric calculation as a date. Keep the original Tasks source and the copied A:H records unchanged.
2. Enter **Practice multiplier** in **K9** and numeric **1.1** in **L9**. Fit K to show the complete label. This assumption is hypothetical; it is not an approved rate or a tax rule.



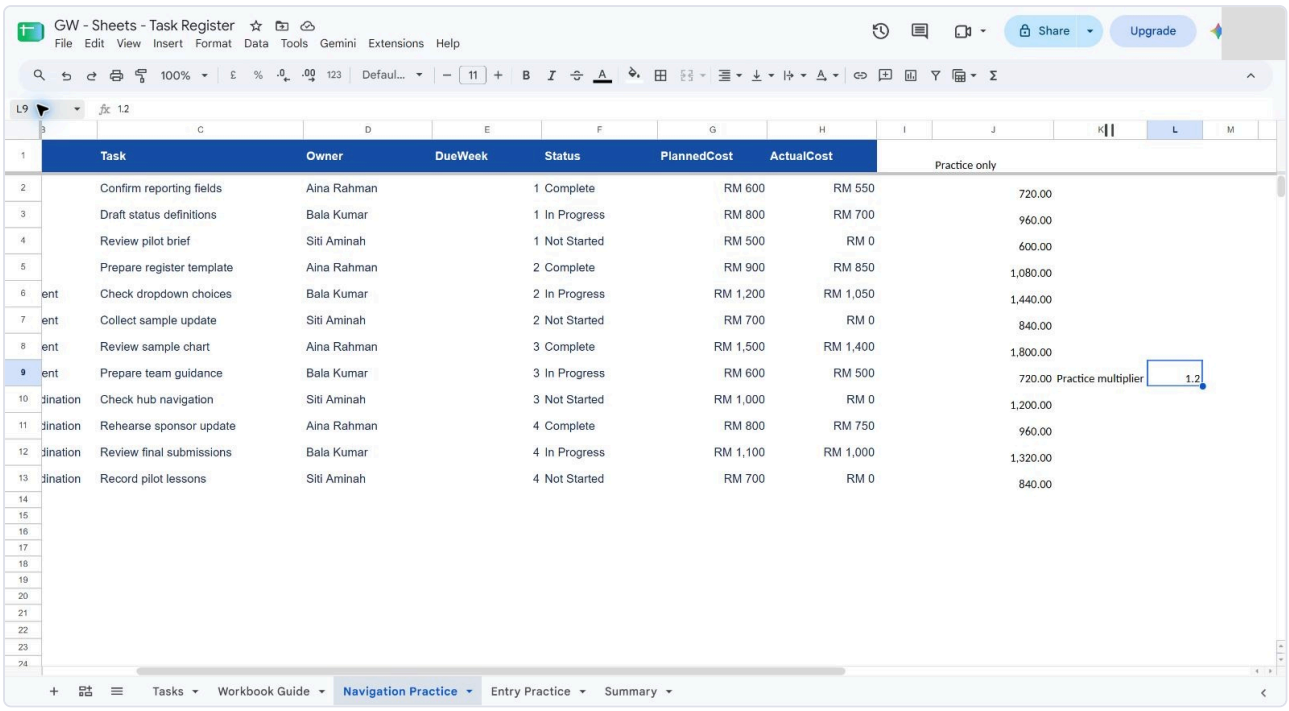
3. In **J2**, enter `=G2*$L$9` and commit it. The first result is **660.00** before filling the remaining rows.



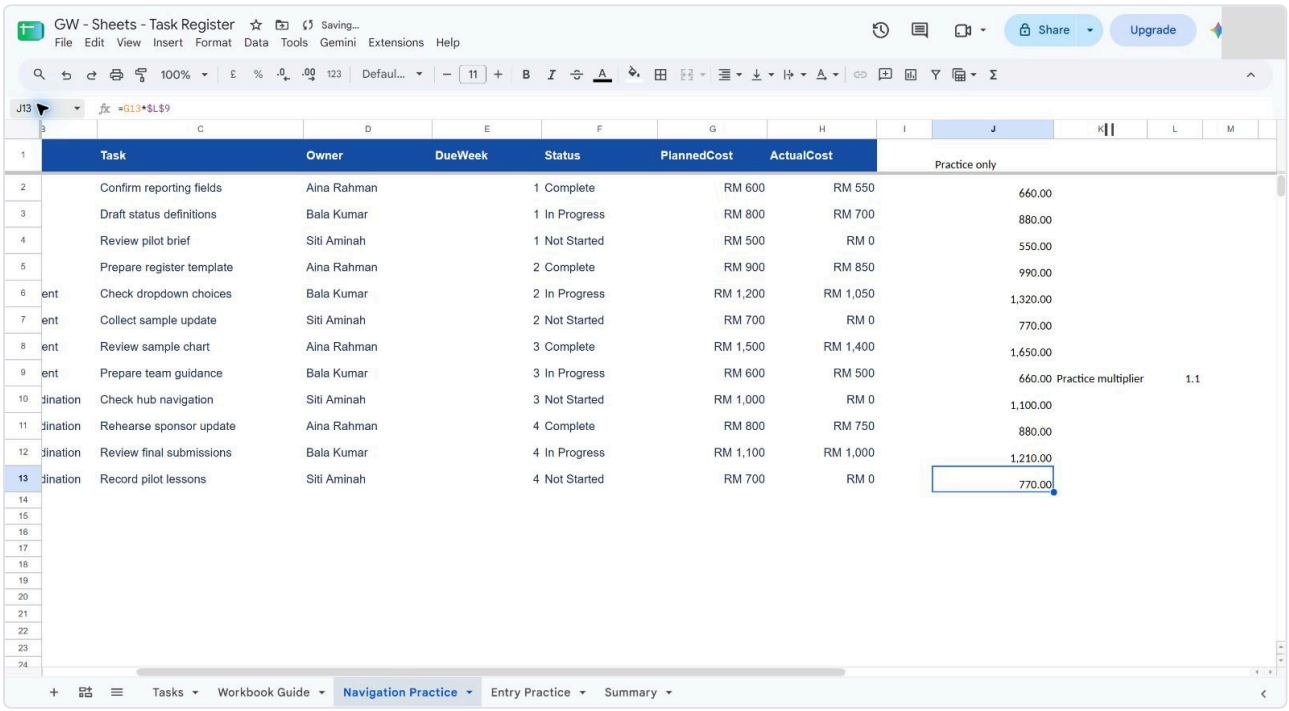
4. Select **J2:J13** and use **Ctrl+D**. Inspect **J13**: `=G13*$L$9` displays **770.00**. G changes with the row; $L$9 stays fixed.



5. Change **L9** to **1.2** and commit it. The first and final results become **720.00** and **840.00**.

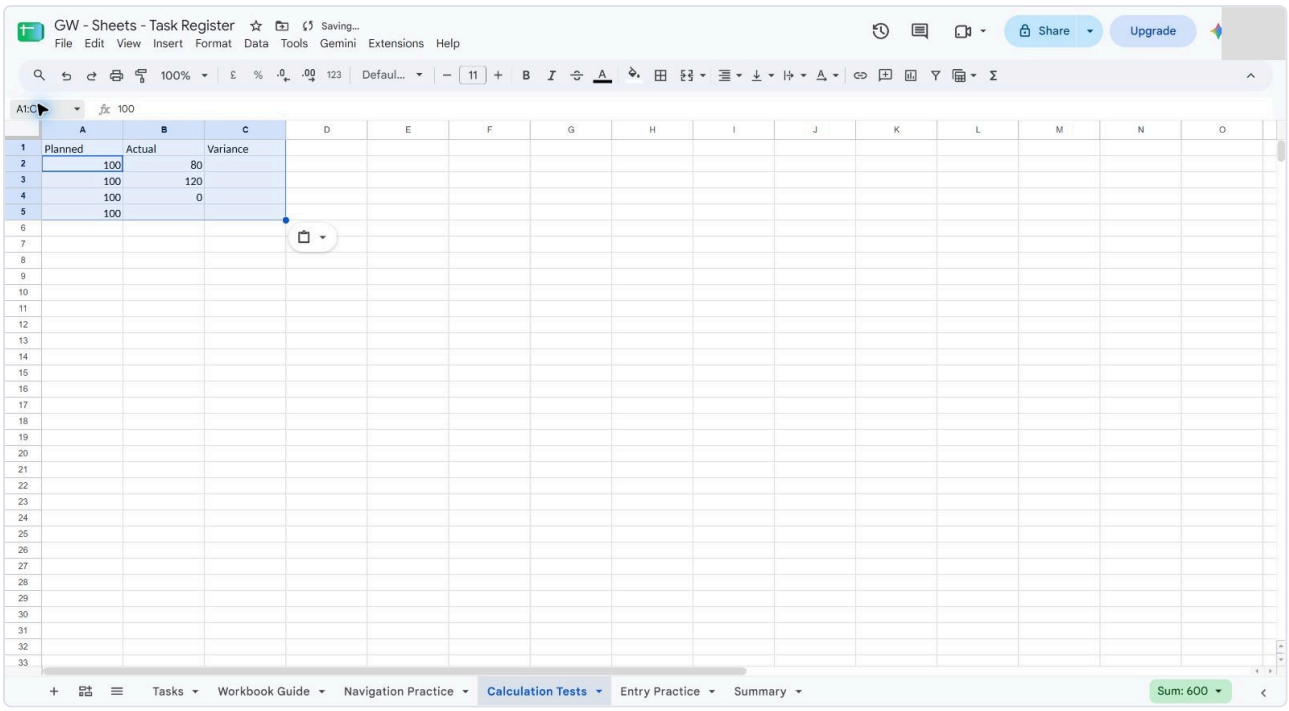


6. Restore **L9** to **1.1**. Reinspect J13's fixed reference and **770.00** result.

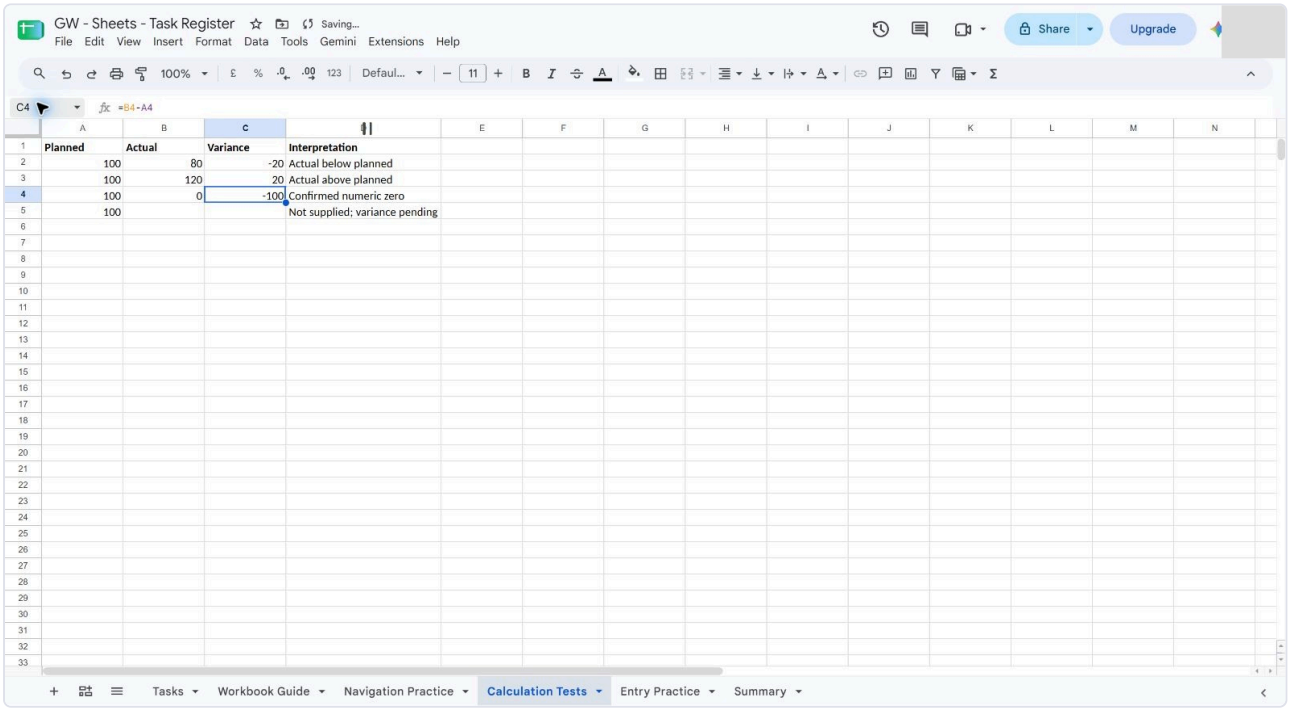


3.4 Test Calculations With Zeros, Blanks and Reference Errors

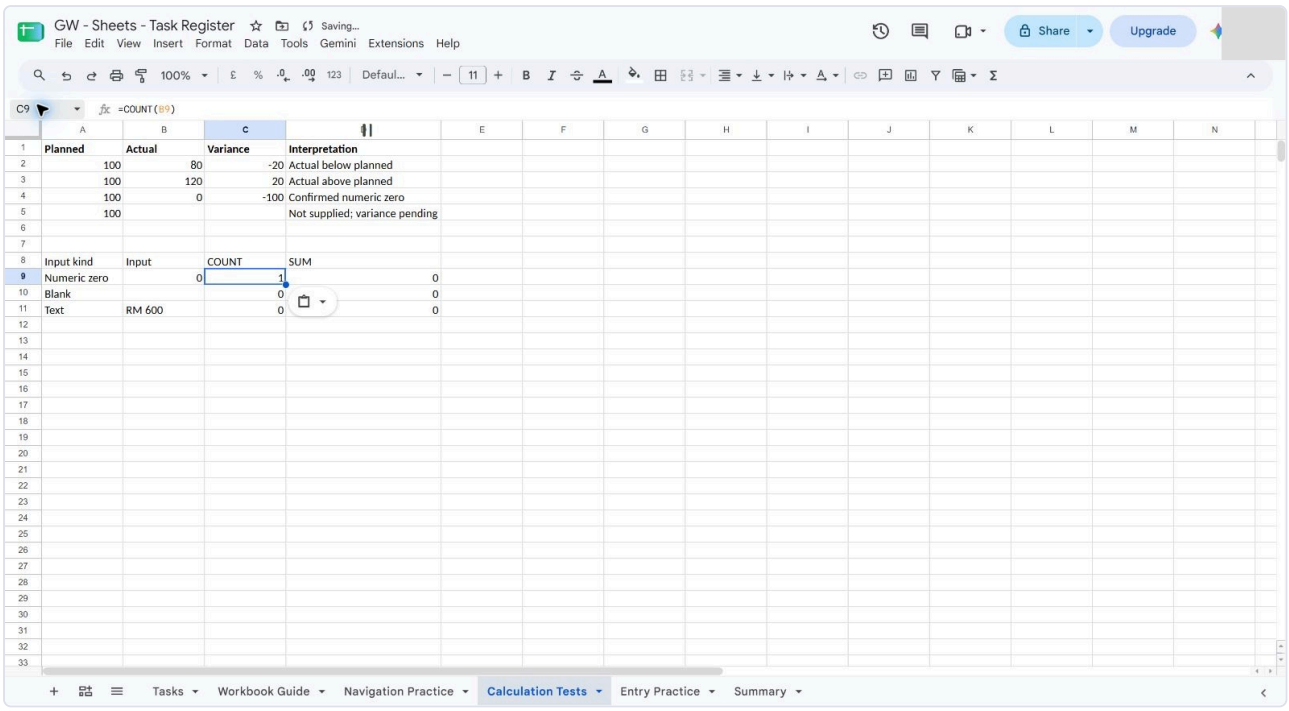
1. Use **Add Sheet**, then double-click the new tab and name it **Calculation Tests**. Enter **Planned / Actual / Variance** in **A1:C1**, and the four planned/actual pairs **100/80**, **100/120**, **100/0**, **100/blank** in rows 2–5. Leave **B5** genuinely empty.



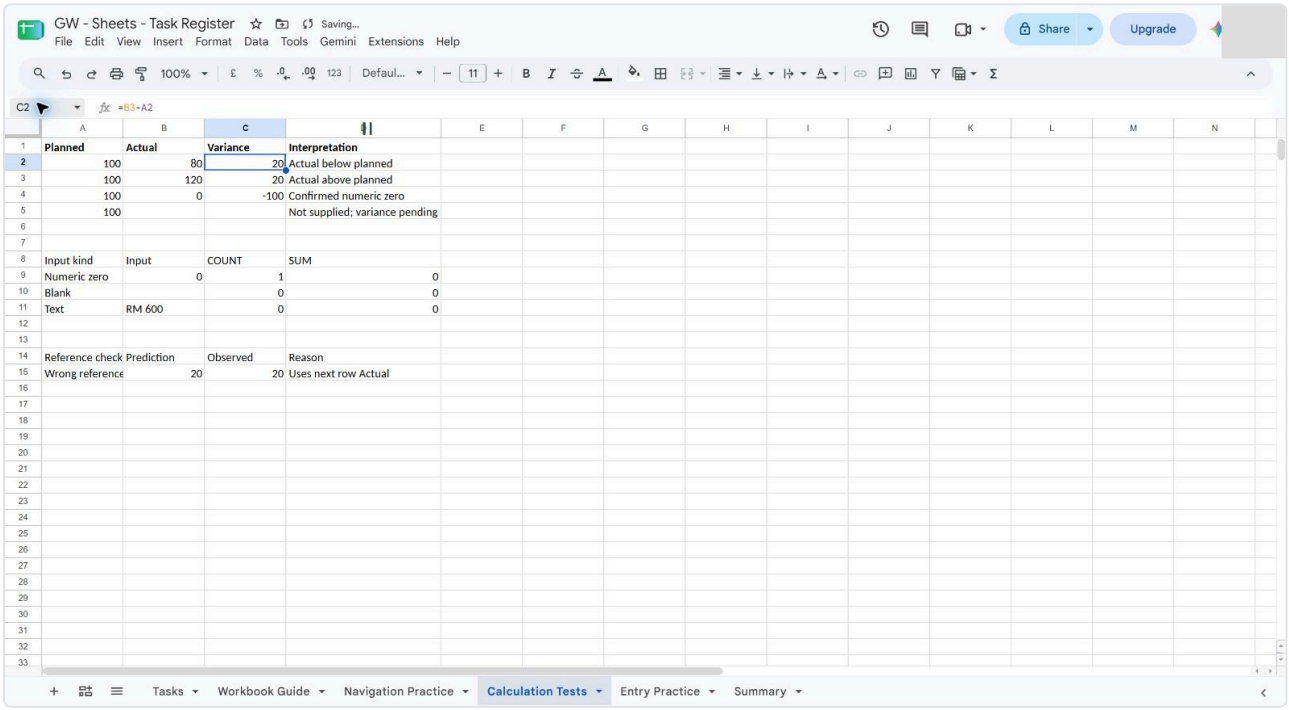
2. Enter **=B2-A2** in C2 and fill **C2:C4**. Results are **-20, 20, -100**. Leave C5 empty because its actual is not supplied. Add the interpretation in D and fit its column. A formula that treats an empty reference as zero would not establish confirmed zero expenditure.



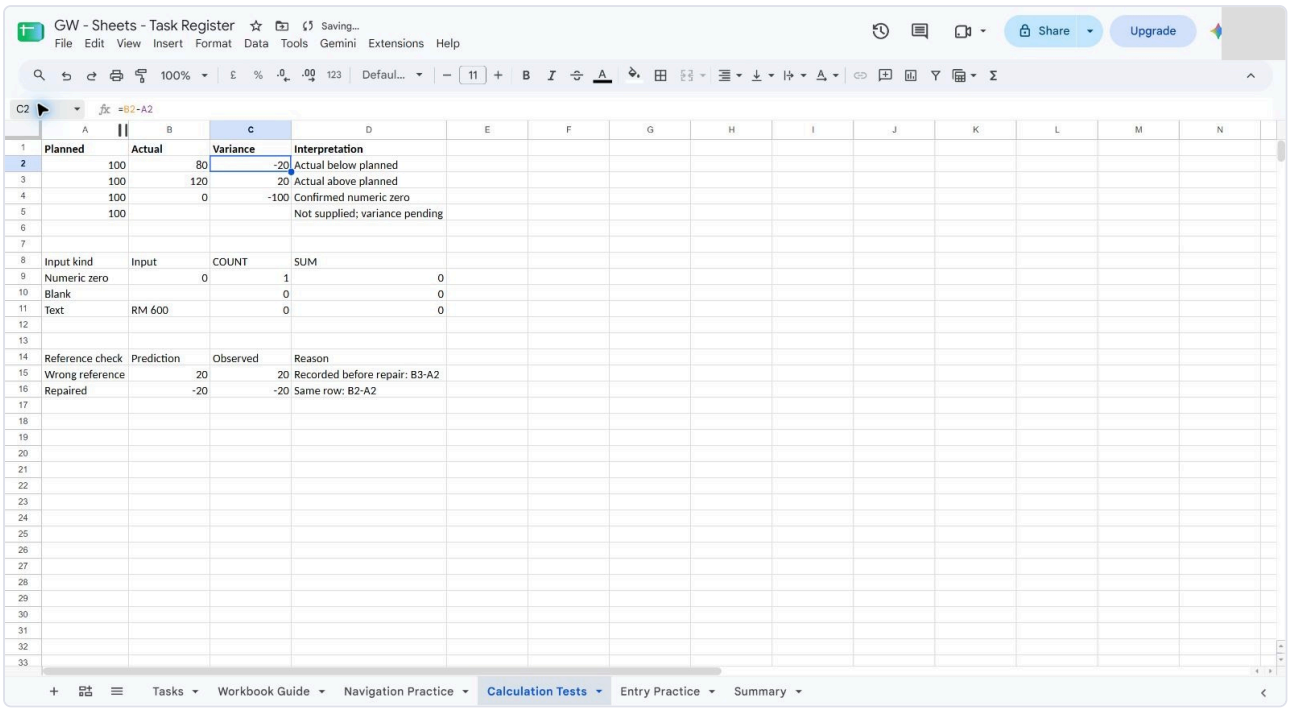
3. In **A8:D11**, create a separate test with headers **Input kind / Input / COUNT / SUM**. B9 is numeric **0**, B10 stays blank, B11 is text **RM 600**. In C9 and D9 enter **=COUNT(B9)** and **=SUM(B9)**, then fill through row 11. COUNT returns **1, 0, 0**; SUM returns **0, 0, 0**. SUM alone cannot distinguish these three inputs.



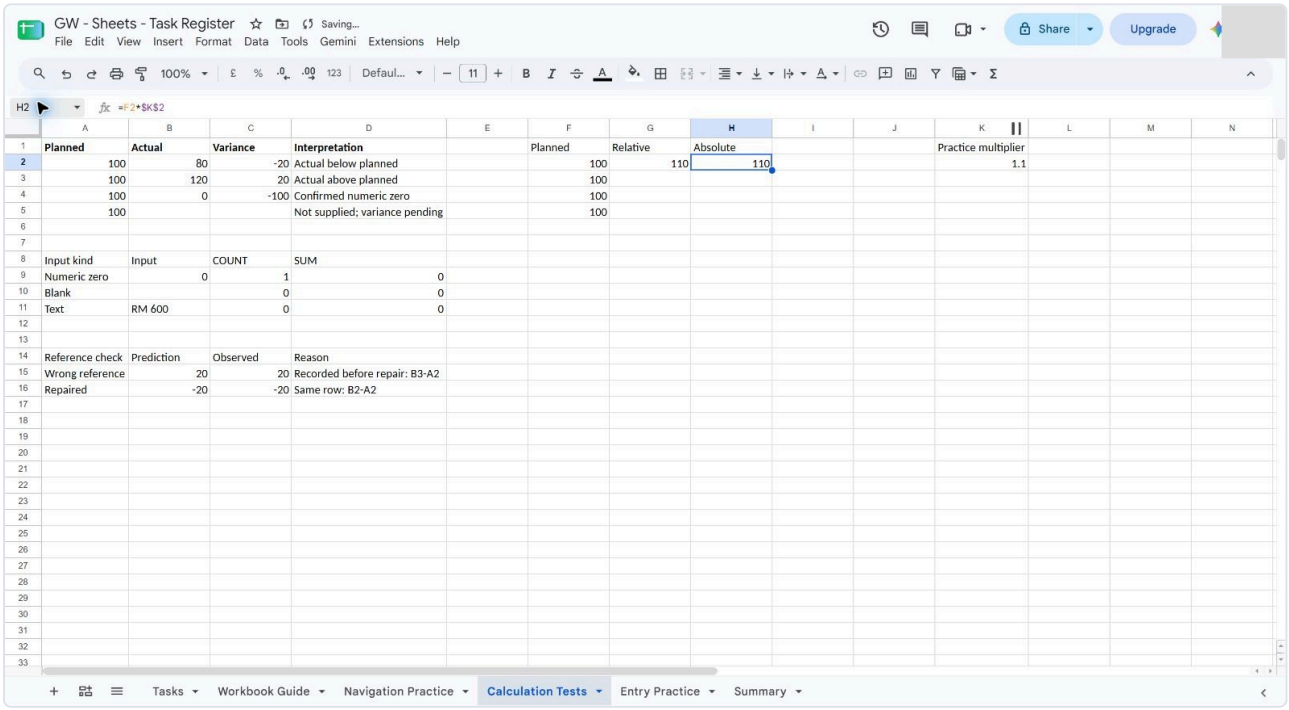
4. Predict that the intentionally wrong **=B3-A2** in C2 gives **20** by using the next row's actual. Enter it, inspect the formula bar and compare it with the first pair's intended **-20**. Record prediction and observation in **A14:D15**.



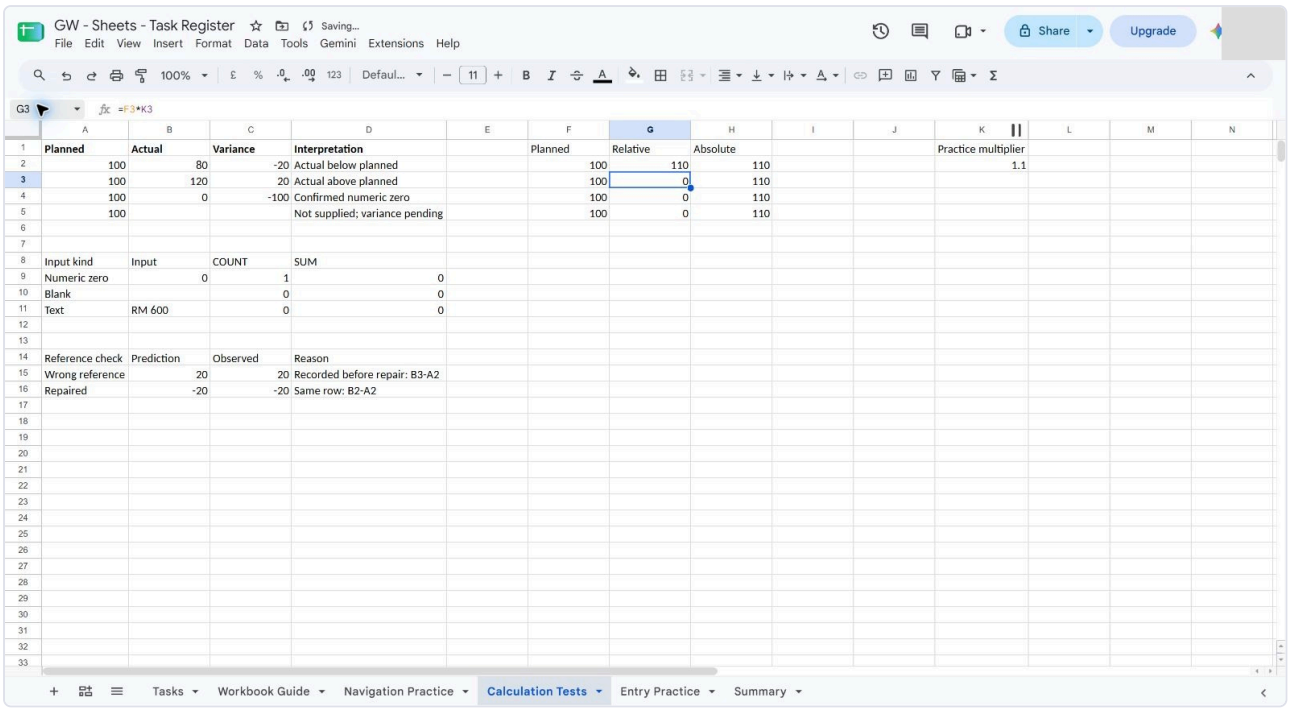
5. Repair C2 to **=B2-A2** and verify **-20**. Keep the wrong test's observed **20** as a labelled historical value, rather than a live link that changes after repair. Add the repaired observation in row 16. The reference uses **=C2** for the current repaired result.



6. Build the reference comparison in **F1:H5**: headers **Planned / Relative / Absolute**, and four planned inputs of **100**. Put **Practice multiplier** in K1 and **1.1** in K2. Enter `=F2*K2` in G2 and `=F2*$K$2` in H2. Both initially return **110**.



7. Fill **G2:H5** using Ctrl+D. Inspect **G3** `=F3*K3` and **G5** `=F5*K5` : both refer to blank multiplier cells and give **0**. Inspect **H3** `=F3*$K$2` and **H5** `=F5*$K$2` : both keep the intended assumption and give **110**. Retain the relative column as a labelled teaching counterexample.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	**Planned**	**Actual**	**Variance**	**Interpretation**		Planned	Relative	Absolute			Practice multiplier			
2	100	80	-20	Actual below planned		100	110	110			1.1			
3	100	120	20	Actual above planned		100	0	110						
4	100	0	-100	Confirmed numeric zero		100	0	110						
5	100			Not supplied; variance pending		100	0	110						
6														
7														
8	Input kind	Input	COUNT	SUM										
9	Numeric zero	0	1	0										
10	Blank		0	0										
11	Text	RM 600	0	0										
12														
13														
14	Reference check	Prediction	Observed	Reason										
15	Wrong reference	20	20	Recorded before repair: B3-A2										
16	Repaired	-20	-20	Same row: B2-A2										
17														
18														
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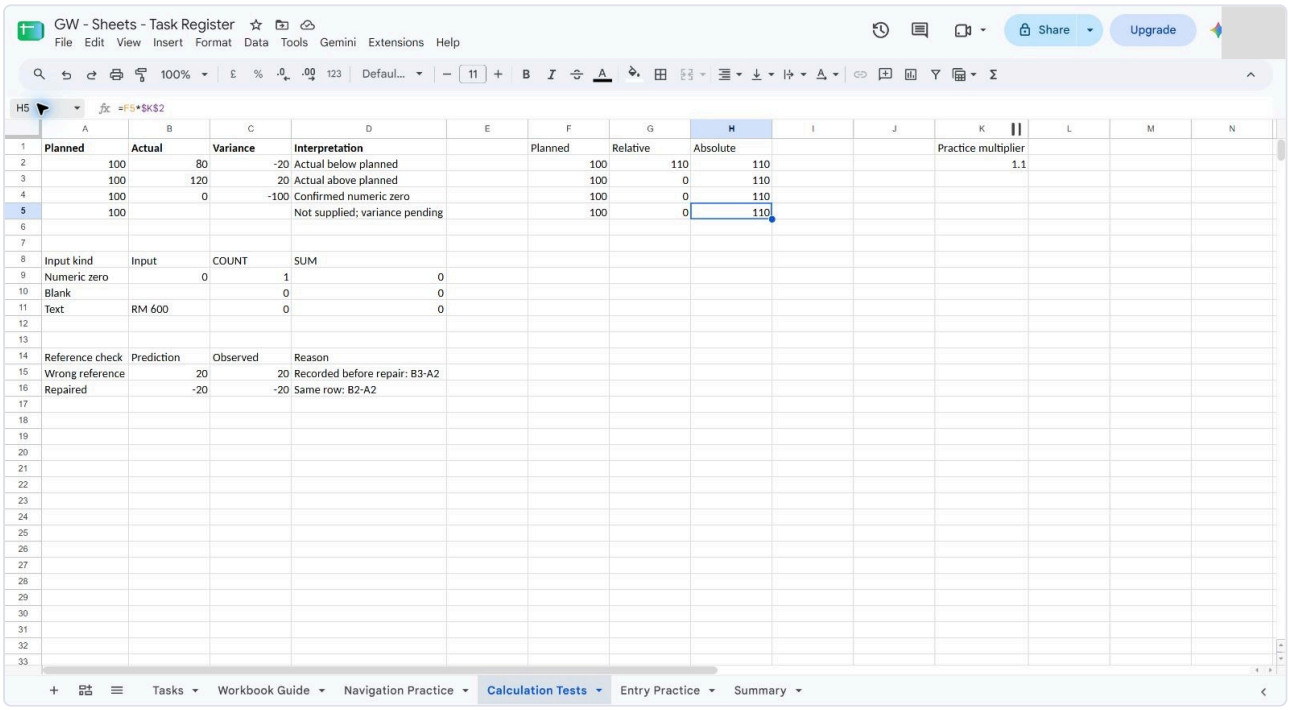
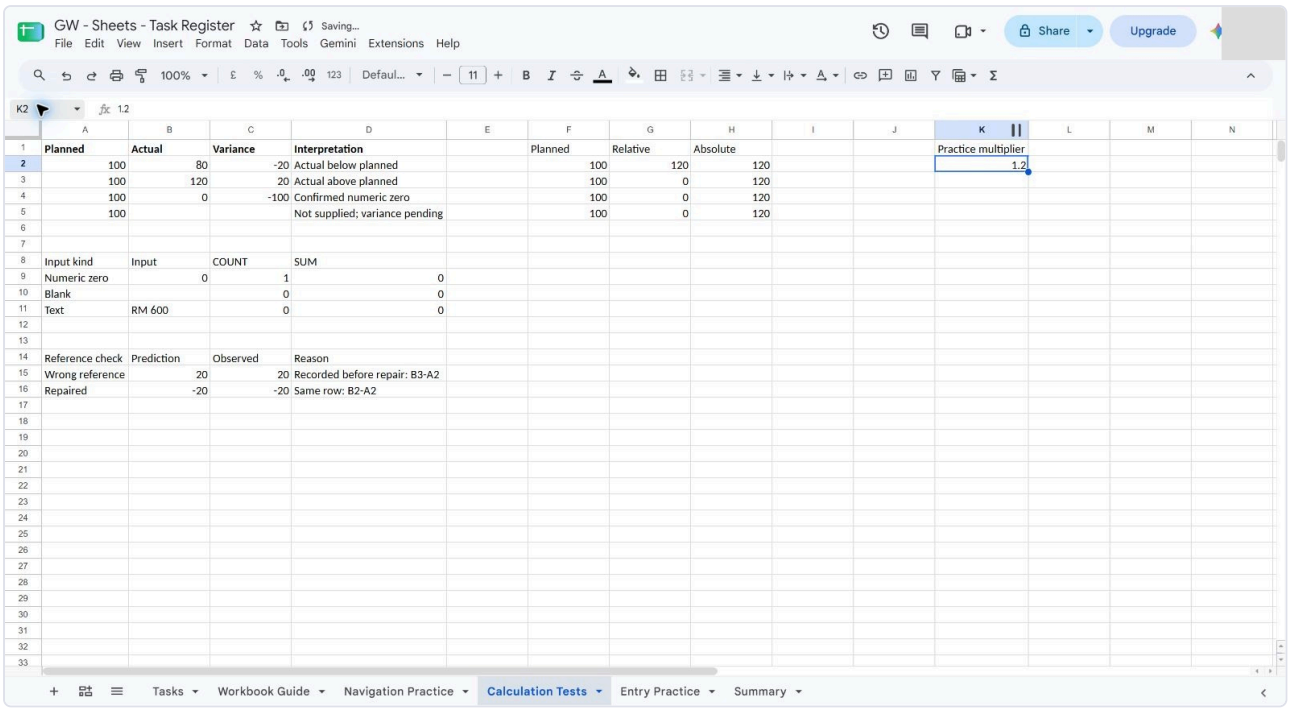
33														

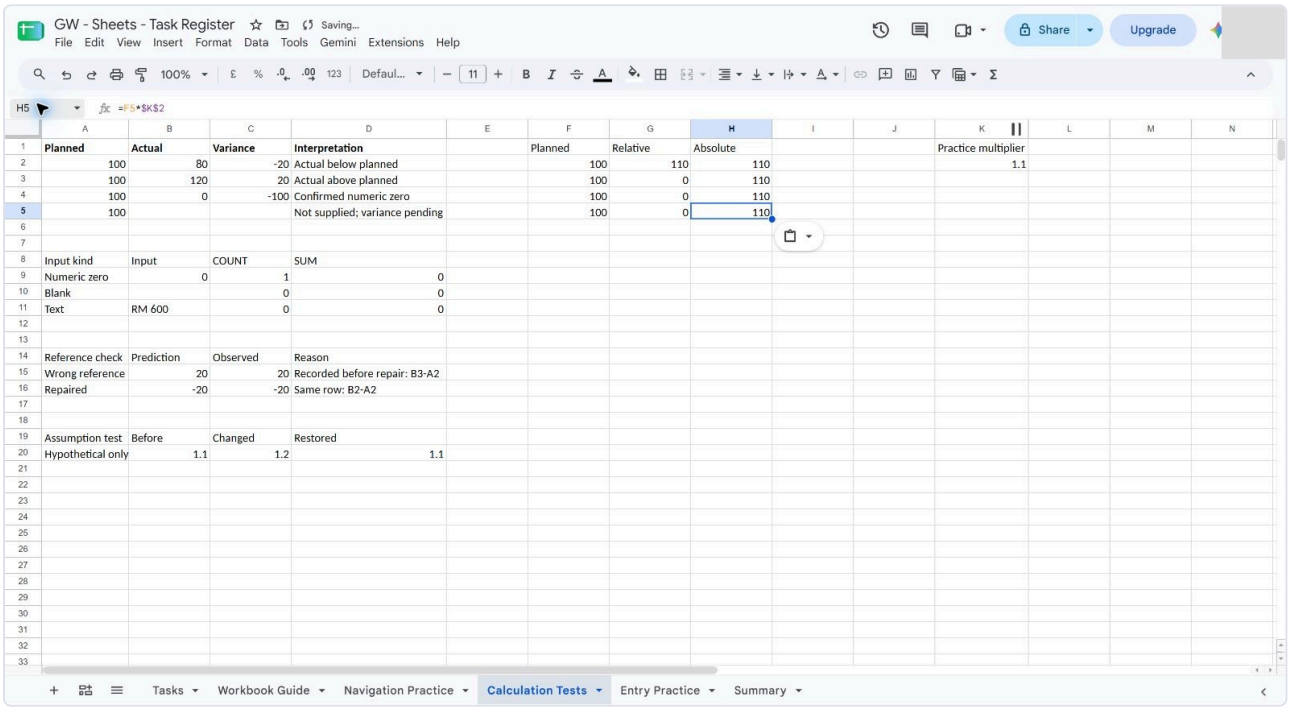


	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	**Planned**	**Actual**	**Variance**	**Interpretation**		Planned	Relative	Absolute			Practice multiplier			
2	100	80	-20	Actual below planned		100	110	110			1.1			
3	100	120	20	Actual above planned		100	0	110						
4	100	0	-100	Confirmed numeric zero		100	0	110						
5	100			Not supplied; variance pending		100	0	110						
6														
7														
8	Input kind	Input	COUNT	SUM										
9	Numeric zero	0	1	0										
10	Blank		0	0										
11	Text	RM 600	0	0										
12														
13														
14	Reference check	Prediction	Observed	Reason										
15	Wrong reference	20	20	Recorded before repair: B3-A2										
16	Repaired	-20	-20	Same row: B2-A2										
17														
18														
19														
20														
21														
22														
23														
24														
25														
26														
27														
28														
29														
30														
31														
32														
33														

8. Change K2 to **1.2**. G2 and all four absolute results become **120**; the later relative results remain **0**. Record that observation, restore K2 to **1.1**, and keep the before/change/restored values in **A19:D20**. This is hypothetical practice only.

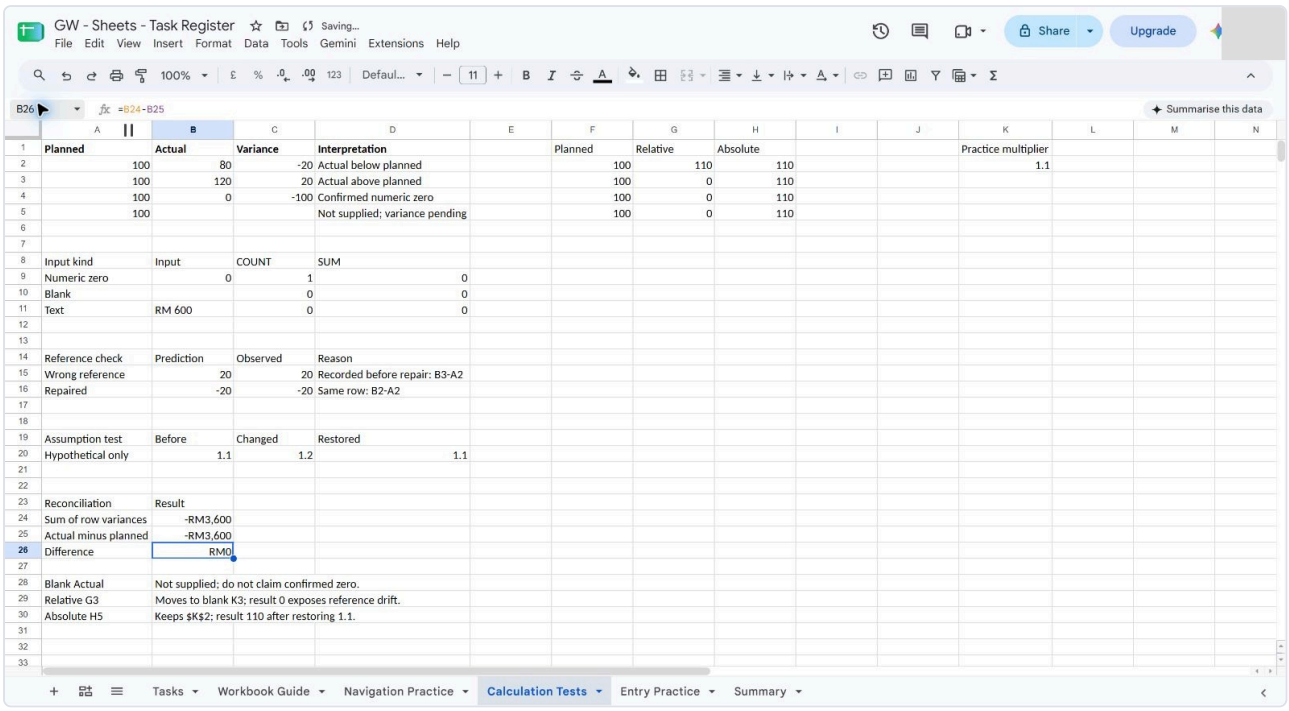


Planned	Actual	Variance	Interpretation		Planned	Relative	Absolute			Practice multiplier
100	80	-20	Actual below planned		100	120	120			1.2
100	120	20	Actual above planned		100	0	120			
100	0	-100	Confirmed numeric zero		100	0	120			
100			Not supplied; variance pending		100	0	120			
Input kind	Input	COUNT	SUM							
Numeric zero	0	1	0							
Blank		0	0							
Text	RM 600	0	0							
Reference check	Prediction	Observed	Reason							
Wrong reference	20	20	Recorded before repair: B3-A2							
Repaired	-20	-20	Same row: B2-A2							



Planned	Actual	Variance	Interpretation		Planned	Relative	Absolute			Practice multiplier
100	80	-20	Actual below planned		100	110	110			1.1
100	120	20	Actual above planned		100	0	110			
100	0	-100	Confirmed numeric zero		100	0	110			
100			Not supplied; variance pending		100	0	110			
Input kind	Input	COUNT	SUM							
Numeric zero	0	1	0							
Blank		0	0							
Text	RM 600	0	0							
Reference check	Prediction	Observed	Reason							
Wrong reference	20	20	Recorded before repair: B3-A2							
Repaired	-20	-20	Same row: B2-A2							
Assumption test	Before	Changed	Restored							
Hypothetical only	1.1	1.2	1.1							

9. In **A23:B26**, reconcile the original source with **=SUM(Tasks!I2:I13)**, **=SUM(Tasks!H2:H13)-SUM(Tasks!G2:G13)** and the difference **=B24-B25**. Both methods give **-RM3,600** and their difference is **RM0**. Fit A to show its labels. Record the blank-information and reference-drift explanations separately below. All original **Tasks!A1:H13** values remain unchanged.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Planned	Actual	Variance	Interpretation		Planned	Relative	Absolute			Practice multiplier			
2	100	80	-20	Actual below planned		100	110	110			1.1			
3	100	120	20	Actual above planned		100	0	110						
4	100	0	-100	Confirmed numeric zero		100	0	110						
5	100			Not supplied; variance pending		100	0	110						
6														
7														
8	Input kind	Input	COUNT	SUM										
9	Numeric zero	0	1	0										
10	Blank		0	0										
11	Text	RM 600	0	0										
12														
13														
14	Reference check	Prediction	Observed	Reason										
15	Wrong reference	20	20	Recorded before repair: B3-A2										
16	Repaired	-20	-20	Same row: B2-A2										
17														
18														
19	Assumption test	Before	Changed	Restored										
20	Hypothetical only	1.1	1.2	1.1										
21														
22														
23	Reconciliation	Result												
24	Sum of row variances	-RM3,600												
25	Actual minus planned	-RM3,600												
26	Difference	RM0												
27														
28	Blank Actual	Not supplied; do not claim confirmed zero.												
29	Relative G3	Moves to blank K3; result 0 exposes reference drift.												
30	Absolute H5	Keeps $K$2; result 110 after restoring 1.1.												
31														
32														
33														

> **Validation boundary:** these are author UI reference checks. The independent estimate table, deliberately omitted total range, independent diagnosis and learner timing pilot have not been completed by this reference run.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Create an independent three-row estimate table with at least one zero actual cost and one positive variance. Calculate and explain the results without copying a finished answer column.
2. Shorten one total range on the practice table, then use a second calculation to detect and repair the omission.
3. Write a short explanation of negative variance, numeric zero, blank information and a fixed reference, each supported by an actual test cell.

Evidence to Hand Over

- Four-case calculation test with a documented blank-information limitation.
- Reference-error and omitted-range diagnosis with repaired formulas.
- Reconciled baseline totals and independent estimate table.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check

A plausible total hides a copied-row error	Inspect row references and compare sum of variances against the difference of totals.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

You calculated and reconciled the register using visible source values and references.

Check yourself: Which part of a copied formula should move, and which should stay fixed?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
Variance
```

```
=H2-G2
```

```
=SUM(G2:G13)
```

```
=SUM(H2:H13)
```

```
=SUM(I2:I13)
```

```
=AVERAGE(G2:G13)
```

```
=COUNT(G2:G13)
```

```
Practice multiplier
```

```
=G2*$L$9
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Calculation pairs Planned/Actual: 100/80; 100/120; 100/0; 100/blank. The final blank means
“not supplied” for this test, not a confirmed zero. Use this to distinguish calculation
mechanics from data meaning.
```

Captured Reference Ranges

Tasks: I1 Variance; I2 **=H2-G2**, fill I2:I13. Summary K1:L7: L2 **=SUM(G2:G13)** →10400; L3 **=SUM(H2:H13)** →6800; L4 **=SUM(I2:I13)** →-3600; L6 **=AVERAGE(G2:G13)** →866.666… (display RM866.67); L7 **=COUNT(G2:G13)** →12.

Navigation Practice: clear prior Lab 2 examples J2:K6 and reset J2:J13 to Number. K9 Practice multiplier; L9 1.1; J2 `=G2*$L$9`, fill to J13. Change 1.1→1.2→1.1; first/final results 660/770→720/840→660/770.

Calculation Tests: A1:D5 four pairs and interpretation, C2:C4 **=B2-A2** filled down gives -20,20,-100; B5/C5 stay empty. A8:D11 zero/blank/text tests give COUNT 1/0/0 and SUM 0/0/0. C2 deliberately wrong **=B3-A2** gives20, repaired **=B2-A2** gives-20. A14:D16 retains the observed test history. F2:F5 are100, G2 `=F2*K2`, H2 `=F2*$K$2`, fill G2:H5. K2 1.1→1.2→1.1; G3/G5 drift to empty K3/K5 while H3/H5 stay anchored. A19:D20 records restoration. B24 **=SUM(Tasks!I2:I13)** and B25 **=SUM(Tasks!H2:H13)-SUM(Tasks!G2:G13)** both give-3600; B26 **=B24-B25** gives0.

The source twelve records remain unchanged. Independent estimate/omitted-range challenge and classroom timing are pending. Lab 2 values-only target-format preservation remains a separate unresolved check.

04

CHAPTER 4

Clean and Organise Records

In this chapter you will find and replace, trim text, sort whole records and freeze the header row.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 50 minutes: 10 explanation/demonstration, 20 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 40 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A cleaned text example, whole-record sort and named cost range.

> **Status:** partial author reference with 22 staged application captures through 4.4. Raw text preservation, TRIM, scoped case repair, whole-record sorts, frozen header, named-range reconciliation and immediate Undo of an unsafe sort verified. The reference used manual raw-value entry; CSV upload/import, independent boundary challenge and learner timing remain unverified.

What You Will Learn

- Clean text without altering source evidence.
- Sort whole records.
- Keep headers visible while scrolling.

Before You Begin

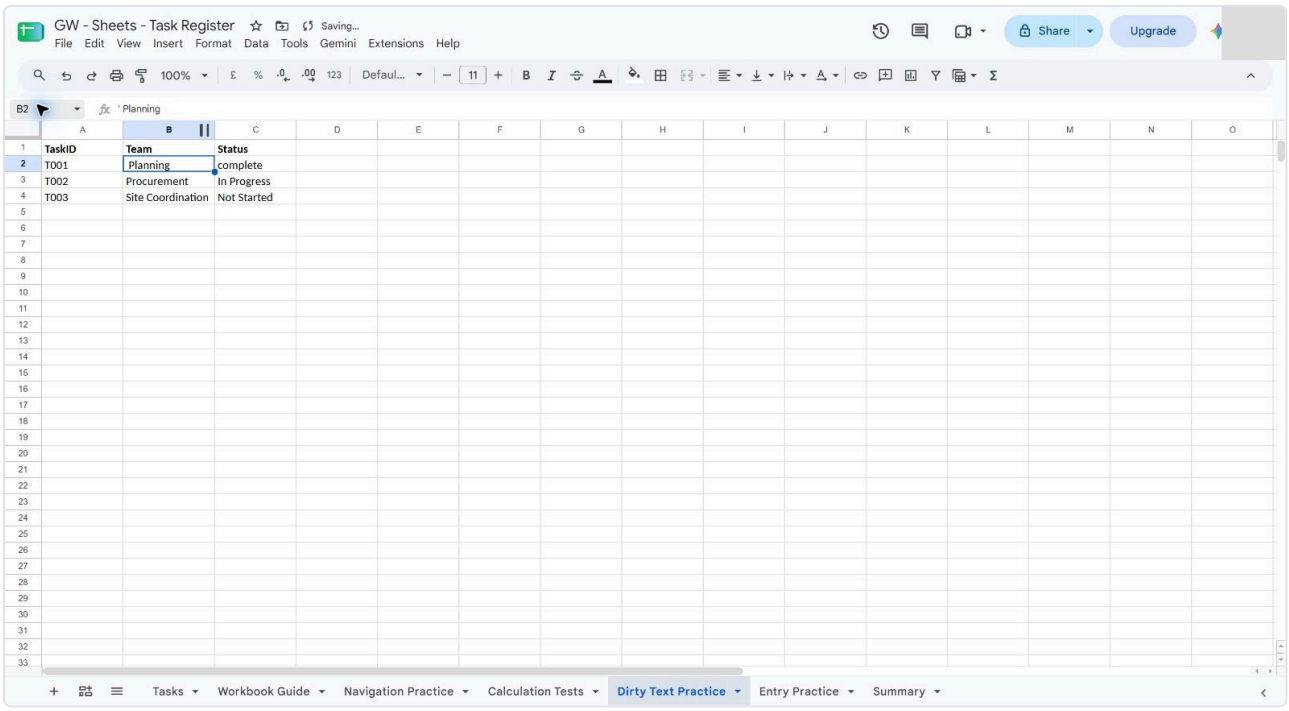
Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Tasks and download DirtyText.csv. Keep cleaning practice separate from the original register.

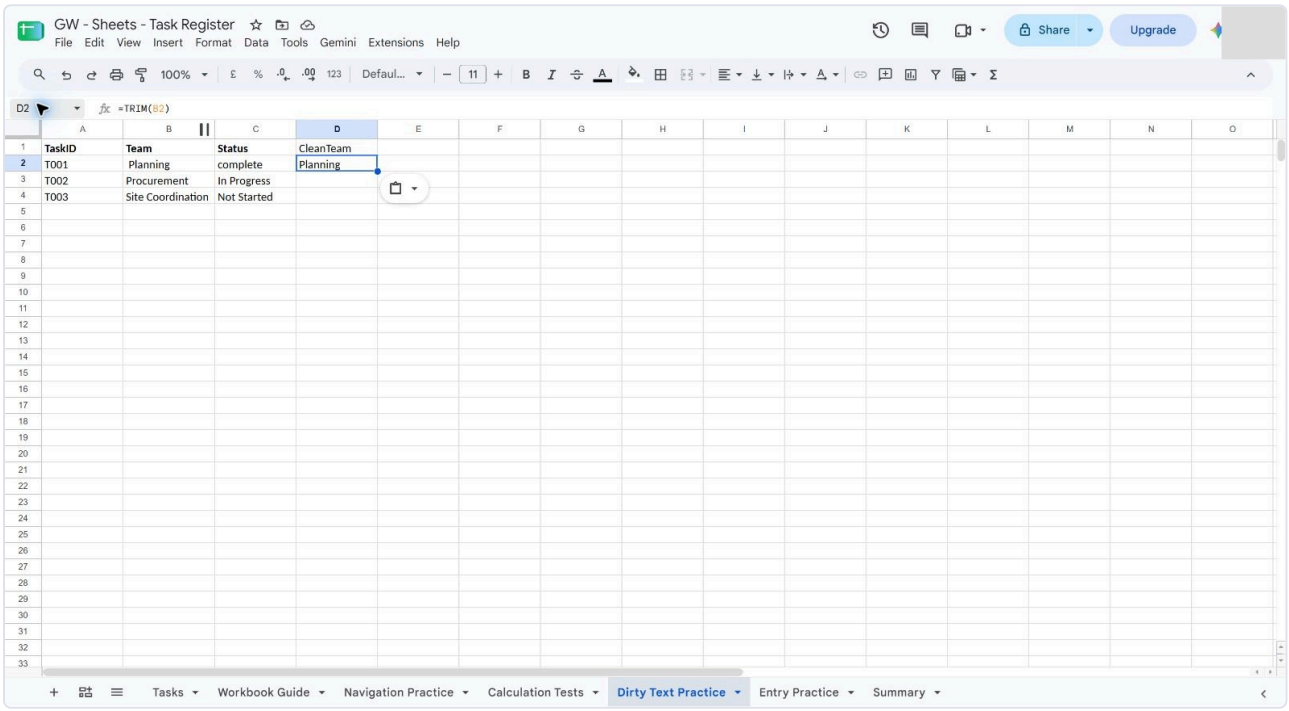
4.1 Clean a Small Import

1. Import **DirtyText.csv** into a separate practice tab, keeping its headings **TaskID / Team / Status**. The captured reference uses a **Dirty Text Practice** tab and manual entry of the supplied values because browser file upload remained unavailable. It does **not** validate the CSV import flow.

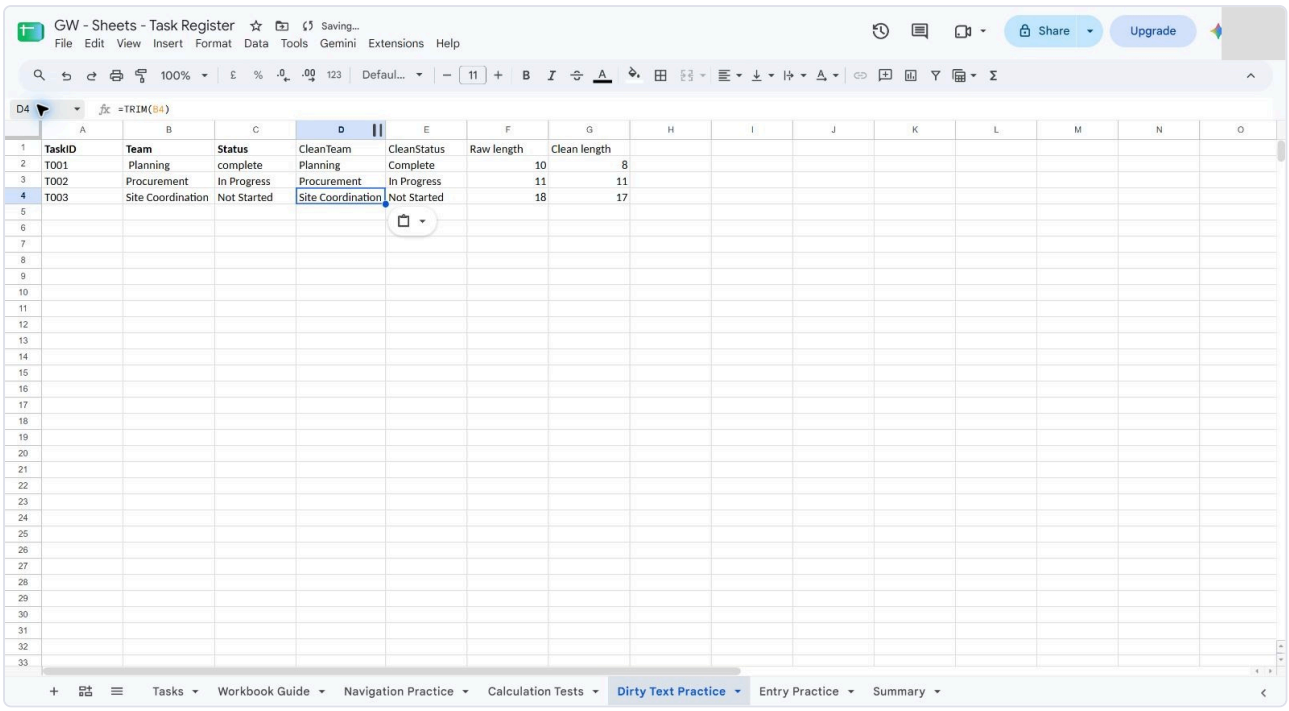
For that fallback, enter the three records from the copy-paste page into A1:C4. Check the raw spaces: B2 must contain one leading and one trailing space around Planning, and B4 one trailing space after Site Coordination. The controlled paste stripped those spaces, so they were explicitly restored through the cell editor with a leading apostrophe before the literal text. The apostrophe is an input prefix, not part of the stored team. Do not clean the raw cells themselves.



2. Enter **CleanTeam** in D1 and **=TRIM(B2)** in D2. Inspect its formula before filling **D2:D4** with Ctrl+D.

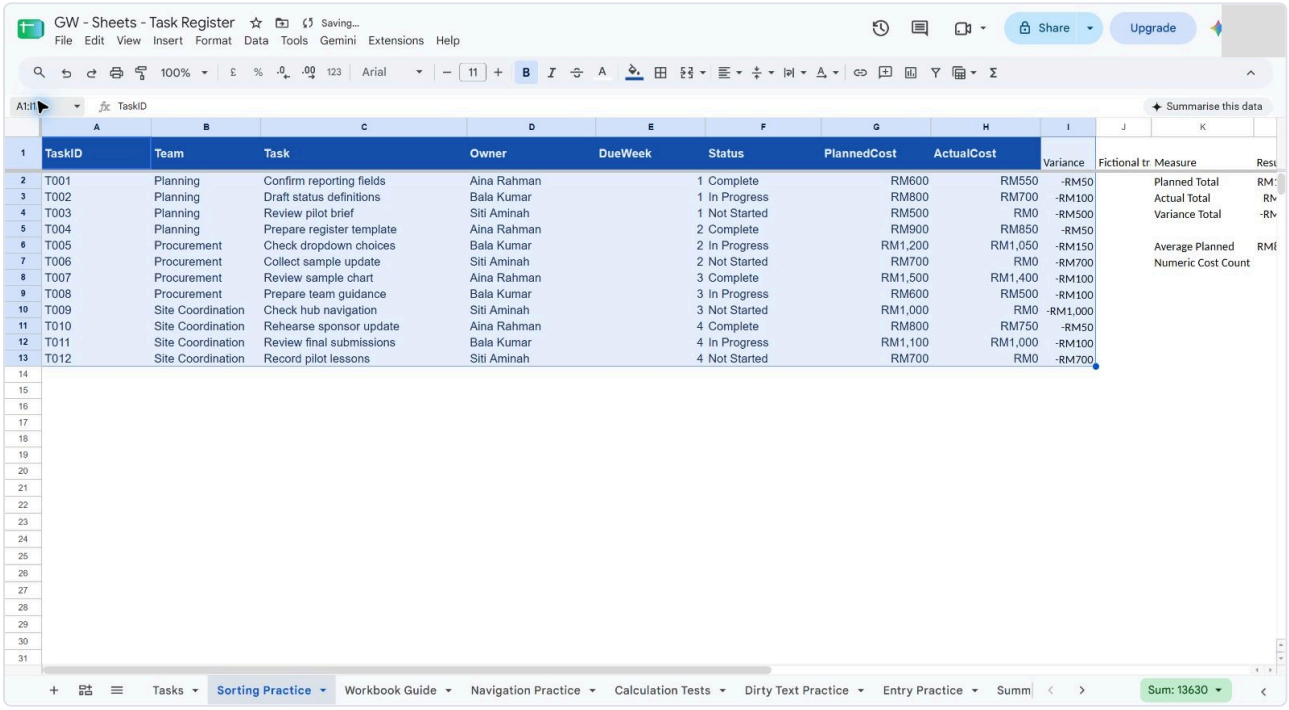


3. Put **CleanStatus** in E1 and the canonical values **Complete / In Progress / Not Started** in E2:E4, retaining the original C column. Fit B and D so complete team names are visible. Add optional **Raw length / Clean length** checks in F:G using **=LEN(B2)** and **=LEN(D2)**, filled through row 4. Planning changes from length **10 to 8** and Site Coordination from **18 to 17**; Procurement remains **11**. These checks make otherwise invisible spaces testable.

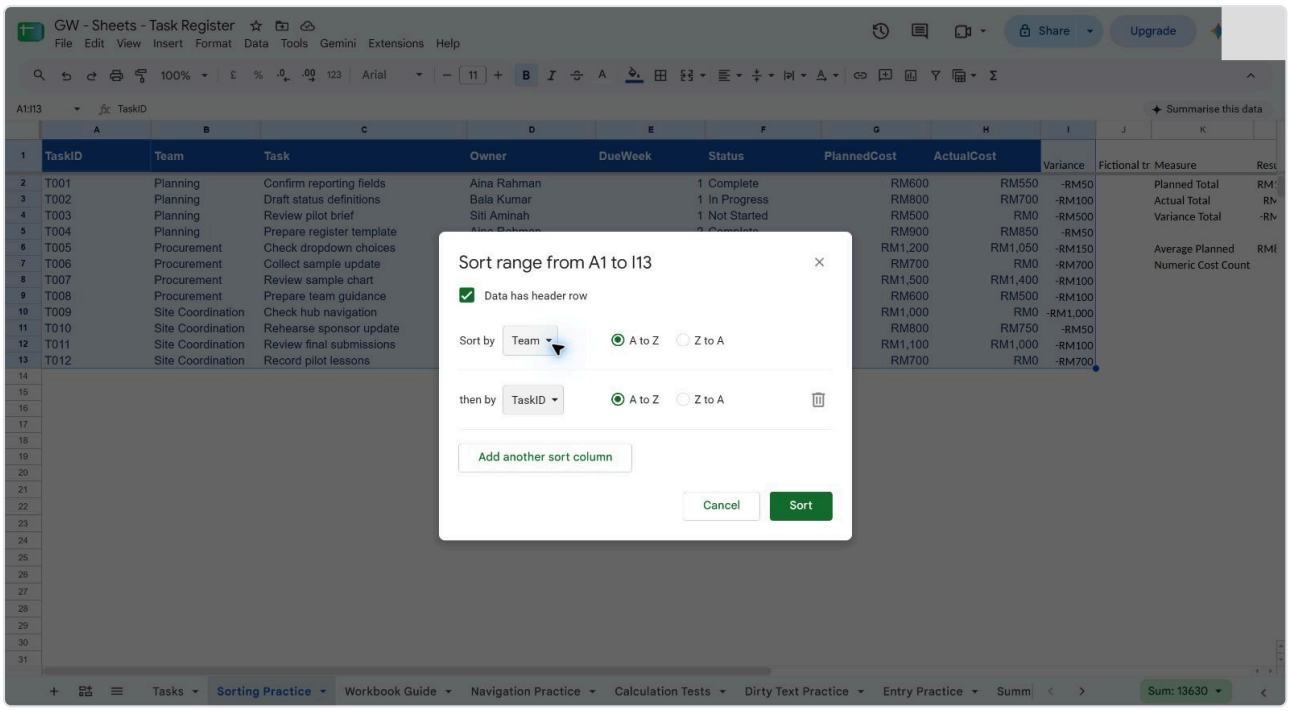


4.2 Sort Whole Records

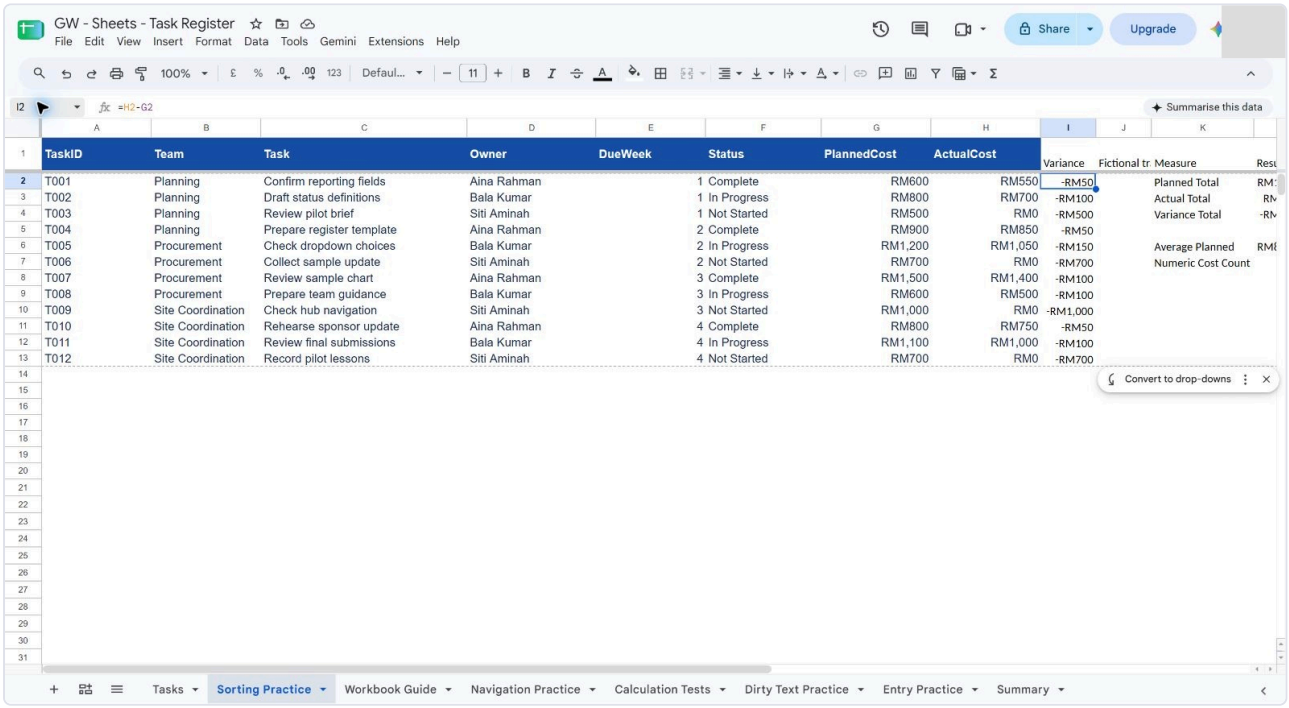
1. Right-click **Tasks > Duplicate**, then rename the copy **Sorting Practice**. Work on the copy to preserve the original register for later chapters. Select **A1:I13**, including the variance column.



2. Open **Data > Sort range > Advanced range sorting options**. Select **Data has header row**. Choose **Team, A to Z**, add another sort column, and explicitly choose **TaskID, A to Z**; the automatic second choice may be a different column. Inspect the full range and both keys before clicking **Sort**.

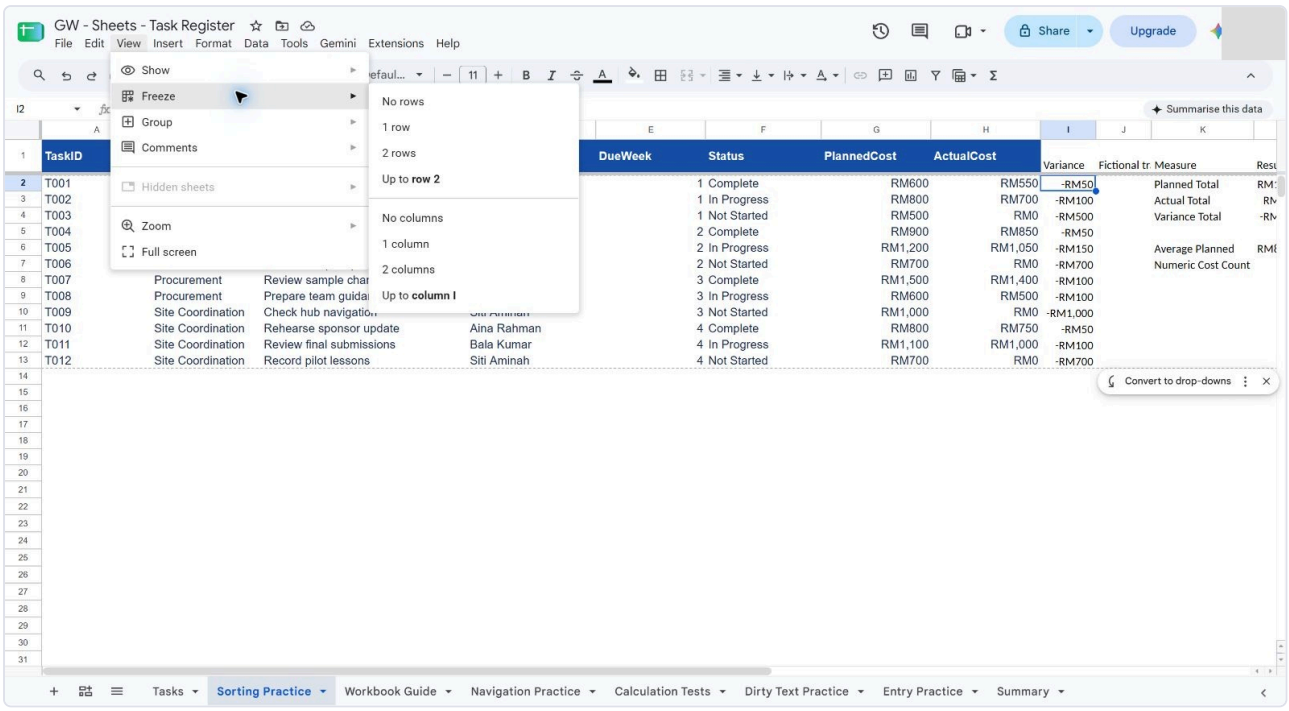


3. Locate **T001** and compare **Aina Rahman**, planned **600**, actual **550**, and **-50 variance** with the source. All twelve record relationships remain intact. This supplied register already follows Team/TaskID order, so this ascending sort legitimately leaves its order unchanged. The intake in 4.4 supplies a visible reorder test.

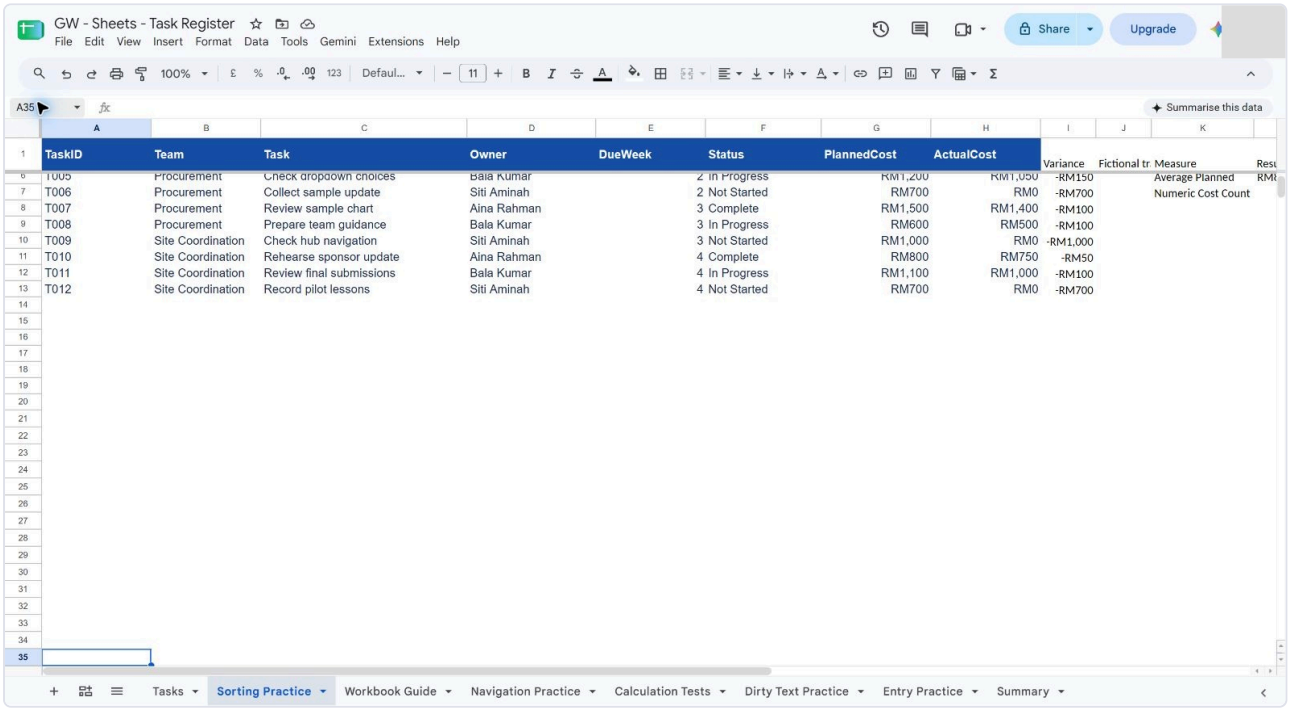


4.3 Freeze and Name a Range

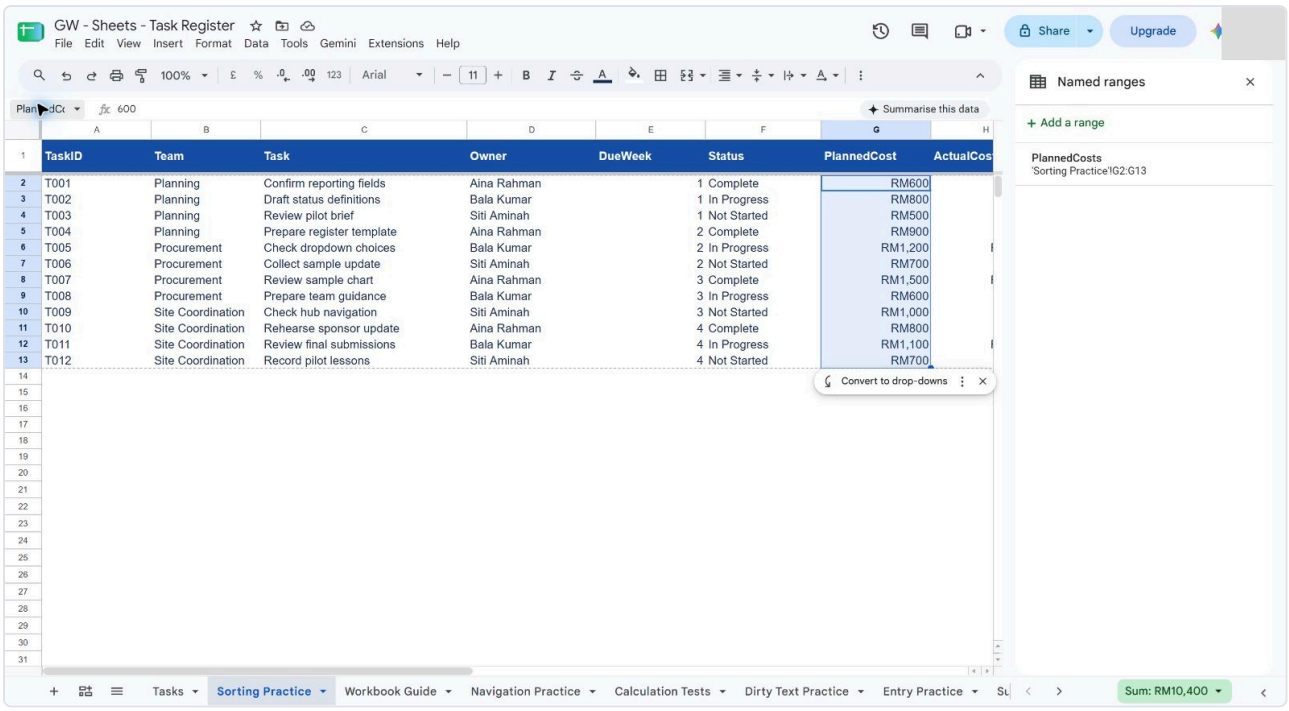
1. On **Sorting Practice**, use **View > Freeze > 1 row**. The copy already inherited a frozen header; retaining or reselecting one row is appropriate. Do not freeze additional rows merely to produce a visible change.



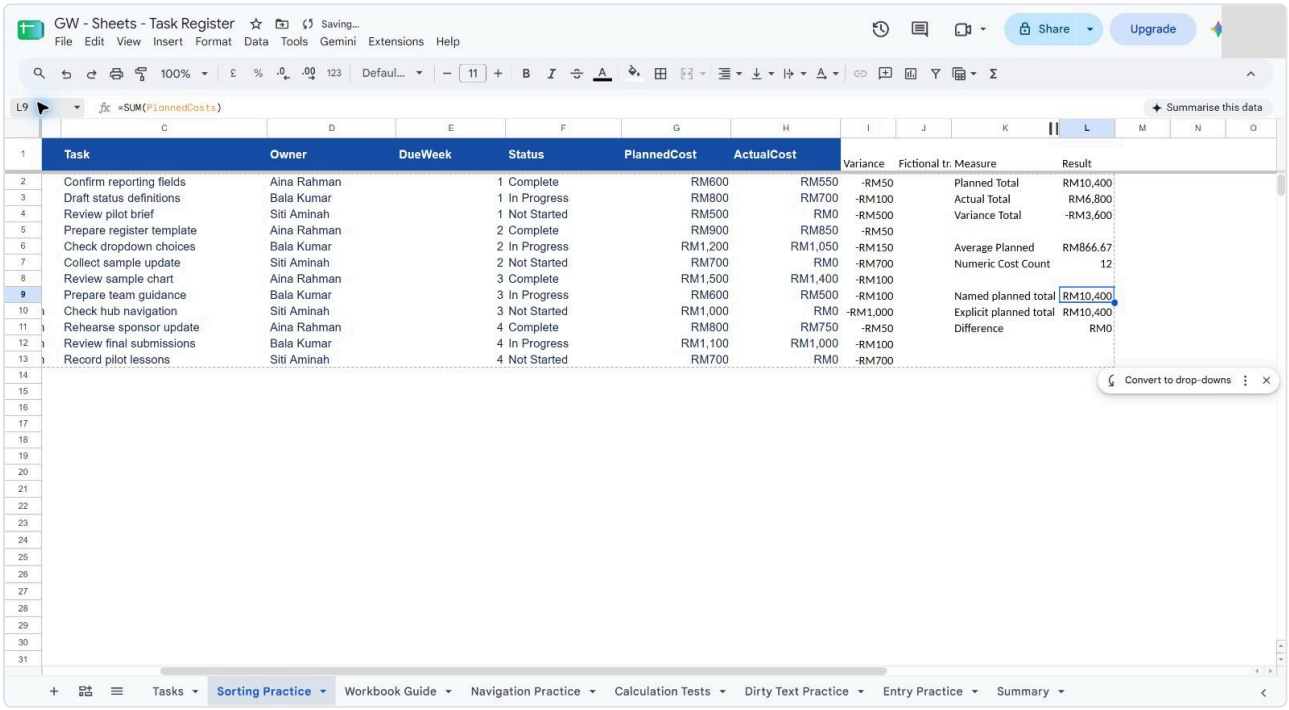
2. Scroll down, or navigate to **A35**, and verify row 1 remains visible above the scrolled rows. Return to the record range afterwards.



3. Select **G2:G13** on **Sorting Practice**, use **Data > Named ranges**, enter **PlannedCosts** and save with **Done**. Inspect the bounds: **'Sorting Practice'!G2:G13**, twelve numeric cost cells. This name deliberately refers to the practice copy, not a changed baseline source.

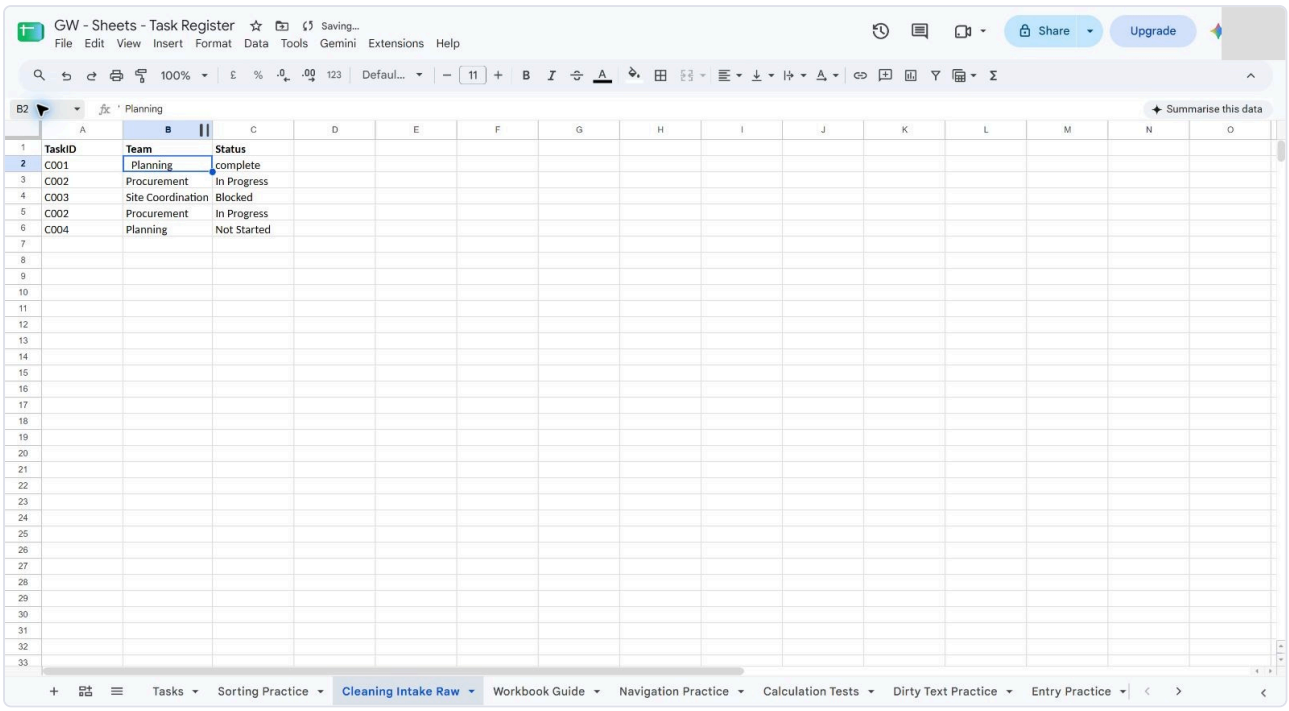


4. In K9:L11 add labels **Named planned total / Explicit planned total / Difference**, with **=SUM(PlannedCosts)**, **=SUM(G2:G13)** and **=L9-L10**. Fit K to show the complete labels and scroll horizontally when needed. Both totals are **RM10,400**; the difference is **RM0**.

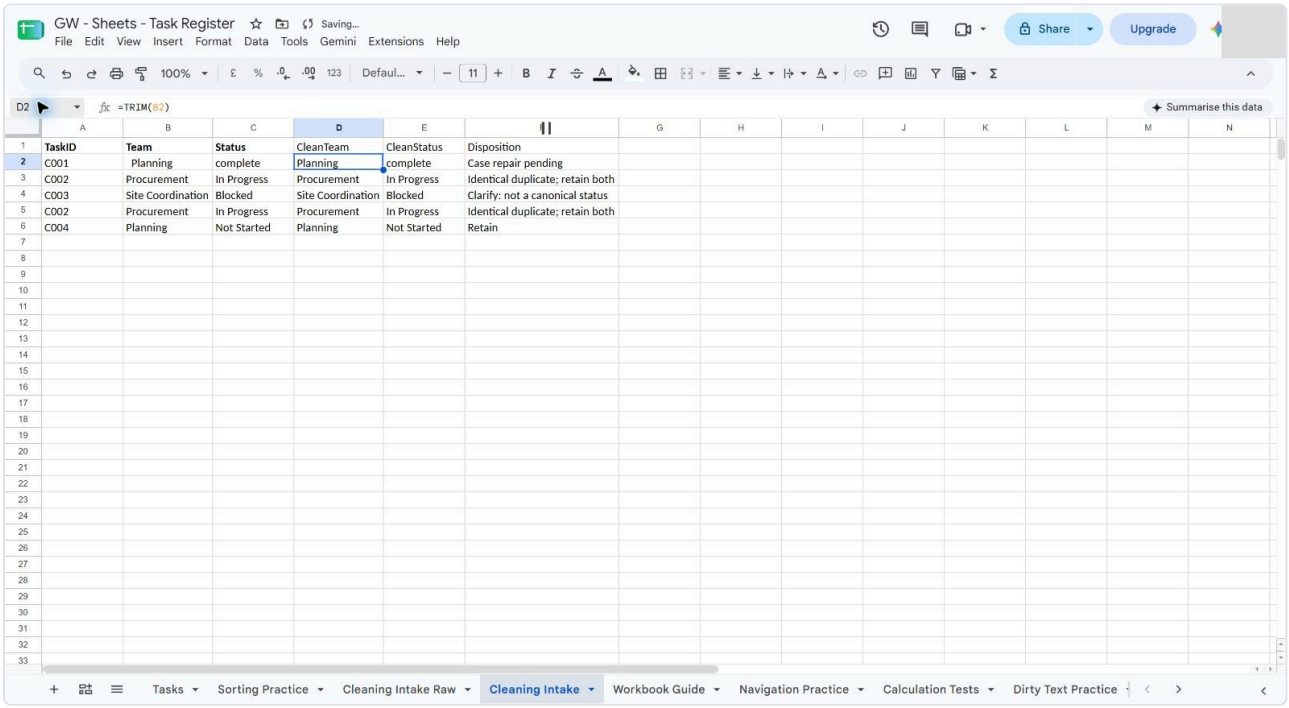


4.4 Clean an Intake and Prove That Records Stay Together

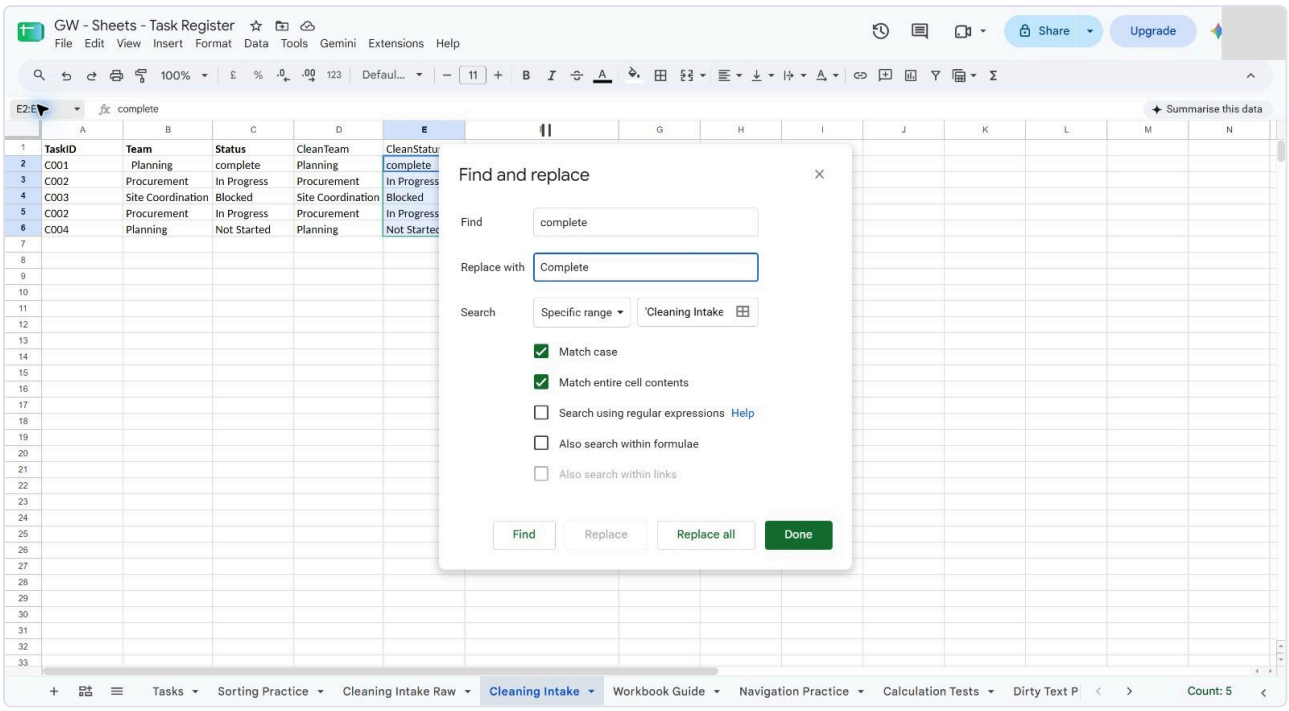
1. Create **Cleaning Intake Raw**, enter the supplied five records in A1:C6, and preserve the surrounding spaces in C001's team. Inspect the raw cell value, not only its appearance. The reference uses two leading and two trailing spaces around Planning. Duplicate this raw tab and rename the working copy **Cleaning Intake**. Keep the raw tab unchanged.



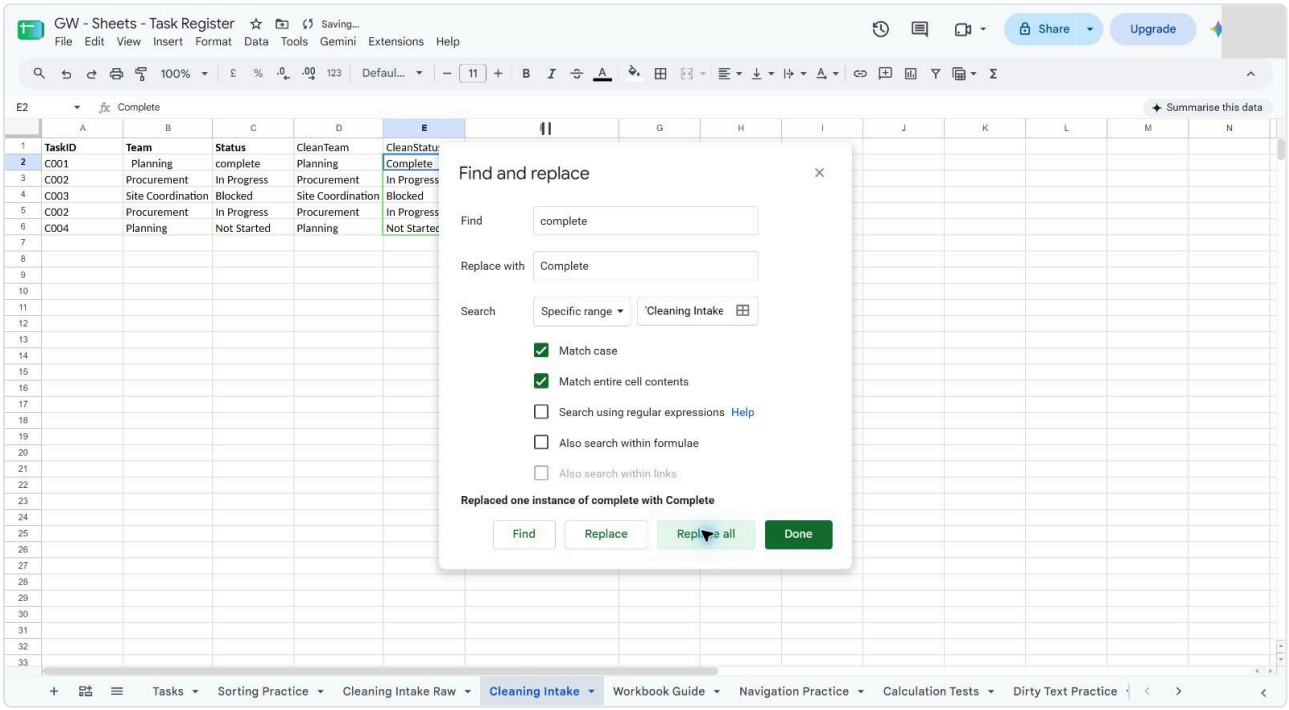
2. Add **CleanTeam / CleanStatus / Disposition** in D:F. Use **=TRIM(B2)** through D6. Initially copy the status values into E, so C001's cleaned status is still lowercase **complete** before the controlled repair. Compare both C002 records across all three supplied fields: they are identical. Keep both and label their disposition **Identical duplicate; retain both**. Retain **Blocked** on C003 and flag it for clarification; no canonical mapping is supplied. No owner or cost fields are invented for this intake.



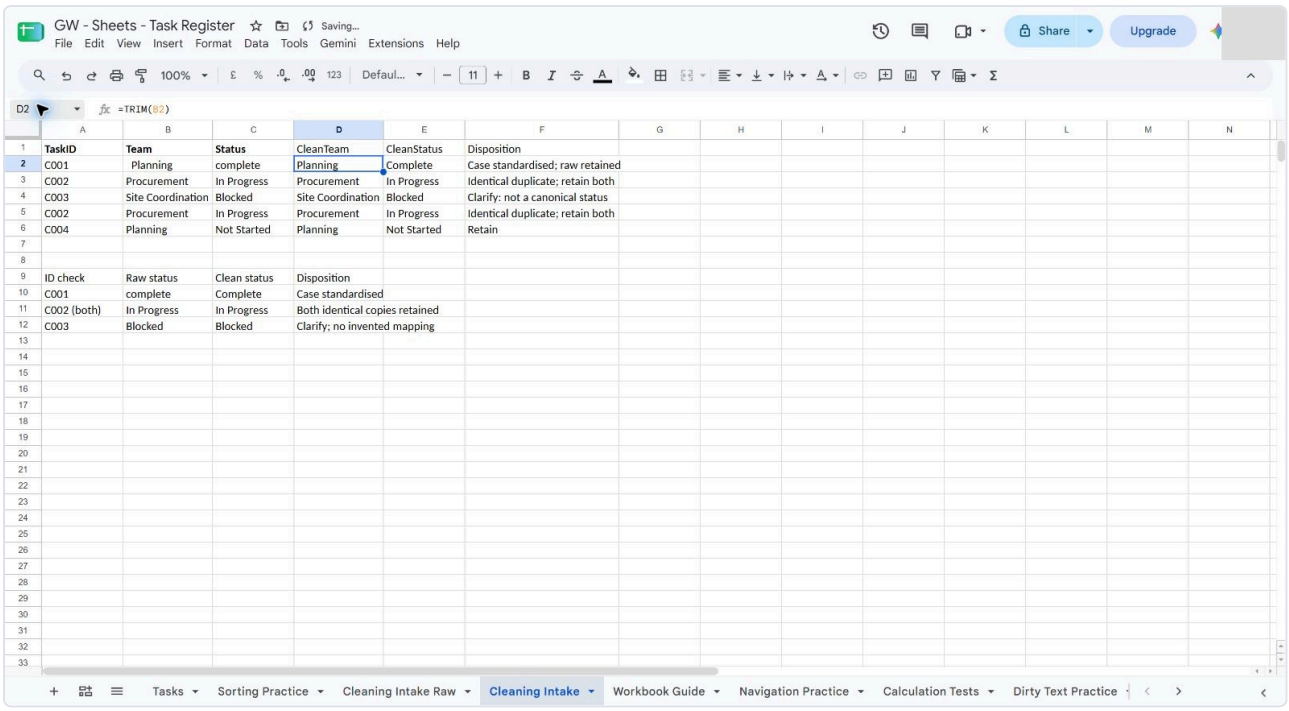
3. Select **E2:E6** and open **Find and replace** with Ctrl+H. Find **complete**, replace with **Complete**, and confirm **Specific range 'Cleaning Intake'!E2:E6**. Enable **Match case** and **Match entire cell contents**; leave regular expressions and formula search off. This is the known case repair supplied by the exercise, rather than an invented spelling error.



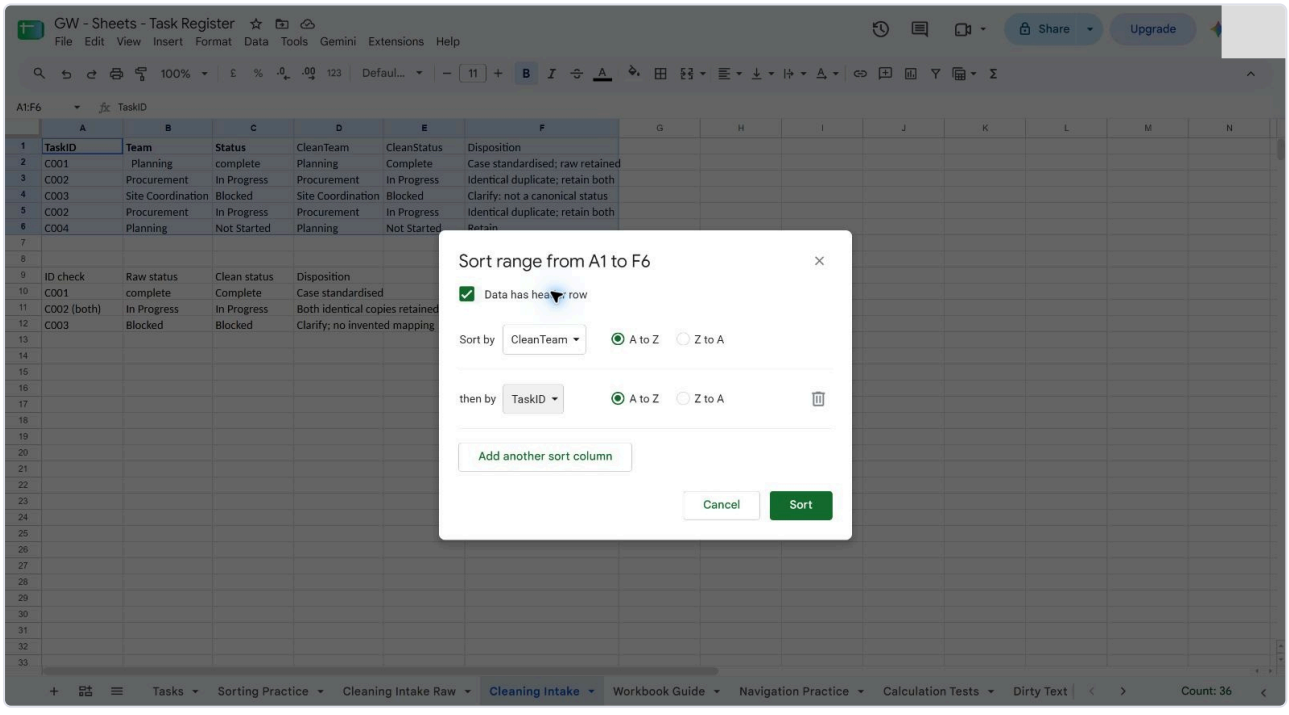
4. Preview with **Find**, then use **Replace all**. Verify **Replaced one instance of complete with Complete**, then **Done**. The raw C2 and raw tab still contain lowercase complete. Update C001's disposition to **Case standardised; raw retained**.



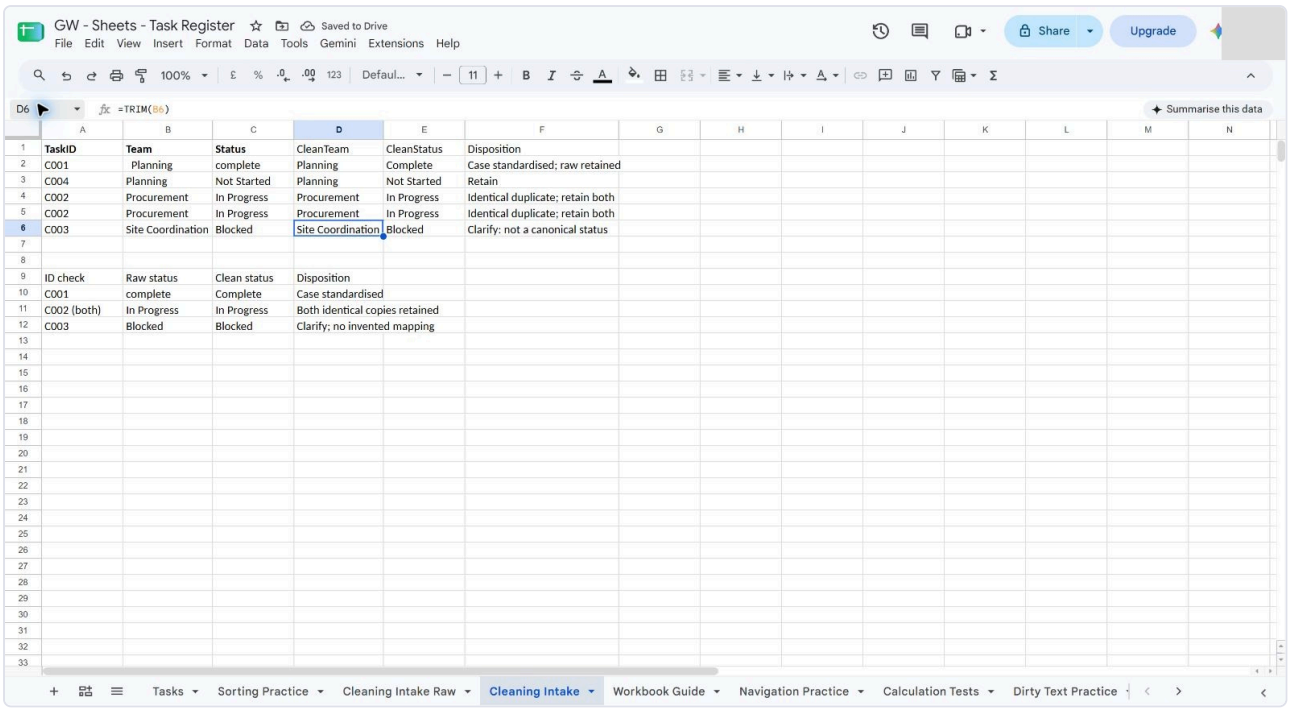
5. Record three TaskID-to-status checks before sorting in A9:D12: C001 complete→Complete, both C002 In Progress→In Progress, and C003 Blocked→Blocked with clarification pending. The supplied intake has no owners, so status relationships provide the identity check.



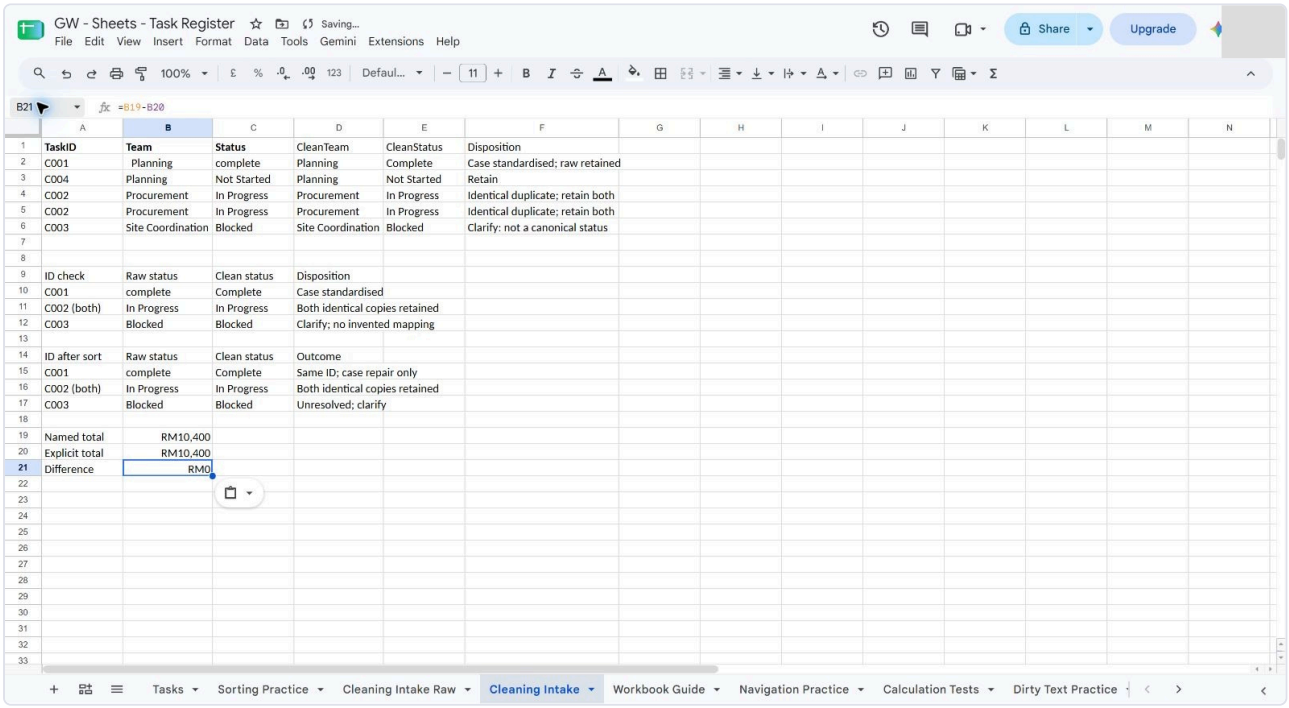
6. Select **A1:F6**, including raw fields, cleaned fields and dispositions. Open advanced sorting, mark the header row, and sort **CleanTeam A to Z**, then **TaskID A to Z**. Using the cleaned team avoids making whitespace part of the sort key. Keep the check table below row 6 outside the sort.



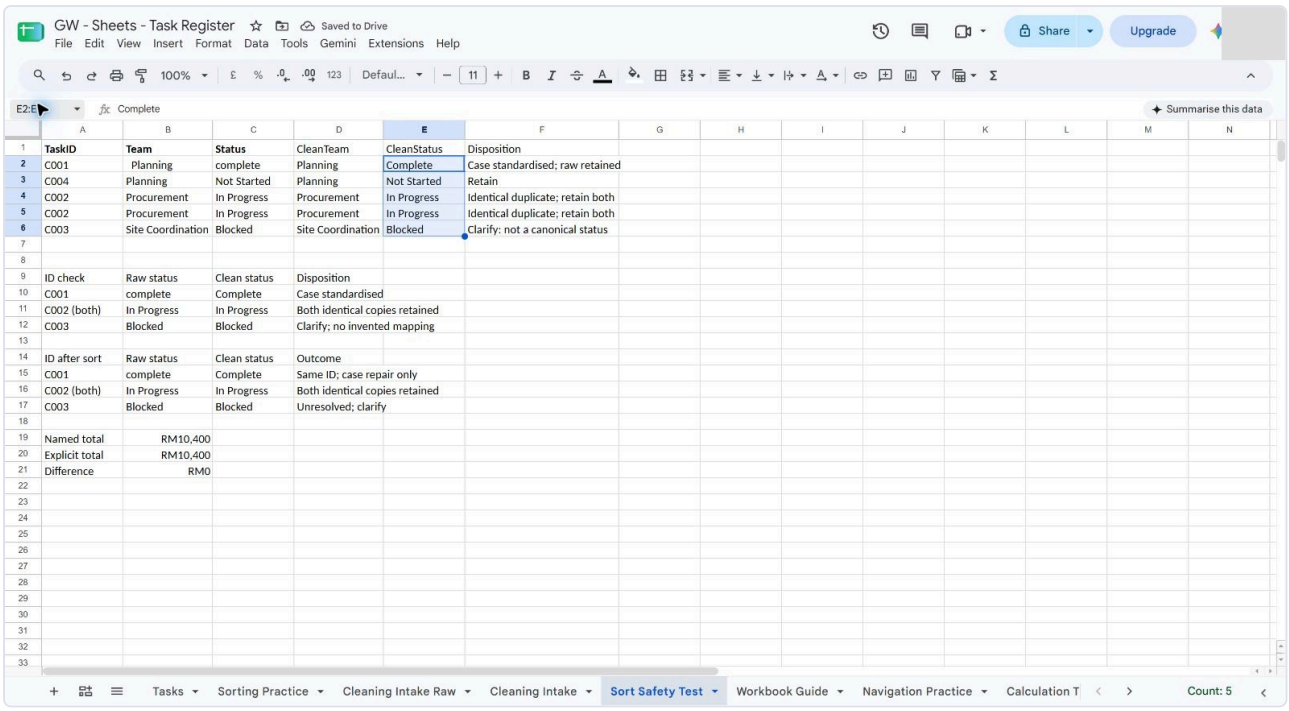
7. Verify the sorted ID order **C001, C004, C002, C002, C003**. Inspect the copied TRIM formulas: each still points to its own row's raw team. Compare the three identity checks again and record them in A14:D17. Both identical C002 rows remain; C003's unresolved status remains attached to C003.



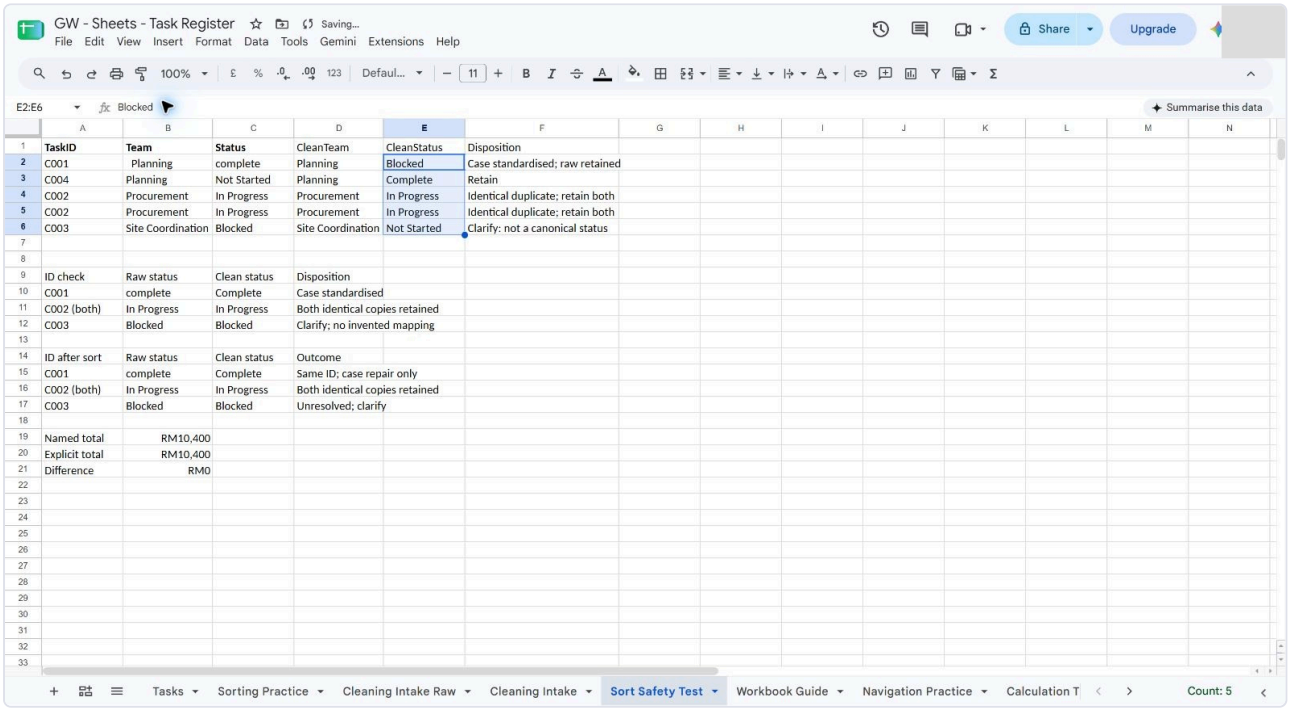
8. Reuse **PlannedCosts**, since this intake has no numeric cost column. In A19:B21 compare **=SUM(PlannedCosts)** with **=SUM('Sorting Practice'!G2:G13)**, then subtract the two results. Both give **RM10,400** and the difference is **RM0**. The exact twelve-cell bounds were inspected in 4.3. This does not test automatic expansion when new rows are added.



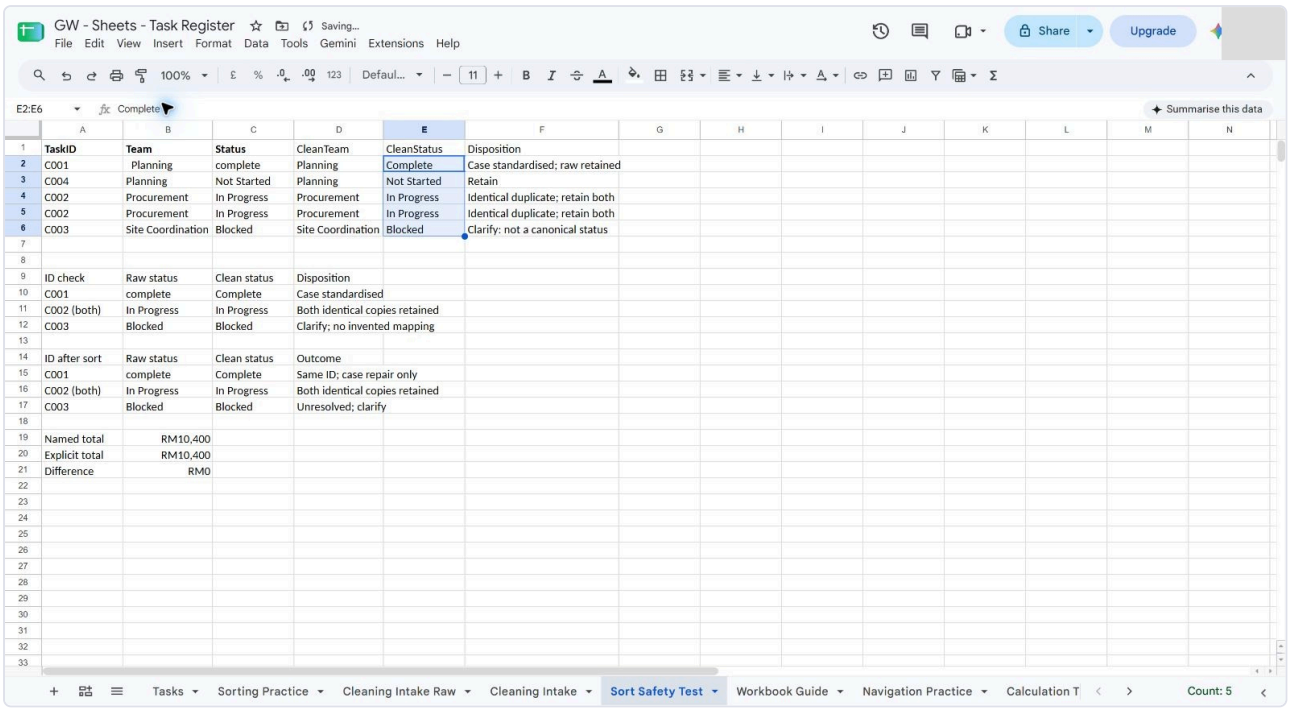
9. Duplicate the cleaned tab as **Sort Safety Test**. Select **only E2:E6** on this disposable copy. Keep the raw and clean intake tabs out of the test.



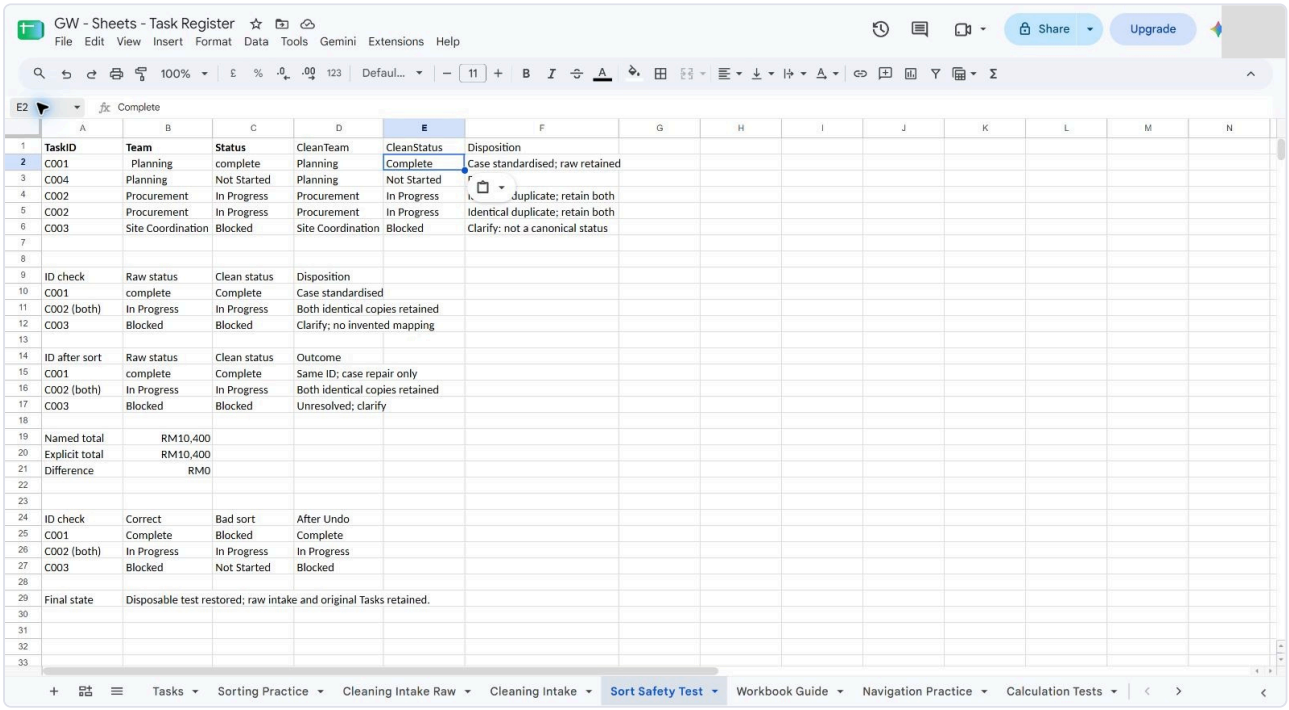
10. Use **Data > Sort range > Sort range by column E (A to Z)** on that isolated selection. Inspect the broken relationship: C001 now incorrectly has **Blocked**, C004 has **Complete**, and C003 has **Not Started**. The two C002 statuses happen to remain In Progress; checking only those would miss the defect.



11. **Undo immediately** with Ctrl+Z, before making any other edit. Compare all six columns and the three ID checks with Cleaning Intake. The reference verified an exact A1:F6 match after Undo.



12. Record the observed correct / bad-sort / after-Undo statuses in A24:D27. C001 is restored to Complete, both C002 to In Progress, and C003 to Blocked. Keep the restored disposable tab as labelled evidence. The original Tasks values/formulas and raw intake remain unchanged.



> **Validation boundary:** the reference verifies the guided and casework operations above. CSV file upload/import, the independent named-range expansion test, an independent cleaner's result and classroom timing remain pending.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Produce a clean intake and an exception list independently. Keep unresolved category and conflicting-duplicate decisions visible.
2. Add one new numeric row only to a named-range practice copy and inspect whether the named range includes it. Repair its bounds and compare the two total methods.
3. Hand over the raw copy, clean result and three row-identity checks so another person can verify the cleaning without trusting your description.

Evidence to Hand Over

- Preserved raw intake, justified clean values and explicit exceptions.
- Whole-record sorting evidence and a restored isolated-column demonstration.
- Named-range boundary test and independently reconciled total.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A clean-looking row is associated with the wrong task	Compare complete TaskID relationships before and after the sort, not only totals.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Cleaning, whole-record sorts and visible headers make the data safer to analyse.

Check yourself: What must stay together when sorting a register?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
CleanTeam
```

```
=TRIM(B2)
```

```
PlannedCosts
```

```
=SUM(PlannedCosts)
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Cleaning intake (TaskID | Team | Status): C001 |  Planning  | complete; C002 | Procurement |
In Progress; C003 | Site Coordination | Blocked; C002 | Procurement | In Progress; C004 |
Planning | Not Started. C001 has surrounding team spaces. The duplicate is identical; Blocked
remains unresolved. These are separate practice records.
```

Exact DirtyText.csv Records

For manual entry, treat each CSV field below as a separate cell and preserve the spaces inside its quotation marks. Do not paste the entire comma-separated block into one cell.

```
"TaskID","Team","Status"
"T001"," Planning ","complete"
"T002","Procurement","In Progress"
"T003","Site Coordination ","Not Started"
```

Manual Raw-Entry Fallback

CSV import remains unverified in the captured browser session. If entry trims the surrounding spaces, use a leading apostrophe in the cell editor to preserve the literal raw text; do not include that prefix in the intended stored value. DirtyText.csv raw team values are `Planning`, `Procurement`, and `Site Coordination`, with lengths10,11,18. TRIM produces Planning/Procurement/Site Coordination, lengths8,11,17. Keep raw A:C and cleaned D:E separate.

Cleaning Intake Raw A1:C6 contains the five supplied C records. The reference explicitly preserves two spaces before and after C001's Planning. Duplicate it as Cleaning Intake. D2:D6 uses TRIM of B on each row; E initially copies raw status. Restrict Find and replace complete→Complete to E2:E6, matching case

and entire cell; one replacement. Both C002 records are identical and retained. C003 Blocked is unresolved, with no invented mapping.

Sort A1:F6 by CleanTeam then TaskID, with the header option: C001,C004,C002,C002,C003. Inspect C001/C002/C003 status relationships before and after. In a disposable Sort Safety Test copy, sorting only E2:E6 A→Z makes C001 Blocked and C003 Not Started; Undo restores Complete and Blocked respectively. C002 remains In Progress through this test.

PlannedCosts bounds: **'Sorting Practice'!G2:G13**, twelve numeric cells. SUM(PlannedCosts)=10400, explicit SUM of that range=10400, difference0. No new-row expansion test is claimed. Tasks A1:L13 values/formulas remain unchanged. Independent challenge and learner timing remain pending.

05

CHAPTER 5

Validate and Filter the Register

In this chapter you will add a status dropdown, identify invalid entries and create a filter view.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 50 minutes: 10 explanation/demonstration, 20 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 40 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A status dropdown and two named filter views.

> **Status:** author reference with 34 staged application captures through 5.4. Strict status validation, actual rejection tests, whole-week boundaries/fractions, clearing limitations, two saved views, visible TaskID checks, ordinary SUM under filtering and restoration verified. Independent coverage/range repair and learner timing remain pending.

What You Will Learn

- Restrict status values.
- Create and name a filter view.
- Check hidden rows before interpreting totals.

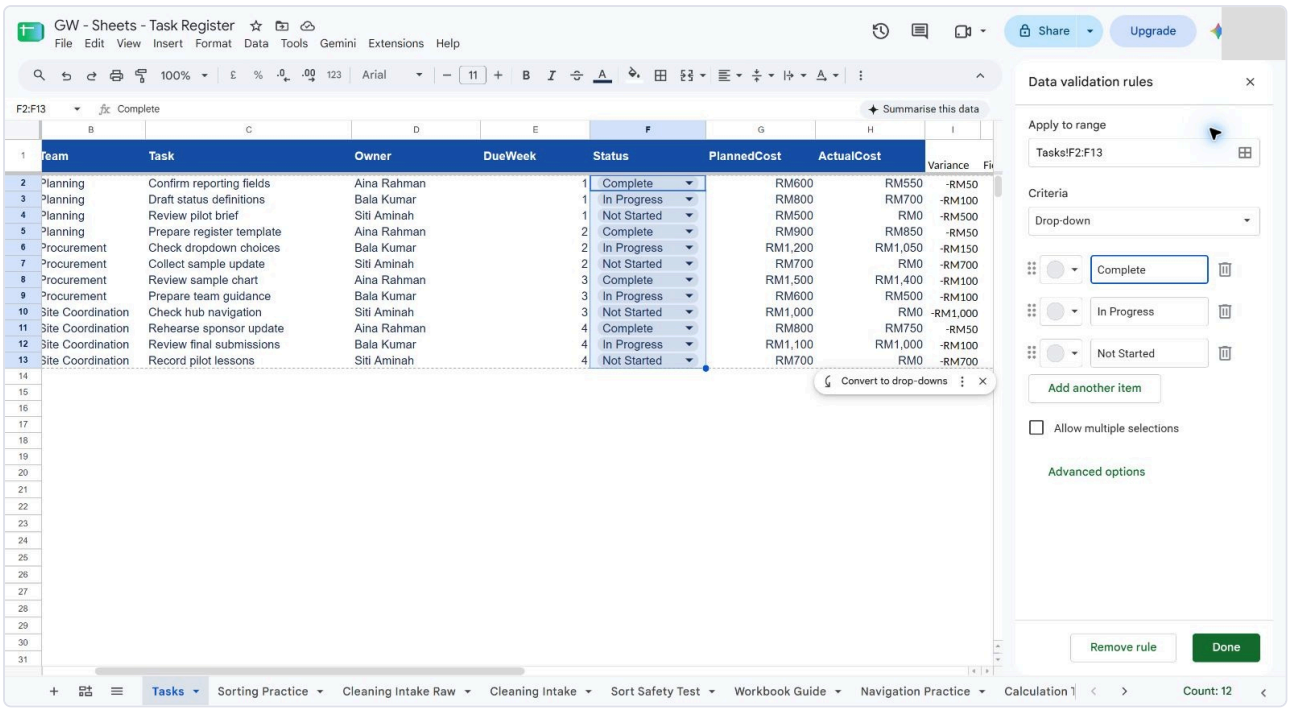
Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

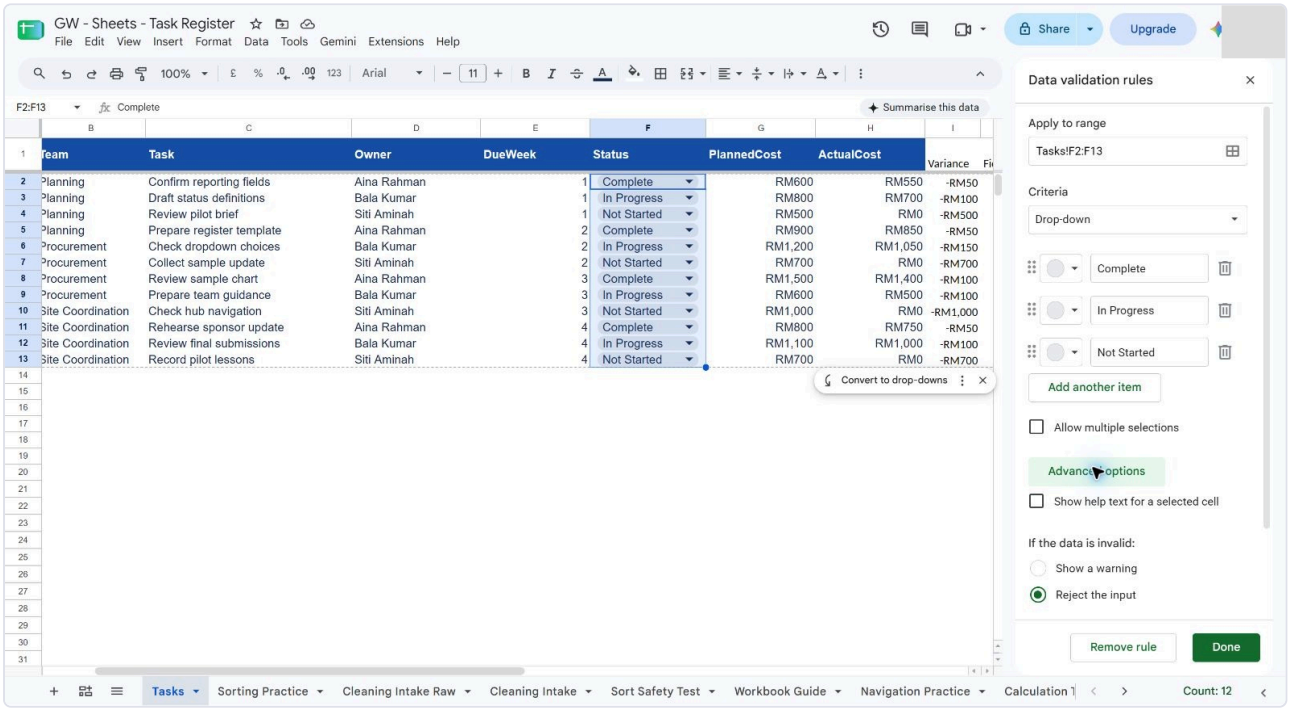
Use the original Tasks table, with all 12 baseline records still present.

5.1 Add a Status Dropdown

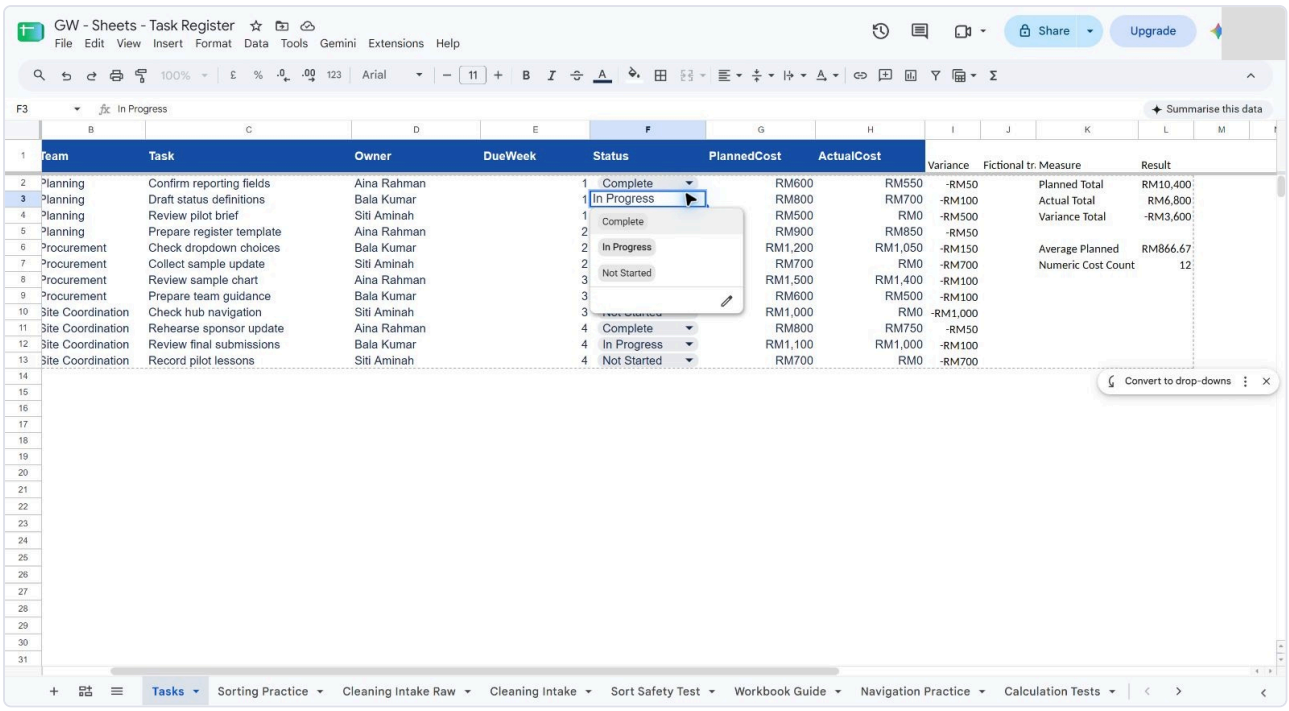
1. On **Tasks**, select **F2:F13** and open **Data > Data validation > Add rule**. Confirm **Apply to range Tasks!F2:F13**. In the current interface the default Drop-down criterion populated its choices from the selected cells; inspect rather than assuming those inferred choices are correct. Use exactly **Complete / In Progress / Not Started**.



2. Open **Advanced options**, confirm **Reject the input**, and keep **Allow multiple selections** off. The reference uses the Chip display. Click **Done**, then close the sidebar.

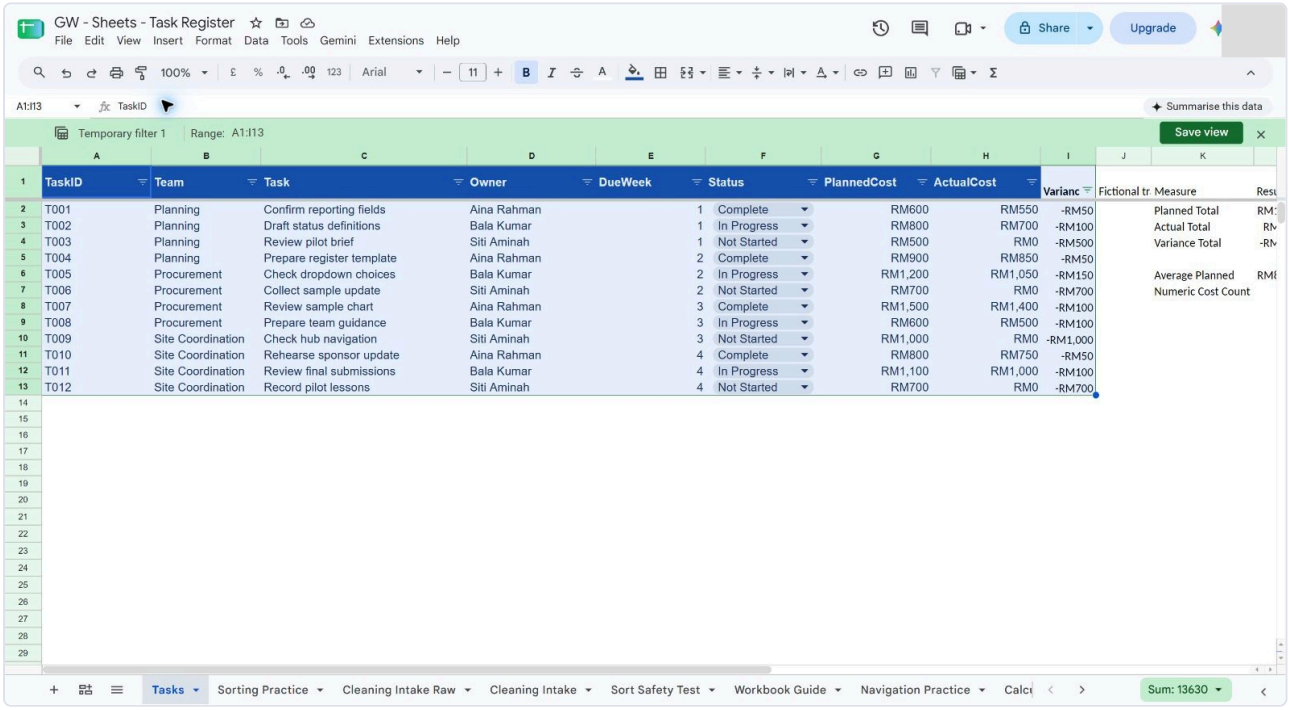


3. Inspect **F3** and open its dropdown. Its existing **In Progress** value remains unchanged, and the three canonical choices are available. Adding a rule must not rewrite the twelve source statuses.

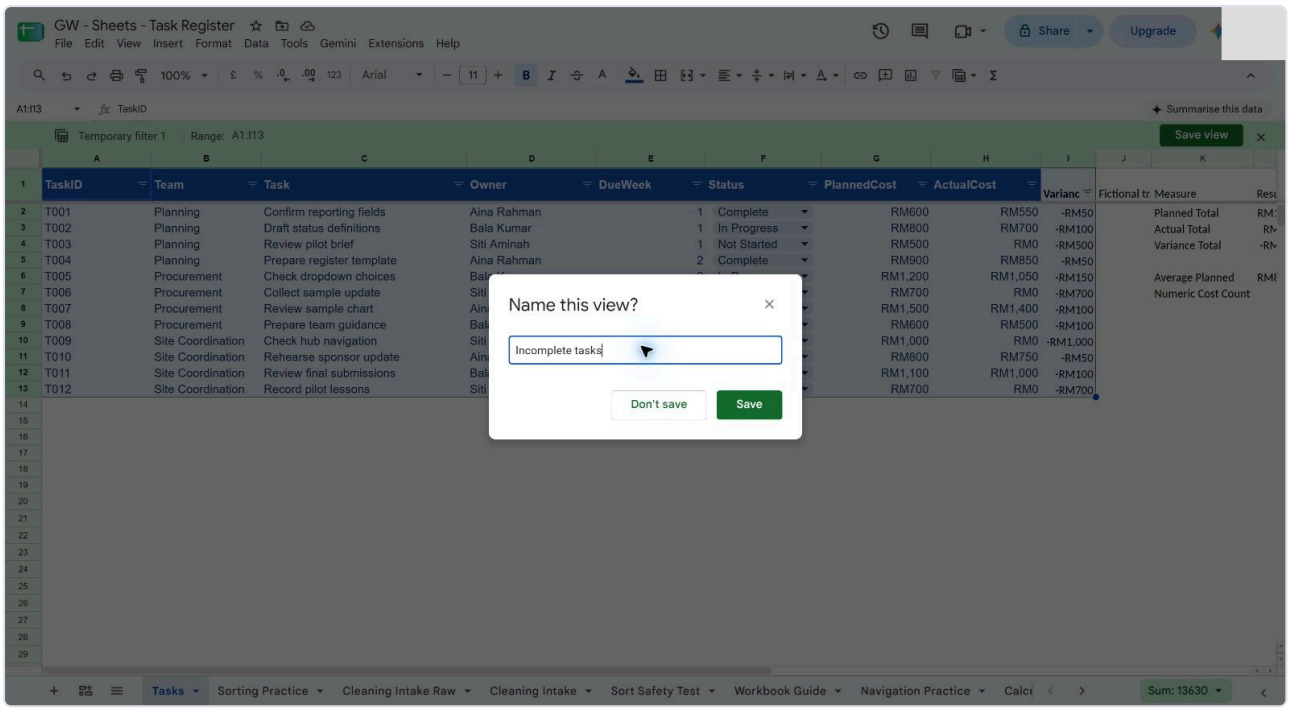


5.2 Create a Filter View

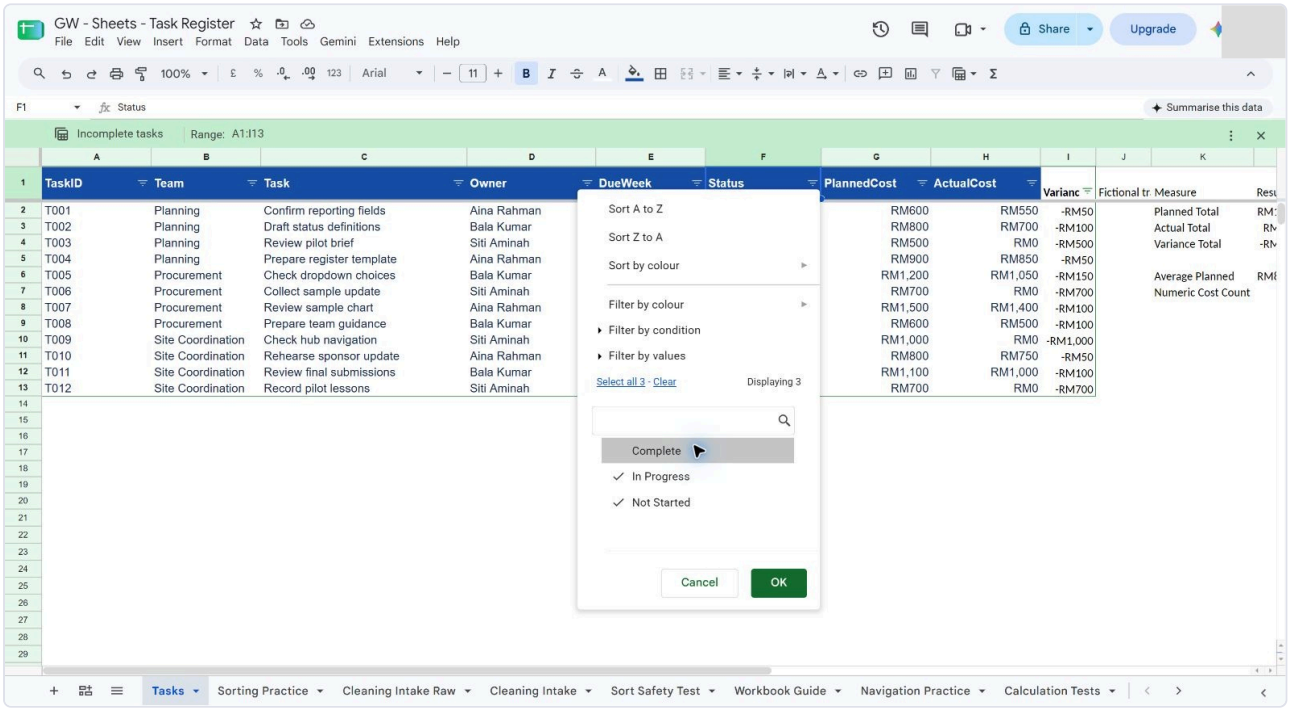
1. Select **A1:I13**, including the header and variance. Use **Data > Create filter view**, or the toolbar's **Filter views > Create filter view**. This interface initially creates **Temporary filter 1**; inspect its range before saving.



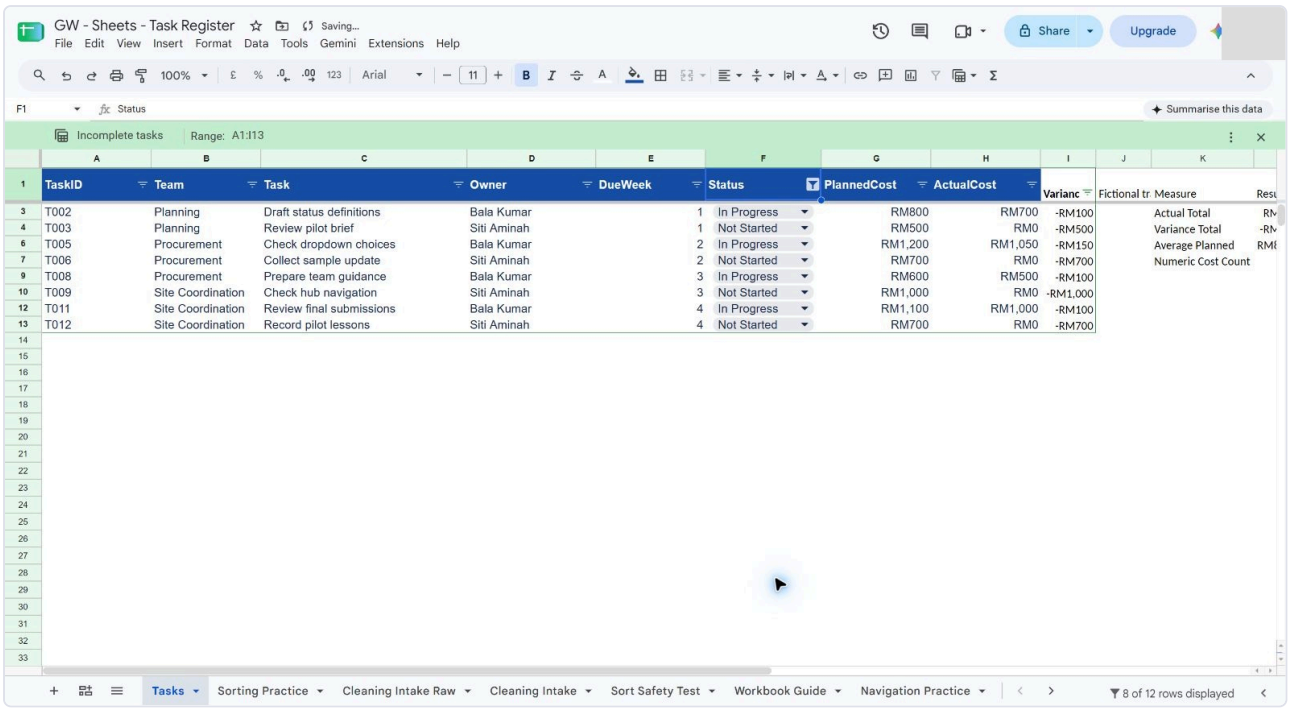
2. Click **Save view**, enter **Incomplete tasks** in **Name this view?**, then **Save**. This save-and-name step replaces the older guide's assumption that a new view is named immediately.



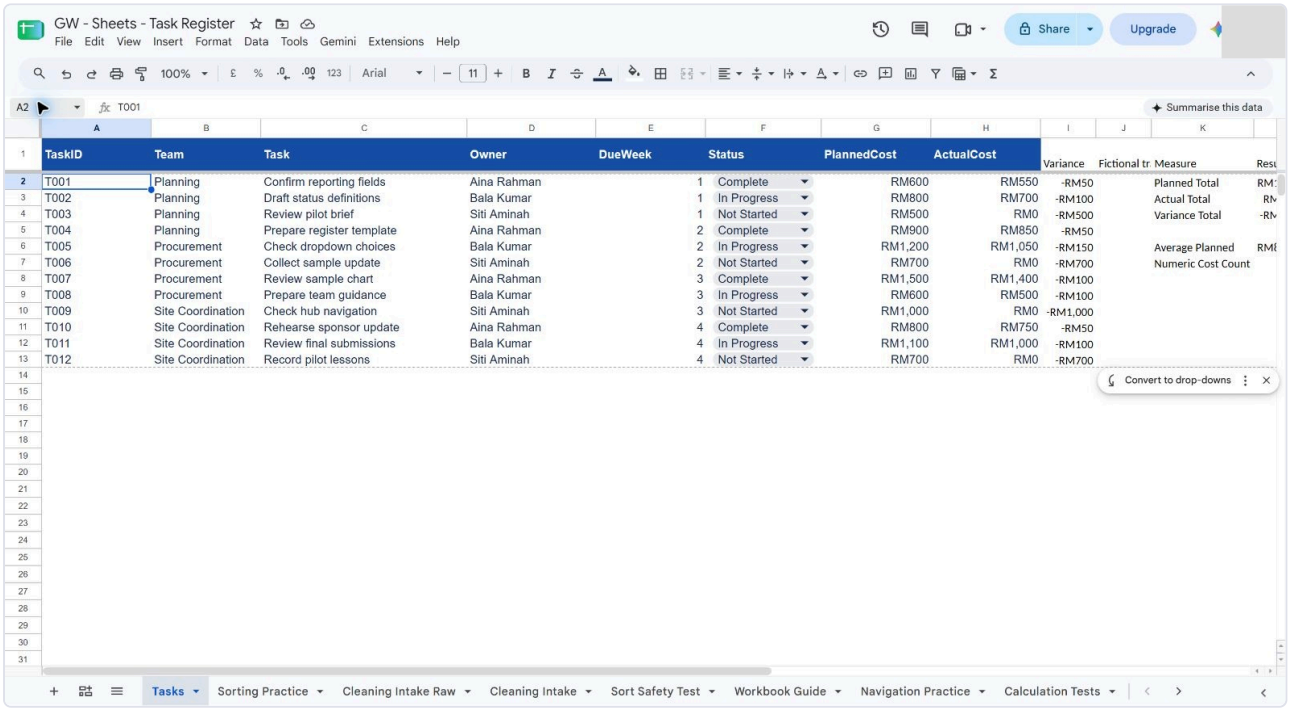
3. Open the Status header filter, uncheck **Complete**, and click **OK**. Leave **In Progress** and **Not Started** selected.



4. Count the visible IDs: **T002, T003, T005, T006, T008, T009, T011, T012**. The lower-right indicator reports **8 of 12 rows displayed**. Compare these with the original source statuses.

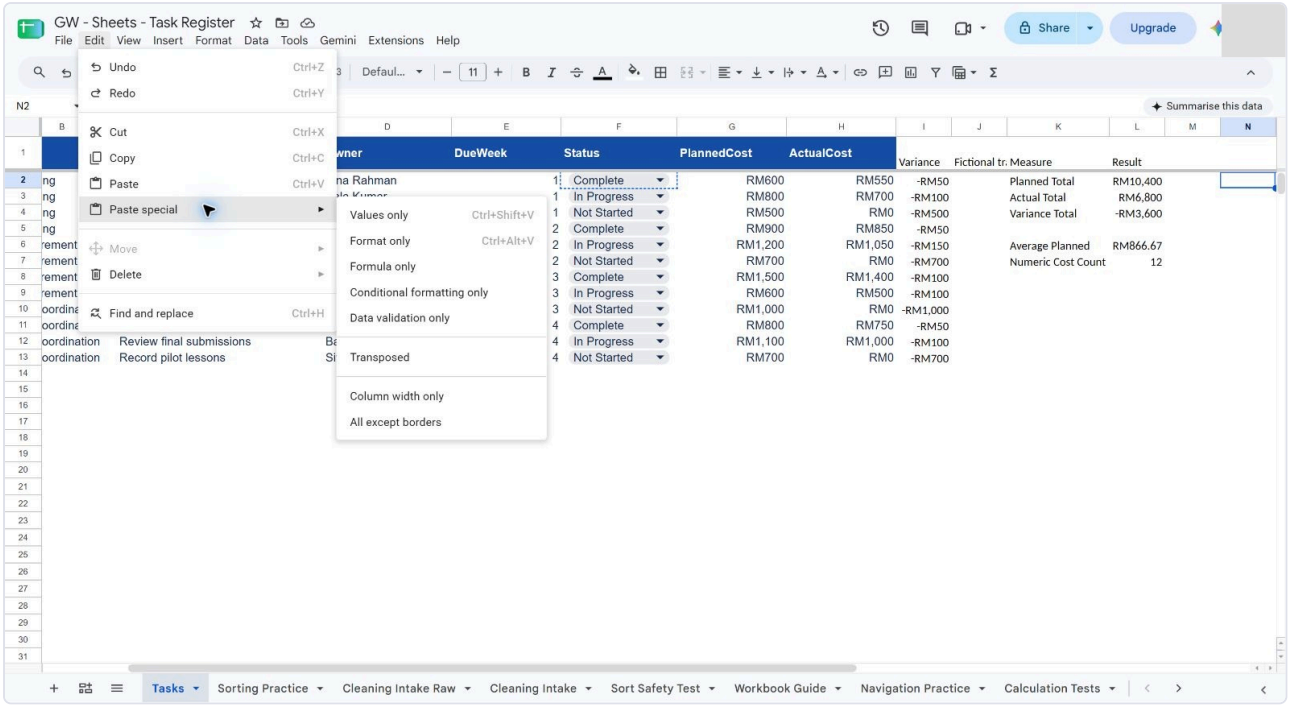


5. Use **Exit** at the right of the green filter-view bar. All twelve records return. Closing the view does not delete its saved definition or any hidden records.

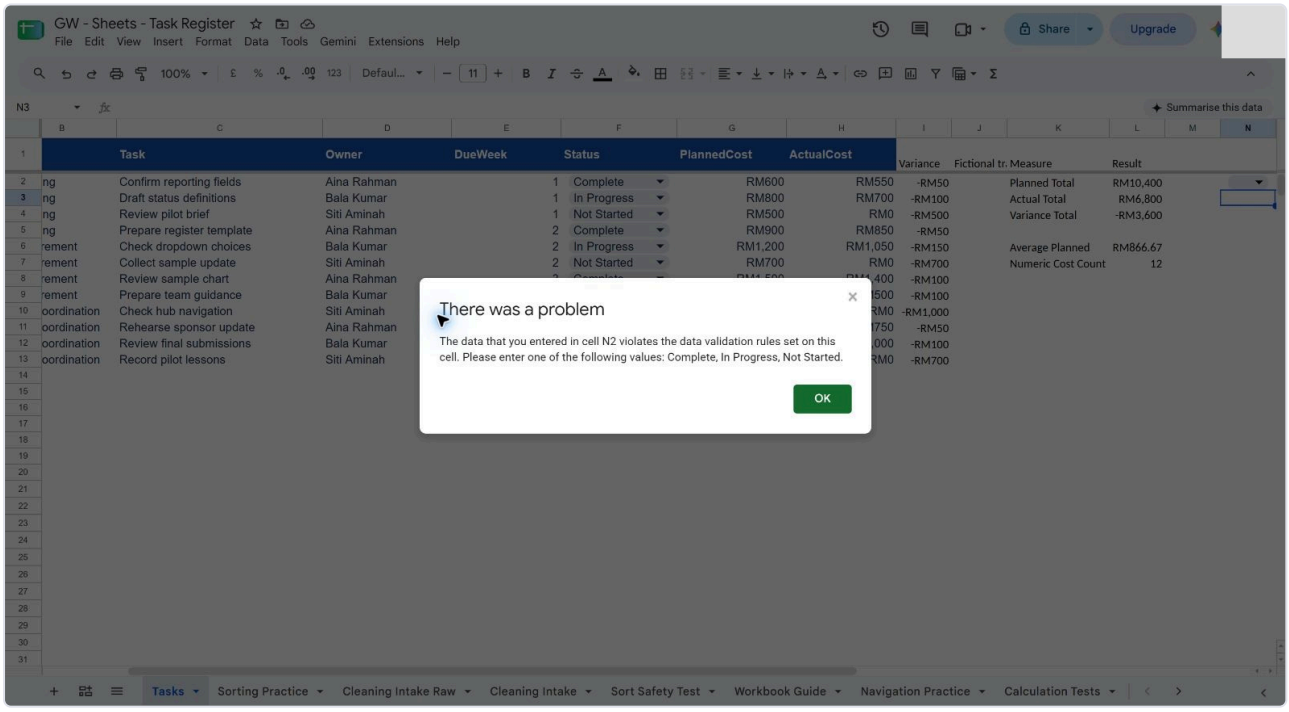


5.3 Test a Rule Safely

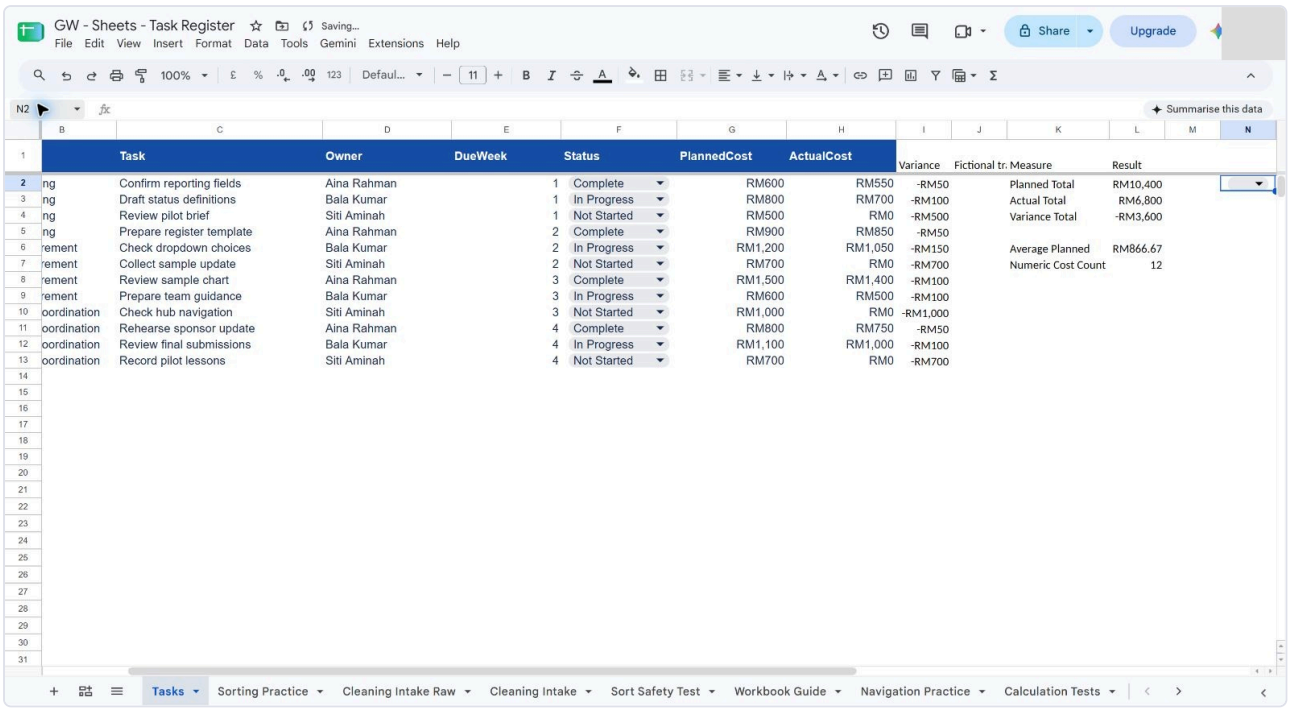
1. Select **F2**, copy it, and select the unused **N2** outside the source register. Use **Edit > Paste special > Data validation only**. This transfers the rule without copying a source status value. Keep the copied clipboard intact while selecting the destination; pasting a cell address would replace it.



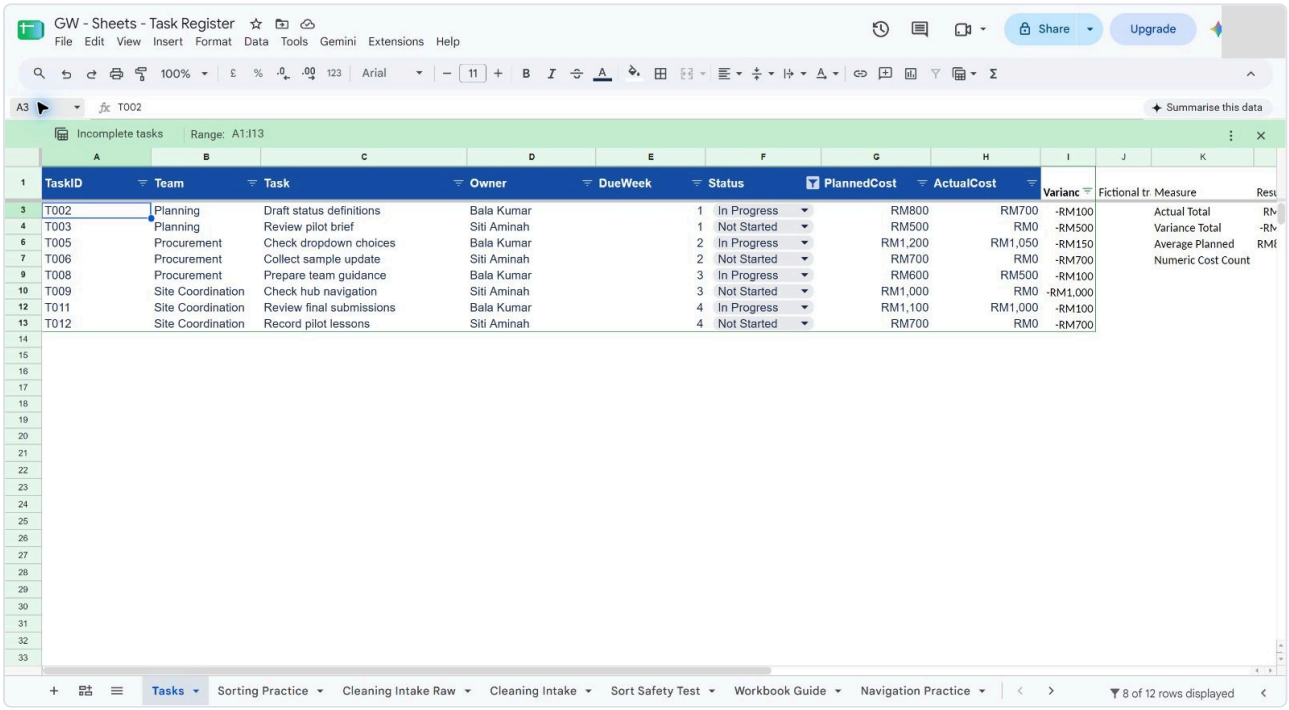
2. Edit N2 with **F2** (the edit-cell key), enter **Done**, and commit. The reference shows **There was a problem** and identifies the allowed values **Complete, In Progress, Not Started**. This is rejection, not a warning that leaves an invalid value stored.



3. Dismiss the message, exit editing if necessary, and clear N2. Its value is empty; its copied validation remains available for safe demonstrations. No baseline status was changed.

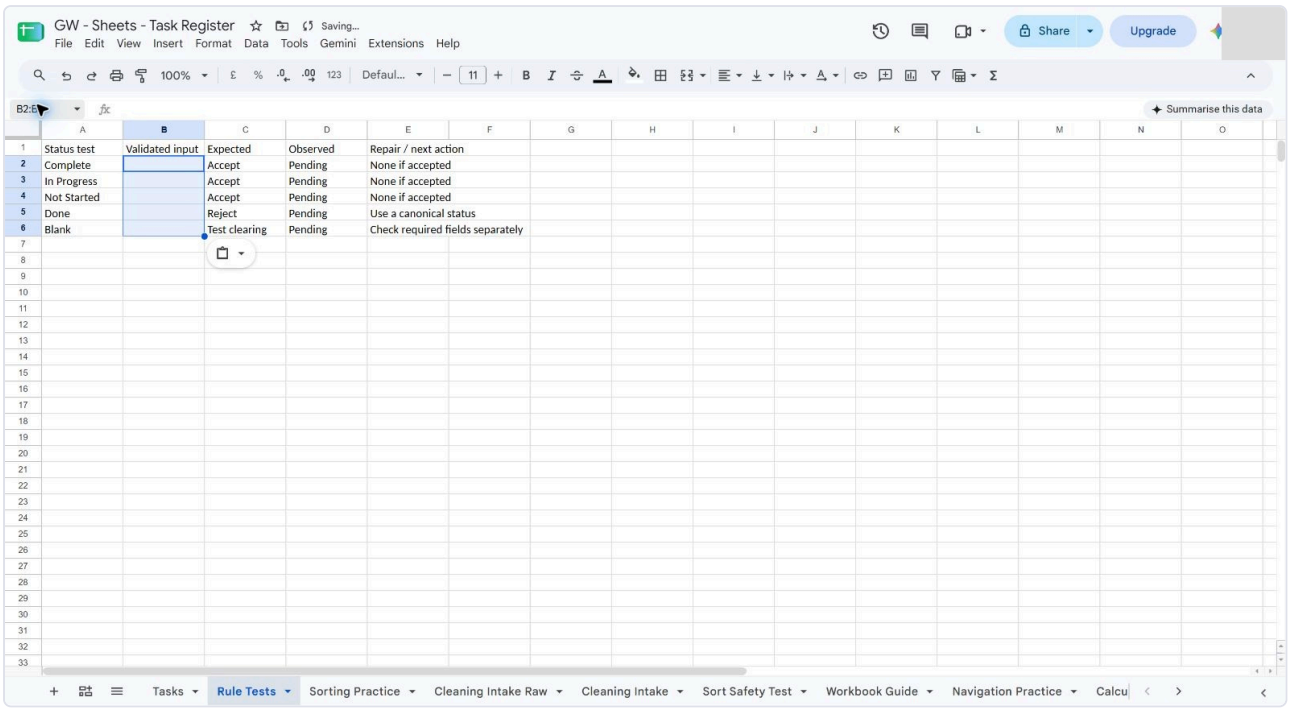


4. Open the toolbar's **Filter views** menu and choose **Incomplete tasks**. It still shows the same eight task IDs. Exit the view before continuing.

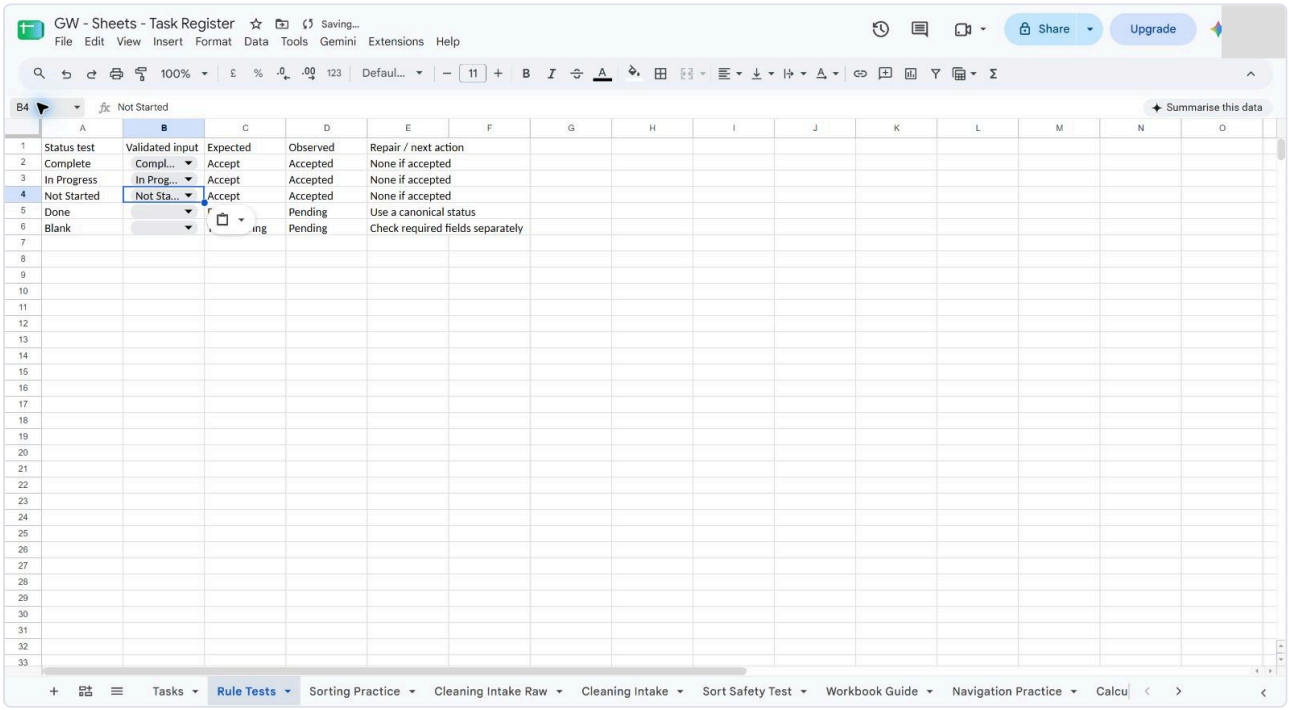


5.4 Design and Verify Entry Rules and Focused Views

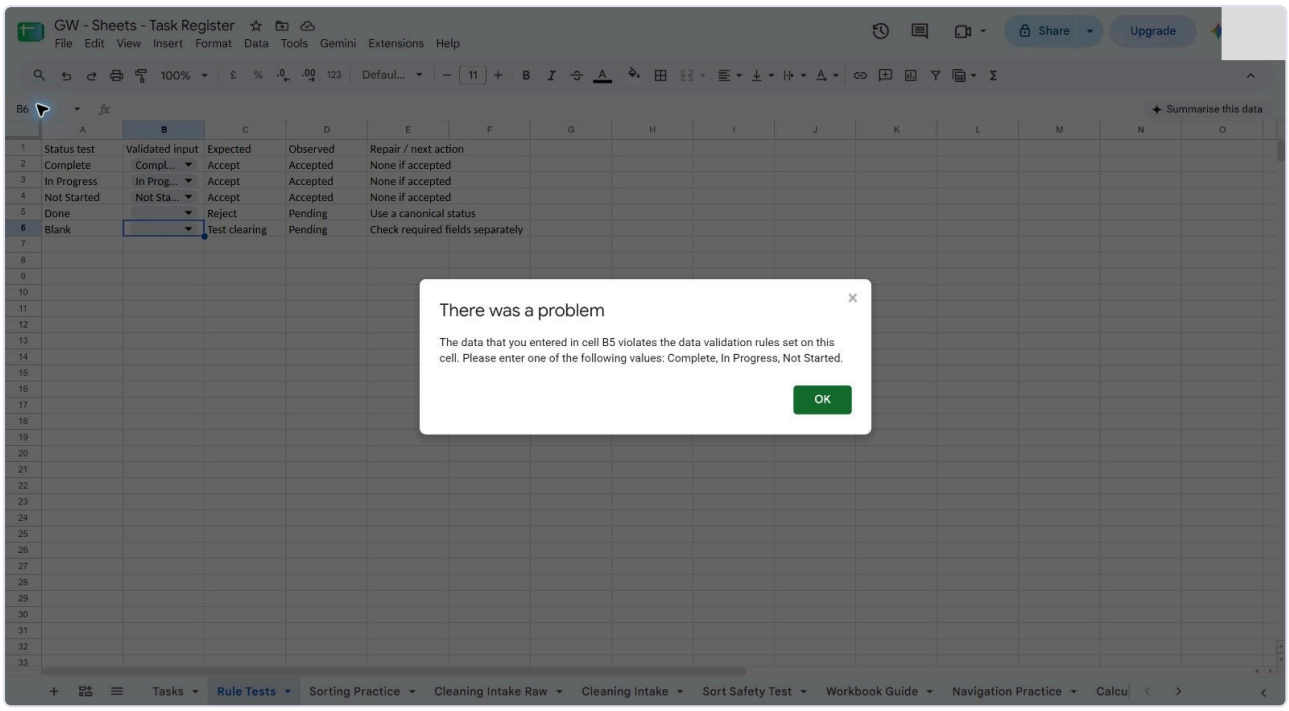
1. Create **Rule Tests** with headers **Status test / Validated input / Expected / Observed / Repair or next action** in A1:E1. Plan **Complete, In Progress, Not Started, Done and Blank** in rows 2–6. Predict acceptance for the canonical statuses and rejection for Done; leave observed results pending. Treat clearing as an actual test, not an assumed required-field guarantee.



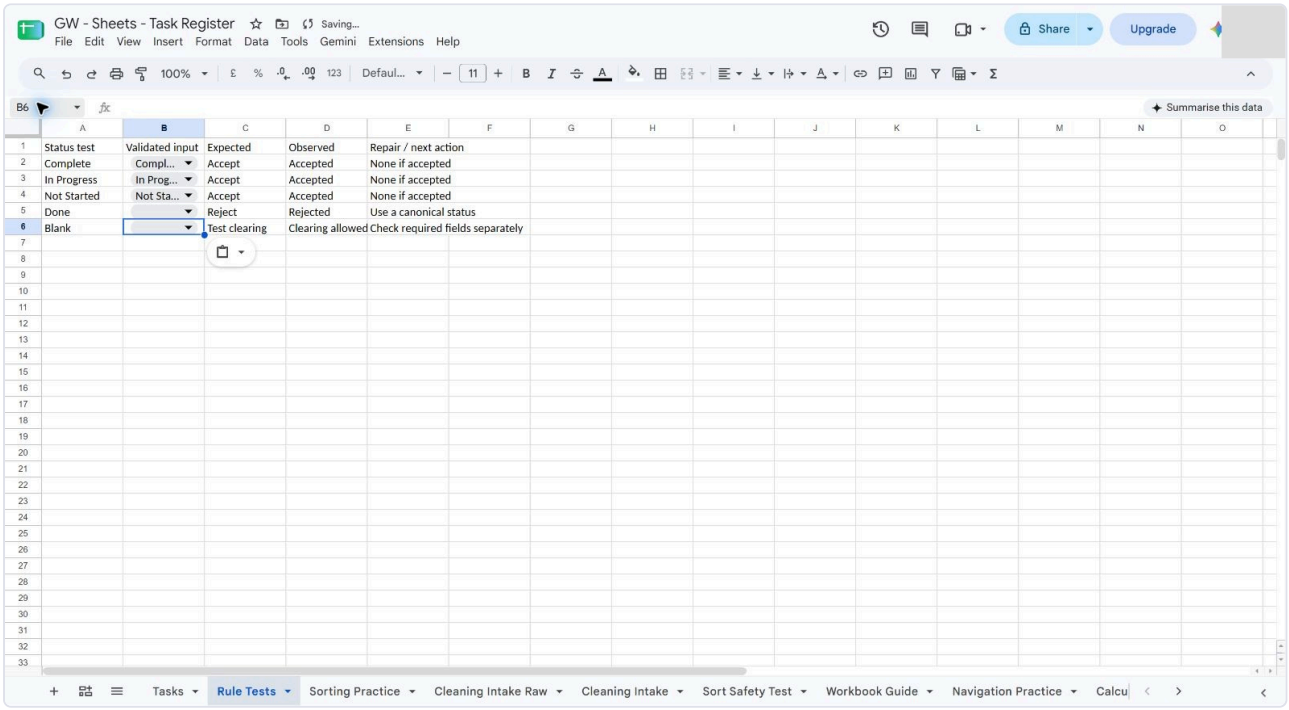
2. Copy validation from Tasks F2 to **Rule Tests B2:B6** using **Paste special > Data validation only**. Enter test values through cell-edit mode, preserving the rule. The three canonical statuses are accepted in B2:B4.



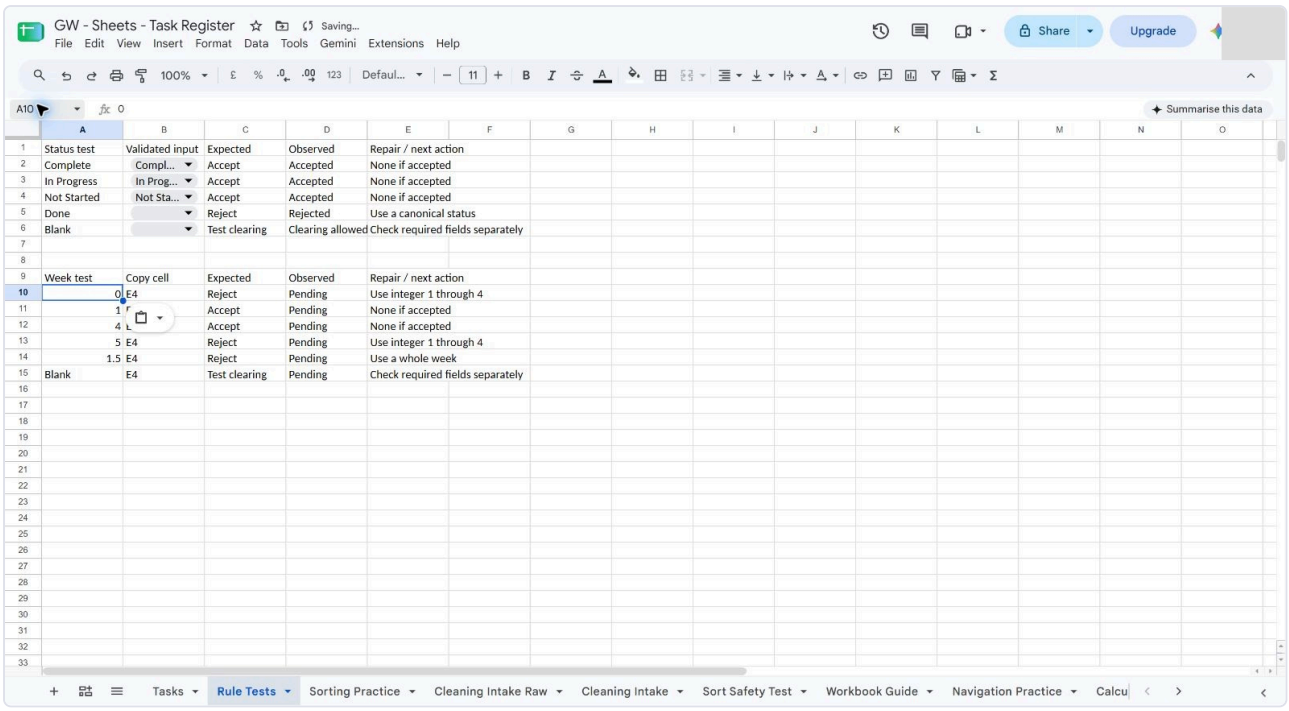
3. Attempt **Done** in B5. It is rejected with the same allowed-value message. Dismiss the message; B5 stays empty. Record **Rejected** separately from the attempted value.



4. Enter a valid status in B6, then clear it with Backspace. Clearing is allowed. Record that observation and keep B6 empty. The tested rule limits nonempty status entries but does not prove that every required field is complete.



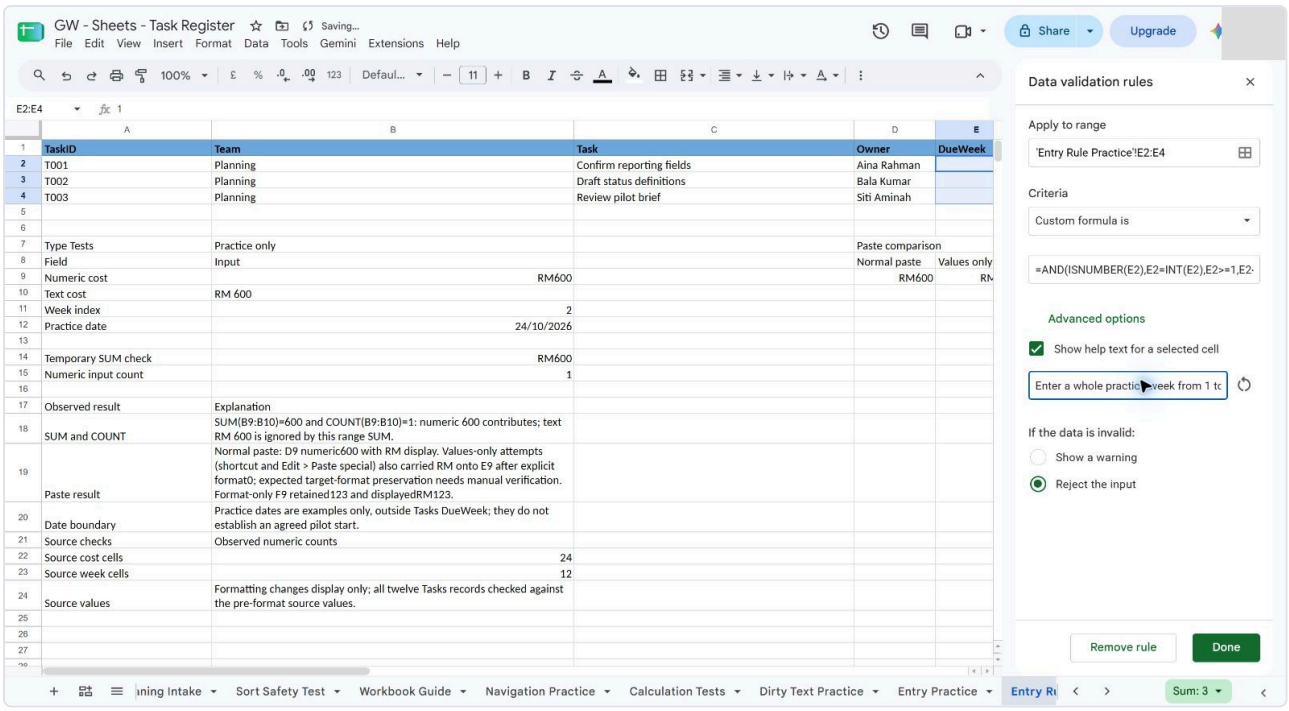
5. Plan the week tests in **A9:E15**: **0 reject / 1 accept / 4 accept / 5 reject / 1.5 reject / blank clearing to test**. Use a copy named **Entry Rule Practice**, created by duplicating **Entry Practice**. The test target is **E4** on that copy; the original three copied records remain unchanged.



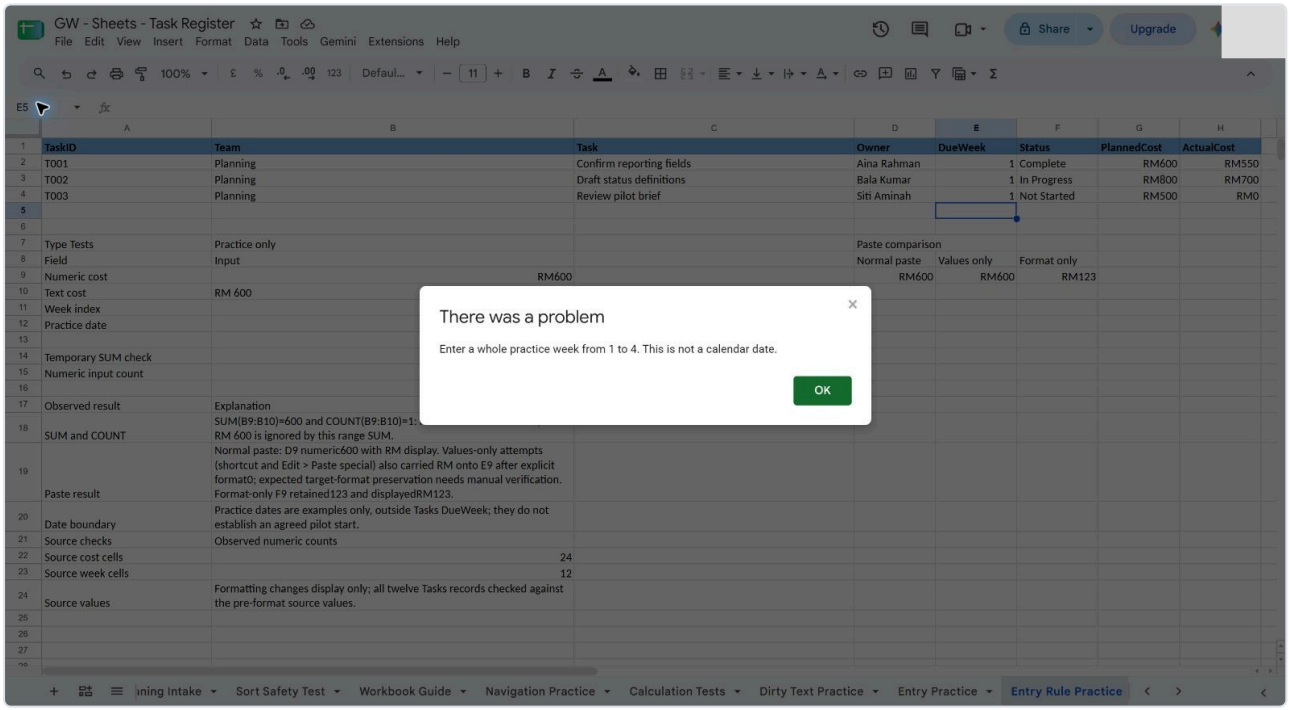
6. On **Entry Rule Practice**, select **E2:E4** only, open **Data validation > Add rule**, choose **Custom formula is**, and enter:

```
=AND(ISNUMBER(E2),E2=INT(E2),E2≥1,E2≤4)
```

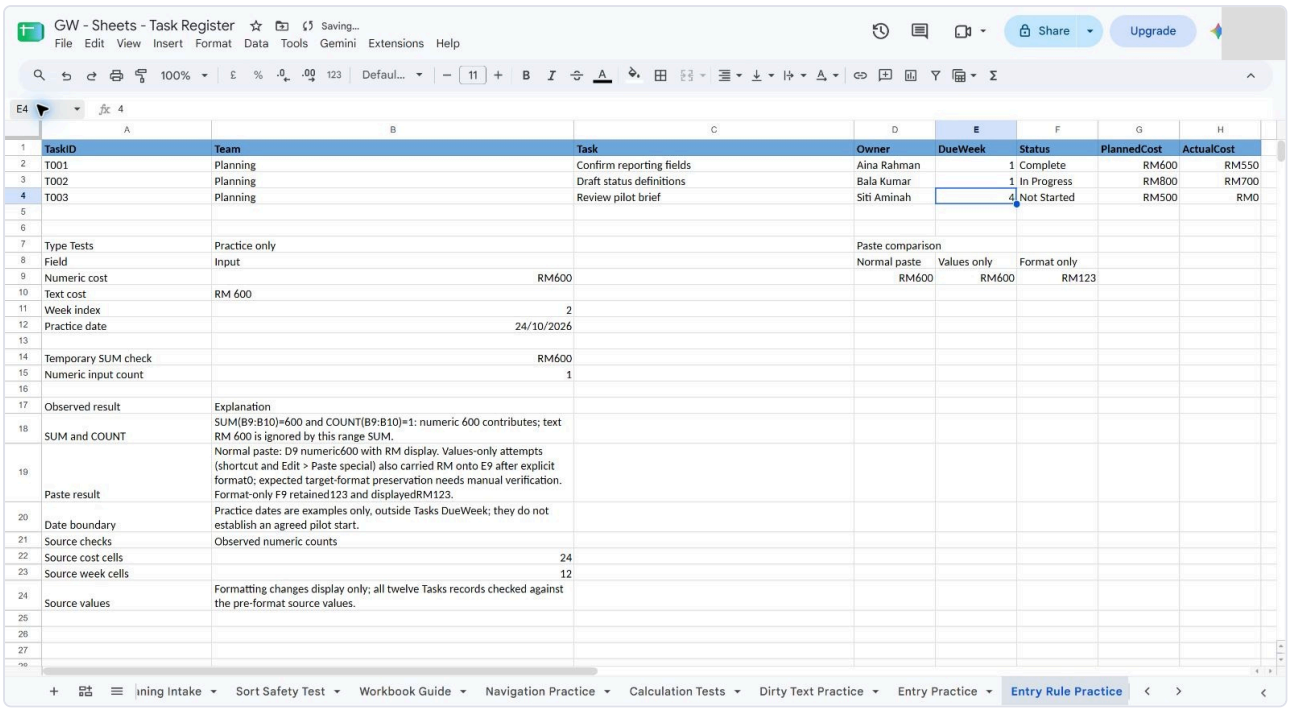
Use the top-left cell E2 as the relative reference. This checks numeric type, a whole number and both bounds; a simple between rule alone would not express the whole-week requirement. Explicitly choose **Reject the input**: the custom criterion initially offered **Show a warning** in this session. Enable help text **Enter a whole practice week from 1 to 4. This is not a calendar date.**, then save and close the sidebar. E2:E4 covers the three-record practice table, not the notes and type tests farther down that tab.



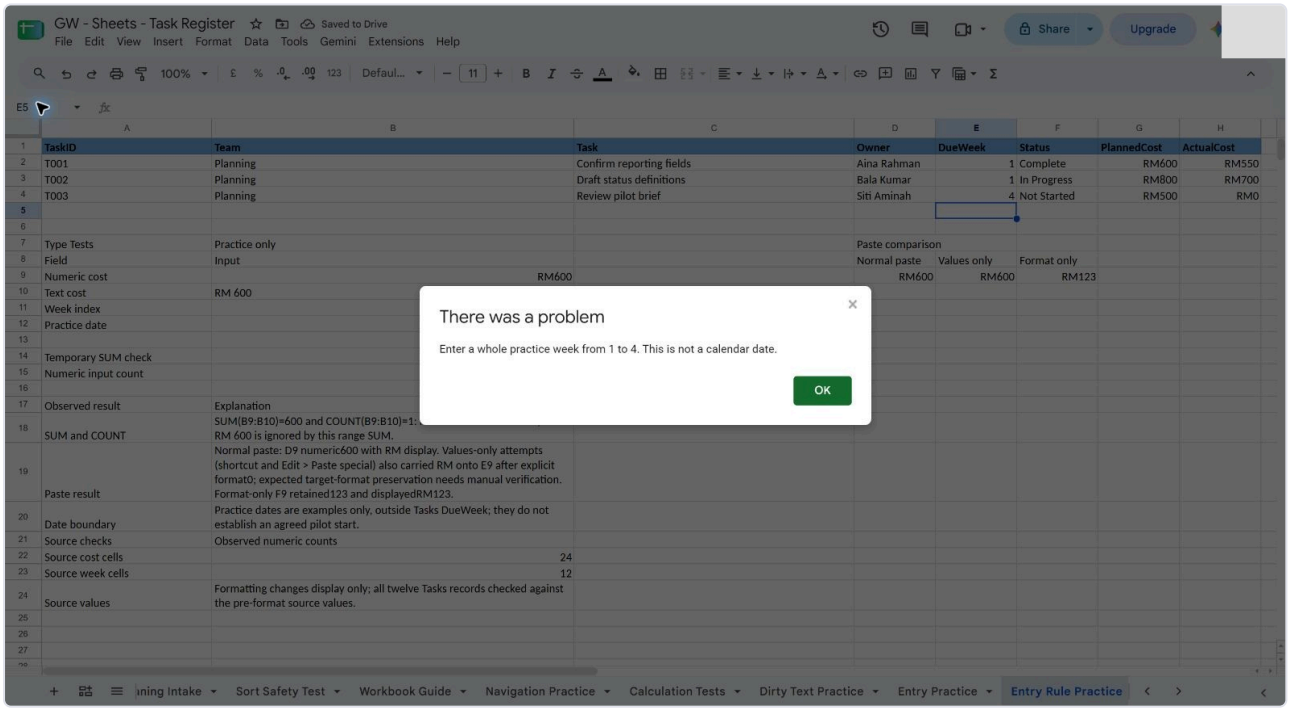
7. Reuse E4 for each planned test. Enter edit mode, select its previous content, type the candidate and commit. **0** is rejected with the configured help text. Dismiss the message before the next test.

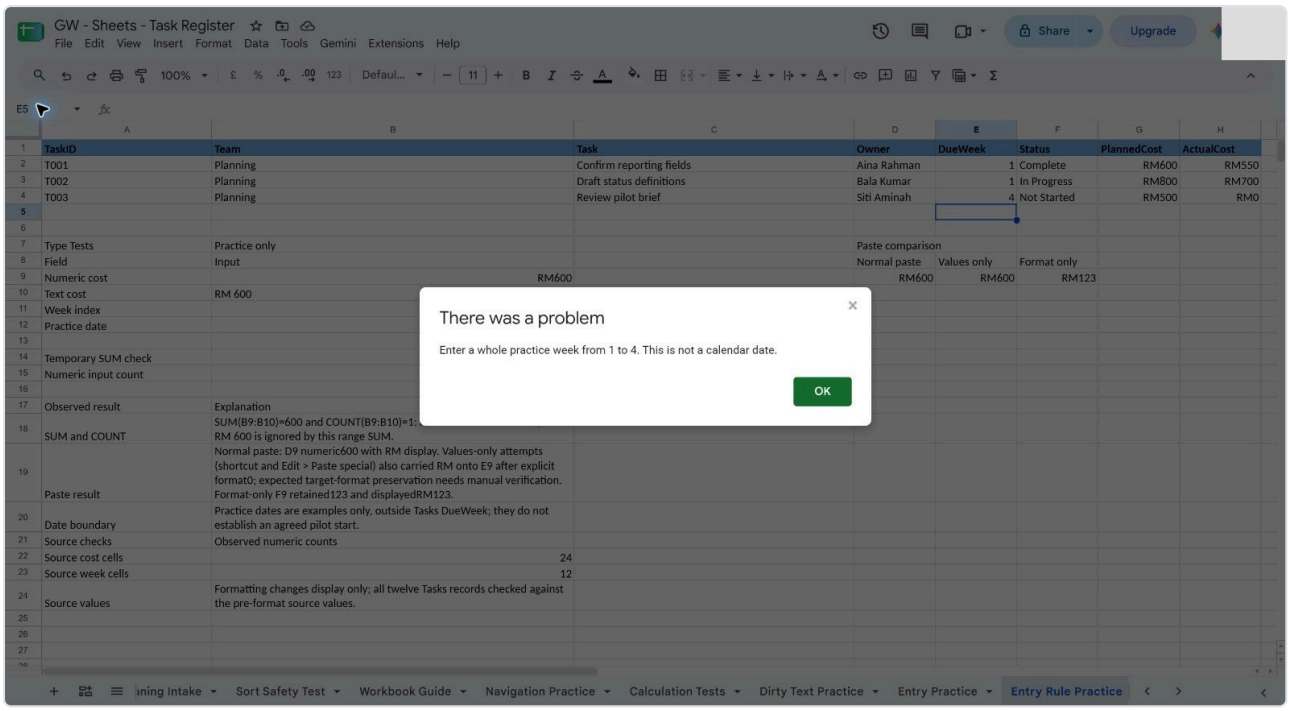


8. Test **1**, then **4**; both are accepted. Inspect the formula bar after committing each input. The captured upper-bound stage shows E4 holding numeric **4**.

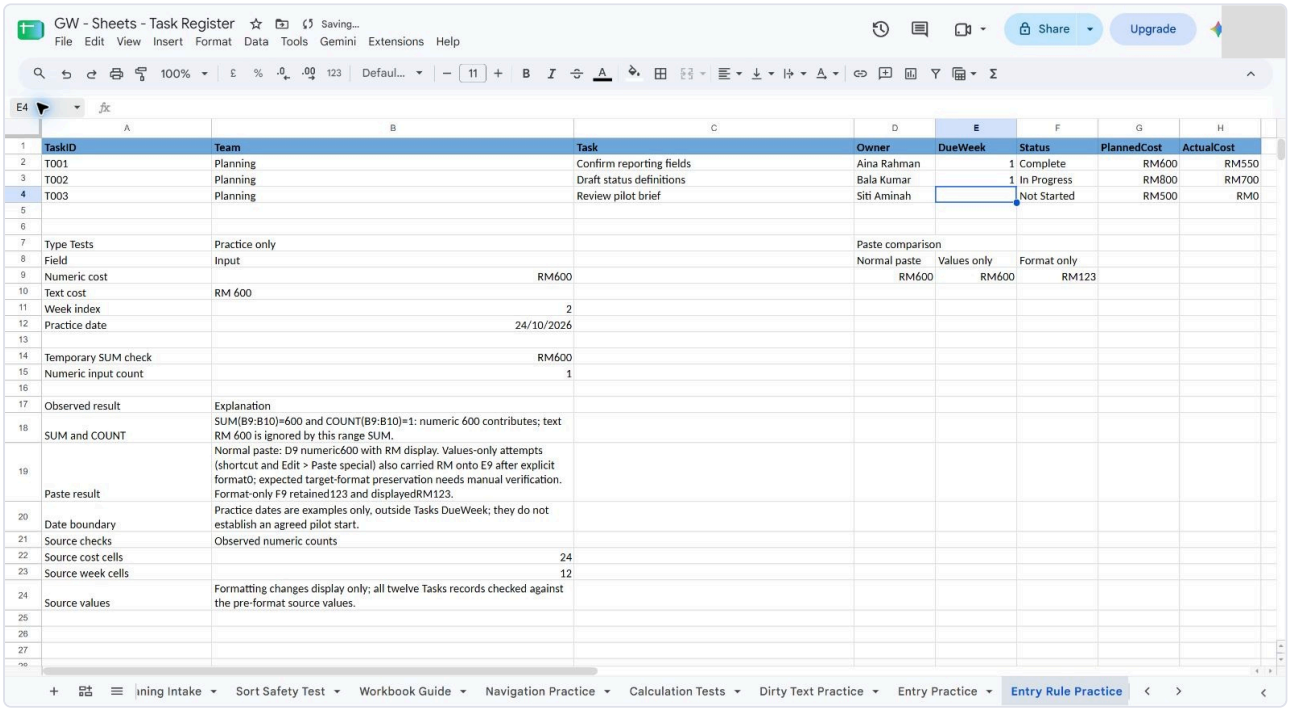


9. Test **5** and **1.5** separately. Both are rejected. A fractional value must not be accepted merely because it lies between the bounds.

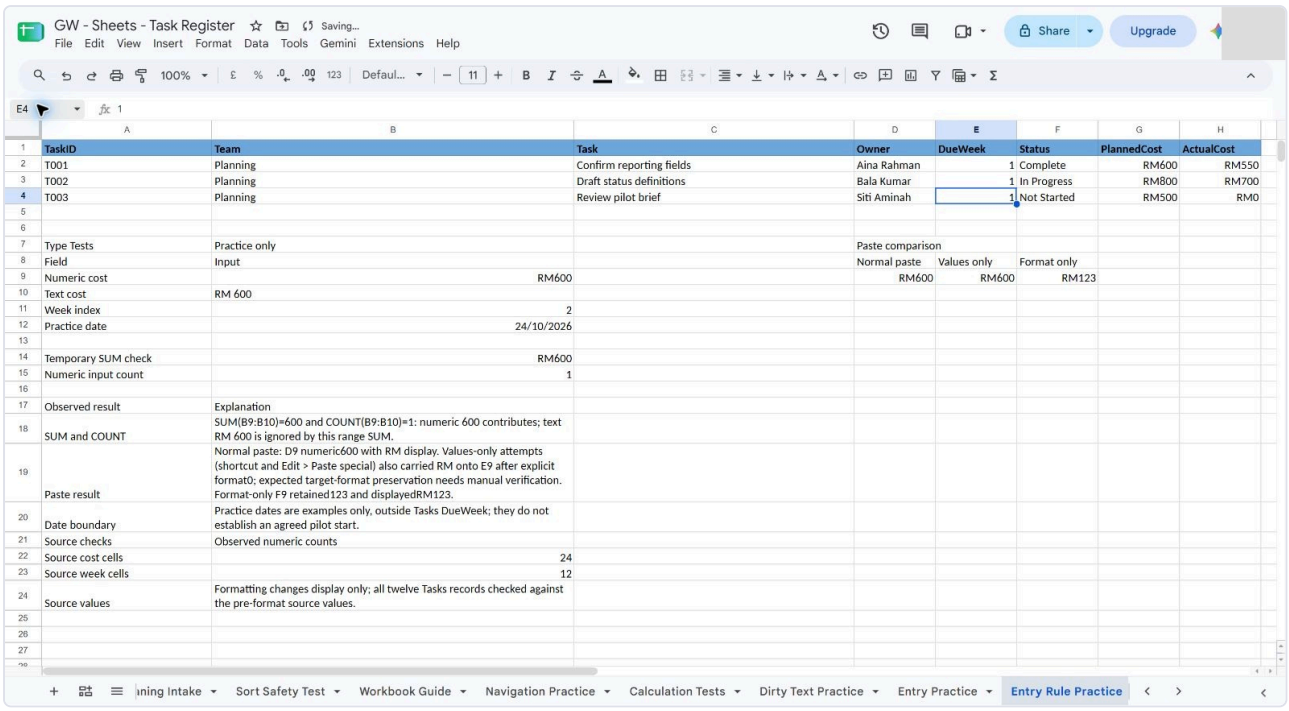




10. Clear the previously valid week and observe that deletion is allowed. This clearing test does not establish acceptance of every possible blank-input mechanism. Required-field completeness still needs an explicit review.



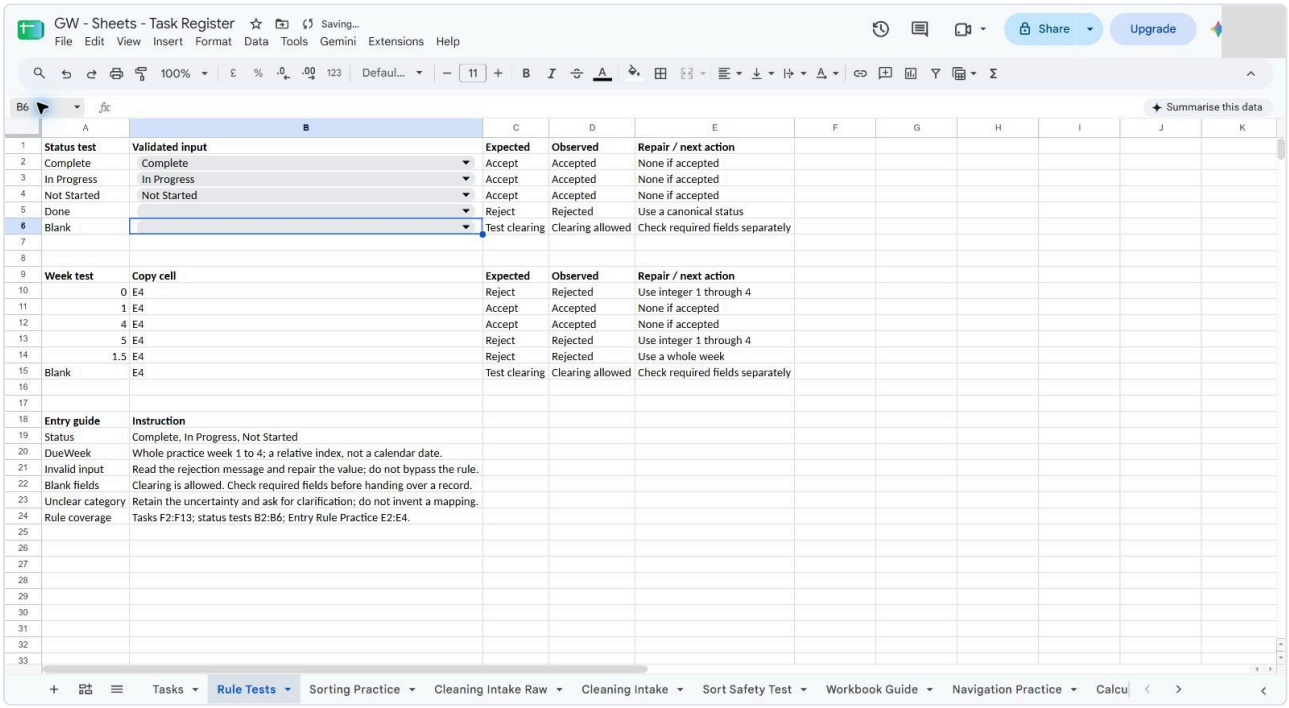
11. Restore E4 to its original numeric **1**. Keep the other practice-record values unchanged.



TaskID	Team	Task	Owner	DueWeek	Status	PlannedCost	ActualCost
T001	Planning	Confirm reporting fields	Aina Rahman	1	Complete	RM600	RM550
T002	Planning	Draft status definitions	Bala Kumar	1	In Progress	RM800	RM700
T003	Planning	Review pilot brief	Siti Aminah	1	Not Started	RM500	RM0

Type Tests	Practice only	Paste comparison		
Field	Input	Normal paste	Values only	Format only
Numeric cost	RM600	RM600	RM600	RM123
Text cost	RM 600			
Week index	2			
Practice date	24/10/2026			
Temporary SUM check	RM600			
Numeric input count	1			
Observed result	Explanation			
SUM and COUNT	SUM(B9:B10)=600 and COUNT(B9:B10)=1: numeric 600 contributes; text RM 600 is ignored by this range SUM.			
Paste result	Normal paste: D9 numeric600 with RM display. Values-only attempts (shortcut and Edit > Paste special) also carried RM onto E9 after explicit format0; expected target-format preservation needs manual verification. Format-only F9 retained123 and displayedRM123.			
Date boundary	Practice dates are examples only, outside Tasks DueWeek; they do not establish an agreed pilot start.			
Source checks	Observed numeric counts			
Source cost cells	24			
Source week cells	12			
Source values	Formatting changes display only; all twelve Tasks records checked against the pre-format source values.			

12. Record all observed results in Rule Tests. Add an entry guide in A18:B24 covering the canonical statuses, whole relative weeks, actual rejection messages, clearing limitation, unclear-category escalation and precise rule ranges. Fit the columns so labels and instructions are readable. Do not invent a mapping for an unclear category.

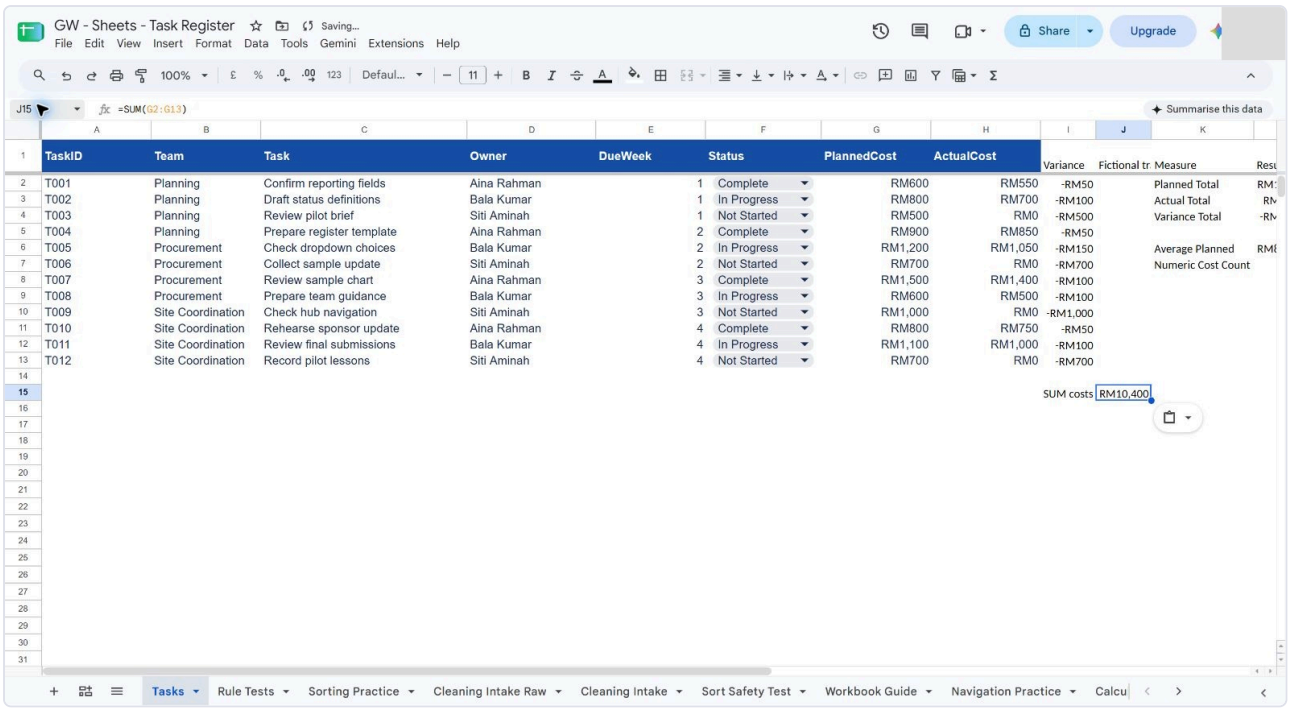


Status test	Validated input	Expected	Observed	Repair / next action
Complete	Complete	Accept	Accepted	None if accepted
In Progress	In Progress	Accept	Accepted	None if accepted
Not Started	Not Started	Accept	Accepted	None if accepted
Done		Reject	Rejected	Use a canonical status
Blank		Test clearing	Clearing allowed	Check required fields separately

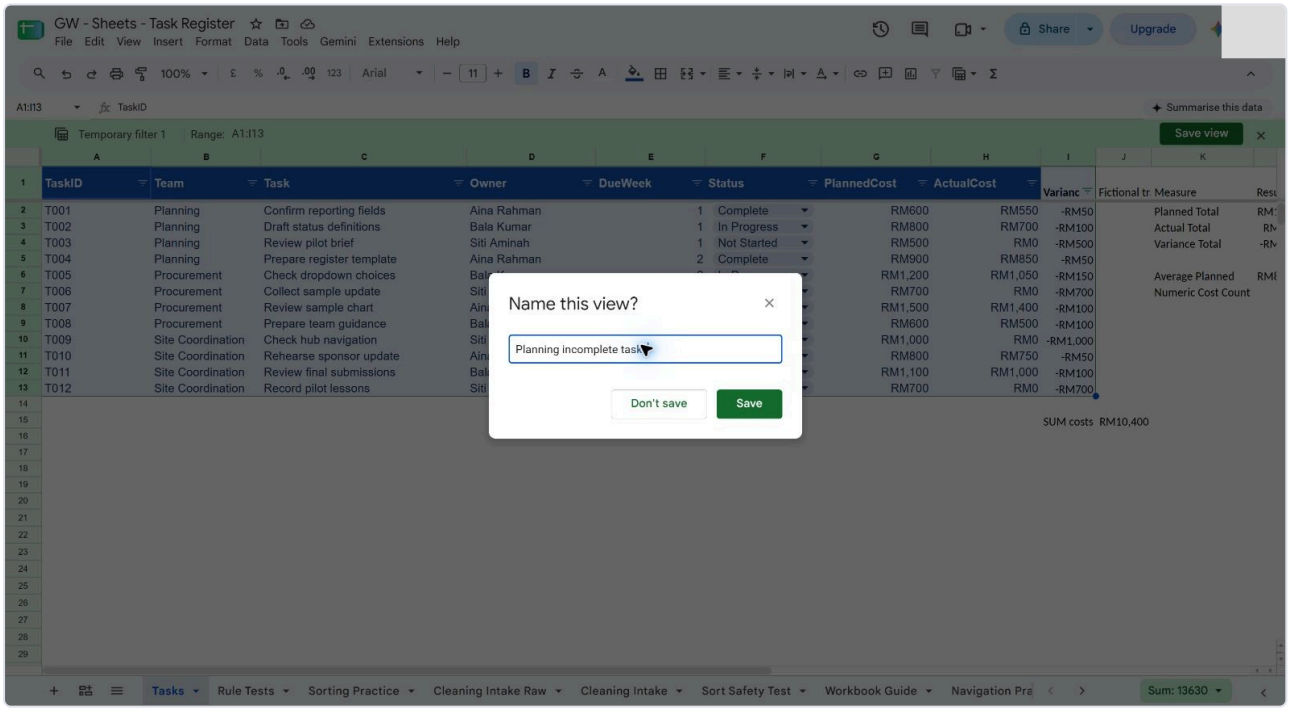
Week test	Copy cell	Expected	Observed	Repair / next action
0	E4	Reject	Rejected	Use integer 1 through 4
1	E4	Accept	Accepted	None if accepted
4	E4	Accept	Accepted	None if accepted
5	E4	Reject	Rejected	Use integer 1 through 4
1.5	E4	Reject	Rejected	Use a whole week
Blank	E4	Test clearing	Clearing allowed	Check required fields separately

Entry guide	Instruction
Status	Complete, In Progress, Not Started
DueWeek	Whole practice week 1 to 4; a relative index, not a calendar date.
Invalid input	Read the rejection message and repair the value; do not bypass the rule.
Blank fields	Clearing is allowed. Check required fields before handing over a record.
Unclear category	Retain the uncertainty and ask for clarification; do not invent a mapping.
Rule coverage	Tasks F2:F13; status tests B2:B6; Entry Rule Practice E2:E4.

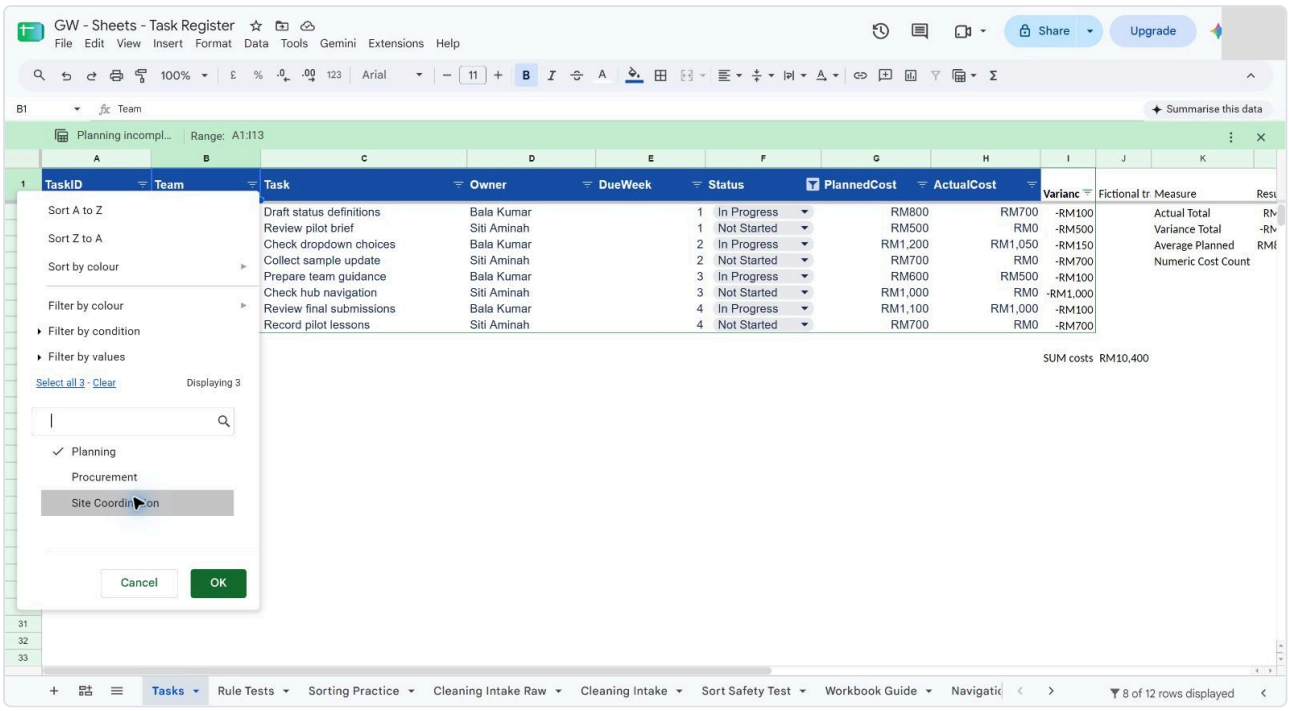
13. On **Tasks**, enter **SUM costs** in I15 and **=SUM(G2:G13)** in J15. This labelled check sits outside the bounded A1:I13 filter range and stays visible while rows are hidden. Before filtering its result is **RM10,400**.



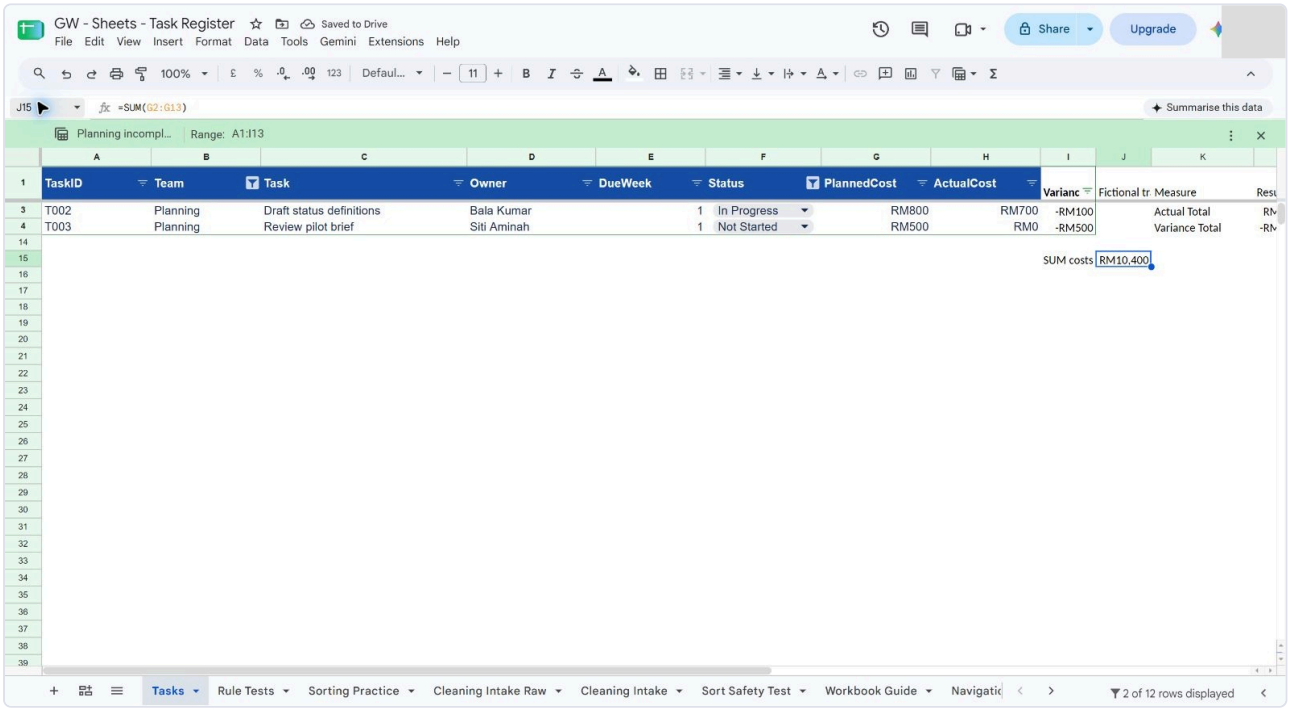
14. Create a second filter view over **A1:I13**, save it as **Planning incomplete tasks**, and exclude Complete in Status as before.



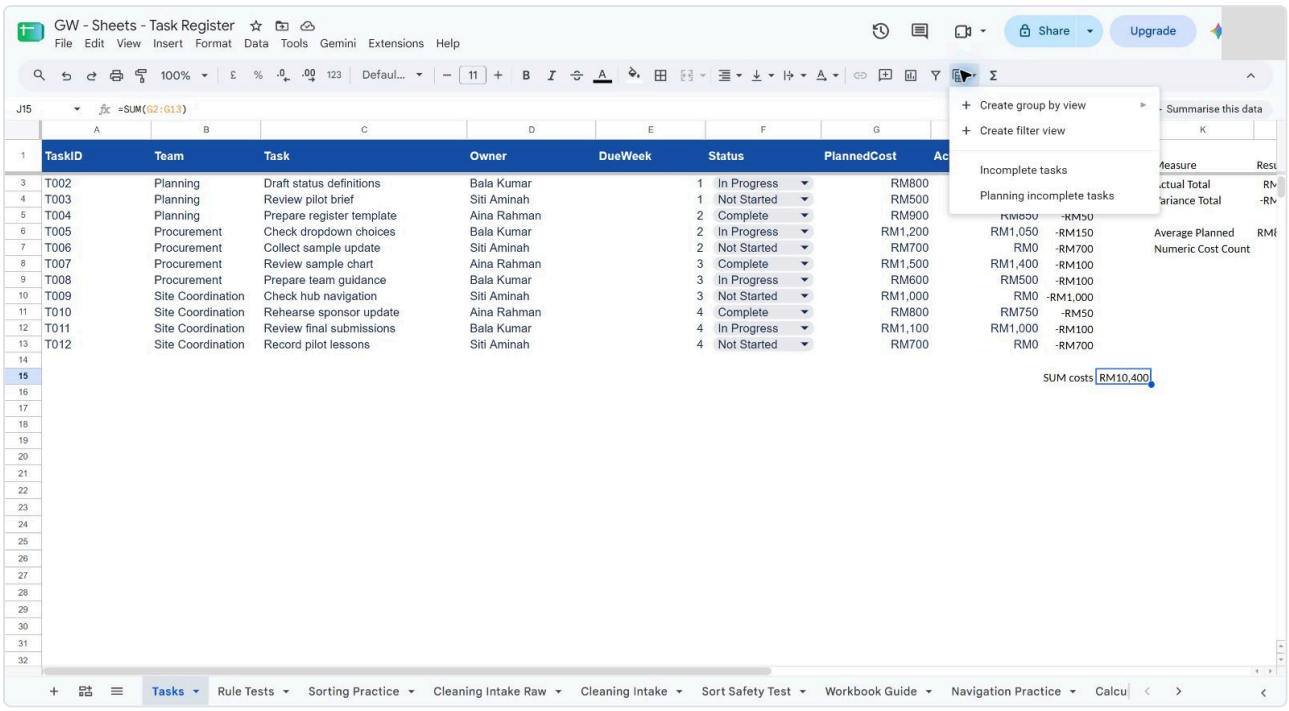
15. In the Team header filter leave only **Planning** selected. Both Team and Status criteria must apply to the same view.



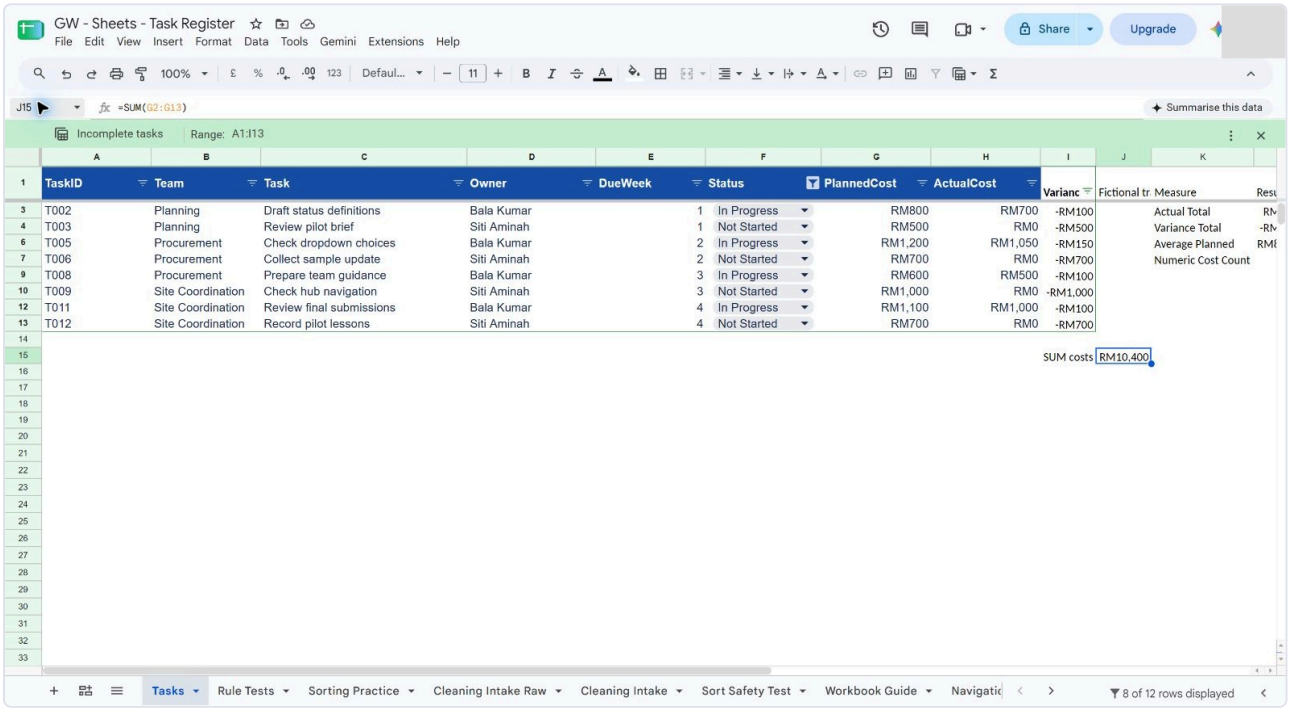
16. Verify **T002 and T003**, and the lower-right **2 of 12 rows displayed** indicator. A manual source scan confirms these are the two Planning records not Complete. J15 still shows **RM10,400** because an ordinary SUM includes values in hidden rows; it is not the total of only the two visible records.

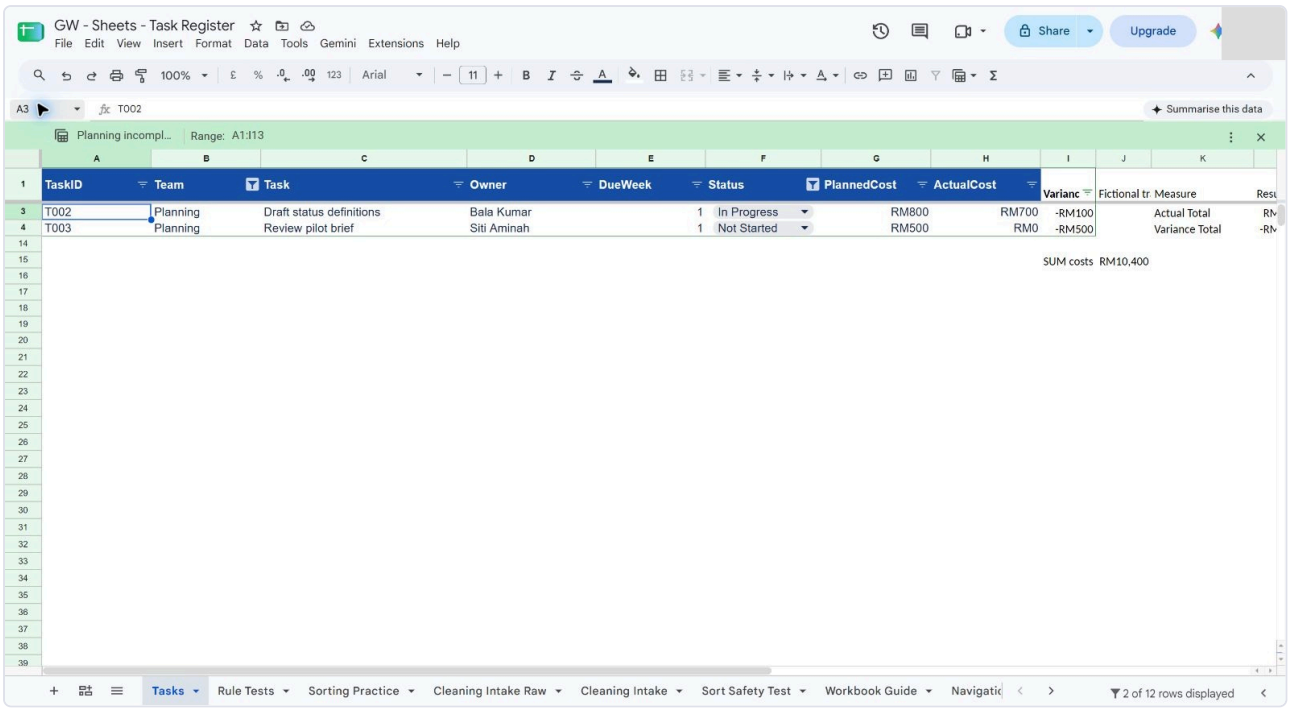


17. Exit the view and inspect the **Filter views** menu. Both saved names remain available.

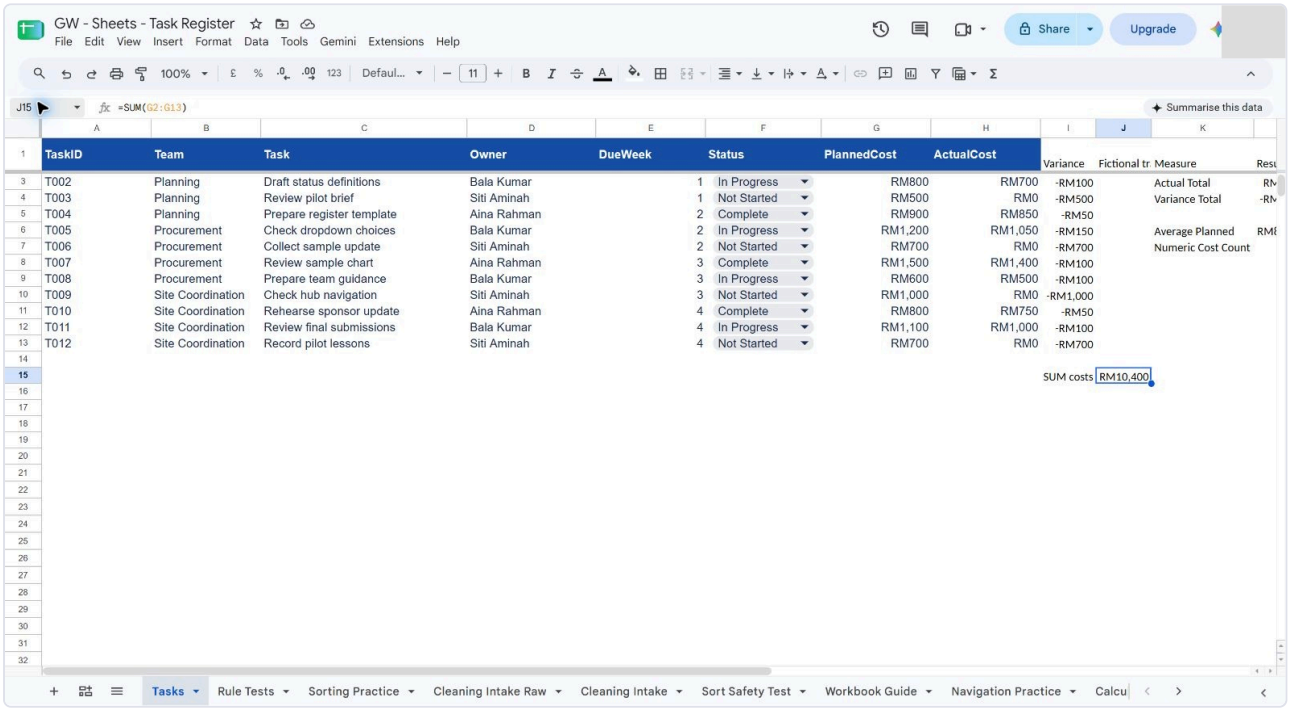


18. Reopen **Incomplete tasks** and verify eight IDs with J15 still **RM10,400**. Exit it, reopen **Planning incomplete tasks**, and verify the same two IDs. Do not use the mere presence of a view name as proof that its criteria persisted.

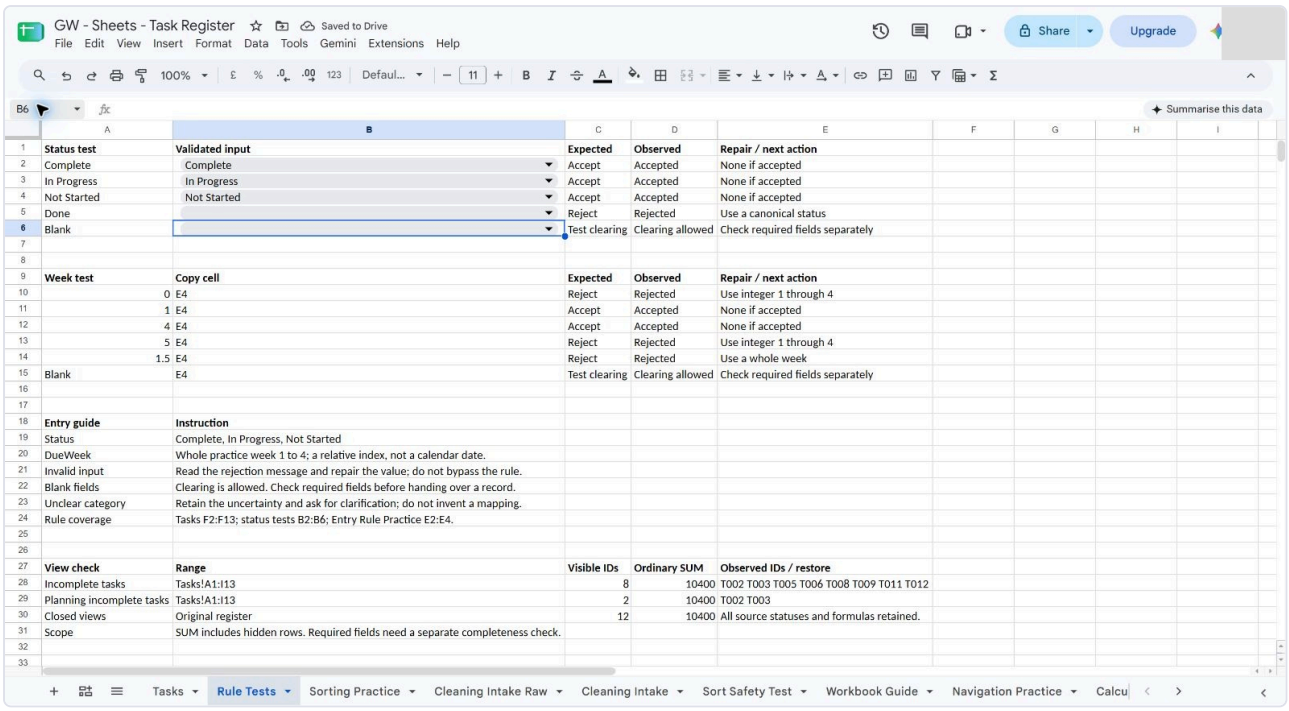




19. Exit the view and leave all twelve source records visible. The source A1:L13 values and formulas remain unchanged; the validation rules and separate J15 control are additions.



20. Record the two views' ranges, visible counts, observed IDs and ordinary SUM in **Rule Tests A27:E31**. All source rows return after closing the views. Inspect rule coverage separately from view coverage: an input outside a bounded rule may escape validation, and hiding a row does not remove it from an ordinary SUM.



> **Validation boundary:** these are author UI reference checks. The independent Procurement view, new-row coverage/range repair challenge, independent entry test and learner timing pilot remain pending. Earlier CSV-import and values-only-format limitations remain separate.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Independently build a small entry test with one valid row, one invalid status and one out-of-range week. Demonstrate the rules rather than merely listing them.
2. Create a named view answering “Which Procurement tasks are not complete?” and list its TaskIDs as evidence.
3. Check whether each new practice row lies inside the rule and view ranges. Repair a missing bound on the practice copy and retest.

Evidence to Hand Over

- Expected-versus-observed rule matrix including week boundaries and blanks.
- Two useful named views with independently checked visible records.
- Entry guide and a repaired coverage/range test.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check

A new row accepts invalid values	Inspect rule coverage as well as the dropdown definition; retest the actual new row.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Validation prevents inconsistent entries and filter views support focused inspection.

Check yourself: Does hiding a row delete it or remove it from an ordinary SUM?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
Incomplete tasks
```

```
Done
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Rule test values: status Complete, In Progress, Not Started, Done and blank; DueWeek 0, 1, 4, 5 and 1.5. DueWeek is a relative index for this training exercise, not an approved calendar schedule.
```

Captured Rule and View Reference

- Tasks status range **F2:F13**; single-selection dropdown Complete / In Progress / Not Started, **Reject the input**. Spare N2 has the copied validation and is empty after the rejected Done test.
- Rule Tests status inputs **B2:B6** use that same strict rule. B2:B4 retain Complete / In Progress / Not Started; attempted Done in B5 was rejected, and clearing B6 was allowed. A2:A6 retain the attempted test values; observations are separate in D.
- Entry Rule Practice **E2:E4**: custom formula **=AND(ISNUMBER(E2),E2=INT(E2),E2>=1,E2<=4)**, reject invalid input. Help text: Enter a whole practice week from 1 to 4. This is not a calendar date. Test E4 with0/1/4/5/1.5, then clearing. Observed reject/accept/accept/reject/reject/clearing allowed. Restore E4 to1. The bounded rule does not cover the copied notes below the three-record table.
- Tasks **I15 SUM costs / J15 =SUM(G2:G13)** displays10400 before and during either view. Ordinary SUM includes hidden rows.
- **Incomplete tasks**, Tasks A1:I13: exclude Complete; eight IDs T002,T003,T005,T006,T008,T009,T011,T012.
- **Planning incomplete tasks**, same range: Team Planning and Status excluding Complete; two IDs T002,T003.
- Both views were reopened and checked, then closed with all12 rows visible. Entry guide and view observations are in Rule Tests A18:B24/A27:E31.

Source Tasks A1:L13 values/formulas and the original Entry Practice records remain unchanged. Testing clearing does not guarantee all required fields are complete. Independent Procurement/new-row coverage challenges and classroom timing remain pending.

06

CHAPTER 6

Summarise and Visualise

In this chapter you will build a simple conditional total and chart; ask Gemini to explain a checked formula if available.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 55 minutes: 10 explanation/demonstration, 25 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 45 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A conditional summary and a source-linked status chart.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Calculate a conditional count and total.
- Choose and label a simple chart.
- Use AI to explain a formula without surrendering the check.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Tasks and the starter Summary tab. Existing baseline status counts are four per category.

6.1 Count Statuses

1. In Summary B2 enter `=COUNTIF(Tasks!$F$2:$F$13,A2)` and fill through B4.
2. Add a total in B5 using `=SUM(B2:B4)` and label A5 Total.
3. In D1:E4 add team labels and use SUMIF over Tasks B2:B13 and G2:G13 to calculate planned cost by team.

What should you see?

Status counts are 4/4/4 and total 12. Planned team totals are Planning 2,800, Procurement 4,000 and Site Coordination 3,600.

6.2 Build a Useful Chart

1. Select Summary A1:B4, excluding the total row. Choose Insert, Chart.
2. Choose a column chart and title it `Task count by status`.
3. Inspect category labels and make sure no blank category or total has become a fourth status.

What should you see?

The chart shows three status categories with equal counts.

6.3 Explain and Check

1. If Gemini is available, paste your COUNTIF formula and ask it to explain the anchored range and relative criterion cell in the panel only.
2. Compare the explanation with the formula in B3.
3. Temporarily change T003 to Complete, observe 5/4/3 and restore it. Confirm the chart returns to baseline.

What should you see?

Both your explanation and the chart's update behaviour have been checked against an actual input change.

6.4 Build a Second Summary and Diagnose a Misleading Chart

1. Add a second chart for planned cost by team using the checked D:E summary. Label currency units and keep task counts separate from cost amounts.
2. Create a small **Summary Checks** block comparing total status counts with the number of TaskIDs and team planned totals with Tasks planned total.
3. On a chart copy deliberately include the status total row as a fourth category. Explain the double-counting risk, then repair the source range.
4. On a summary copy deliberately misspell one status label. Observe how COUNTIF and the chart respond. Repair the spelling rather than overwriting the result with a typed number.
5. Write two supported findings, one about task counts and one about planned cost distribution. Avoid claims about on-time success or realised savings.
6. Test a temporary change to one task's planned cost. Check the row, team summary, grand total and chart; then restore the source value.
7. Record each change and restoration in a check log with expected effect, observed effect and the range or label you inspected.

What should you see?

Two reconciled summaries/charts with distinct units. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Prepare an independent one-page dashboard on a new tab with two labelled charts, source links and two concise supported findings.
2. Diagnose the card’s suspicious chart and summary symptoms using the source cells. Repair the cause, not just the visual appearance.
3. Use a reviewer or fresh read-through to check units, population and total reconciliation, then record one correction or evidence for a no-change decision.

Evidence to Hand Over

- Two reconciled summaries/charts with distinct units.
- Documented total-row and misspelt-category fault repairs.
- Independent dashboard, supported findings and restored source-change test.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A missing count is hidden by typing a result	Repair the criterion or formula so later source changes still recalculate.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

A checked summary and chart communicate the register accurately.

Check yourself: Why should the total row normally be excluded from this chart's categories?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
=COUNTIF(Tasks!$F$2:$F$13,A2)
```

```
=SUM(B2:B4)
```

```
Task count by status
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Troubleshooting card: A status chart shows Complete, In Progress, Not Started and Total.
Another summary says 4, 4 and 0 even though 12 task rows remain. Investigate source range and
exact criterion spelling before altering the chart design.
```

07

CHAPTER 7

Share and Prepare Output

In this chapter you will inspect access, print a selected range and check scaling and page breaks.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 40 minutes: 10 explanation/demonstration, 10 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 30 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: Checked access and readable register and summary print previews.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Inspect sharing without accidental exposure.
- Prepare a readable print range.
- Distinguish printing from public publishing.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use the checked register. No public publishing or email distribution is required.

7.1 Inspect Access

1. Open Share and inspect General access and the list of people.
2. Explain Viewer, Commenter and Editor roles using this fictional file.
3. Close without changing permissions unless the trainer assigns a named test recipient.

What should you see?

The file remains at its existing access level.

7.2 Prepare the Print Range

1. Select Tasks A1:I13. Open File, Print and choose Selected cells.

2. Inspect landscape orientation, fit-to-width scaling and whether the header can repeat.
3. Check every preview page for clipped labels or unreadably small text. Adjust layout rather than shrinking everything to fit one page.

What should you see?

The print preview includes the intended records and readable headers, without unrelated spare calculations.

7.3 Compare Output Options

1. Cancel printing. Inspect File, Share, Publish to web if it is available, then cancel without publishing.
2. Explain that publication can expose a selected sheet or spreadsheet to a broader audience depending on settings.
3. Describe when a PDF snapshot would be more appropriate than an editable or published file.

What should you see?

You can distinguish file access, a print/export copy and a published web view.

7.4 Prepare a Readable, Scoped and Reproducible Output

1. Write an output brief for a contributor who needs working input cells and a sponsor who needs a fixed summary. Identify the intended file and scope for each.
2. Inspect **Share** settings and current inheritance without changing access. Record proposed reader/editor roles separately from actual tested permissions.
3. Select the summary range and choose **File**, **Print**. Compare selected cells, current sheet and workbook scope where offered.
4. Choose orientation and scaling that preserve readable labels. Test repeated frozen rows and page breaks for a multi-page Tasks output when available.
5. Inspect every print-preview page, including the final page. Record whether helper cells, spare tests or unrelated tabs appear.
6. Save a PDF with a clear revision name. Open it and compare count/cost totals and row labels with the maintained source.
7. Change one heading in the working source and demonstrate that the old PDF remains unchanged. Produce a newly named export and update an output manifest.

What should you see?

Inspected access settings and proposed-role plan. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Prepare two independent outputs: a concise sponsor summary and a complete task listing with readable headings. Do not force both into the same one-page layout.
2. Have a reviewer use the manifest to identify the current source and current fixed copy. Repair ambiguous filenames or scope labels.
3. Explain which ranges should be edited, which calculated cells should be reviewed and why a fixed PDF is not a live tracker.

Evidence to Hand Over

- Inspected access settings and proposed-role plan.
- Two scope-checked readable PDFs with the correct final-page content.
- Source-to-export comparison and a revision manifest that identifies the current copy.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
The PDF contains practice/helper content	Inspect selected-cells versus current-sheet/workbook scope and regenerate the intended output.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Checked access and readable output complete the register handover.

Check yourself: Which output would automatically reflect later source changes, and which is a snapshot?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

This chapter uses the supplied files and visible controls. No additional prompt or typed code is required.

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Output brief: Sponsor wants a readable summary of the 12-task baseline and planned costs.
Contributor needs all task rows and labels. Exclude Entry Practice, temporary rule tests and
unrelated helper notes from the sponsor handout.
```

08

CHAPTER 8

Complete the Register Challenge

In this chapter you will add records, repair a formula and reconcile the totals independently.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 45 minutes: 10 explanation/demonstration, 15 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 35 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A separate 13-record challenge copy with reconciled totals and chart.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Extend a register deliberately.
- Repair a formula and reconcile totals.
- Explain a checked chart.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Make a separate challenge copy of your workbook. The original 12-record baseline must stay intact.

8.1 Add a New Record

1. Add T013 for Planning, task `Check the handover`, owner Aina Rahman, DueWeek 4, Status Not Started, PlannedCost 200 and ActualCost 0.
2. Extend validation and copy the variance formula into its row.
3. Identify every bounded count, sum and named range that must include the new row, then extend them.

What should you see?

The challenge copy has 13 unique records and includes the new task in its analysis.

8.2 Repair and Reconcile

1. Deliberately change one copied variance reference in the challenge copy and predict its effect.
2. Repair it and reconcile planned total 10,600, actual total 6,800 and variance -3,800.
3. Check status counts Complete 4, In Progress 4 and Not Started 5.

What should you see?

The extended formulas, summary and chart all include the thirteenth task.

8.3 Explain the Result

1. Show a partner the new record and its contribution to the totals.
2. Explain a relative reference, a fixed reference and a validation rule in your workbook.
3. Leave the challenge copy clearly named and the original source baseline unchanged.

What should you see?

Your explanation is supported by cells and the maintained baseline is preserved.

8.4 Stress-Test the Extended Register and Complete the Handover

1. On the existing 13-record challenge copy, add a **Change Log** naming every range extended for T013: row formula, totals, named range, validation, filter views and summary.
2. Inspect the formula in every variance row and confirm the copied calculation follows its own record after any sort.
3. Create a final control table with record count, planned total, actual total, variance total and status-count sum. Reconcile those by at least two methods.
4. Use the final correction card to change T013 ActualCost from 0 to 150 only on the challenge copy. Predict and check its effects before inspecting the refreshed charts.
5. Prepare a focused view for incomplete Planning tasks and confirm that its record list includes T013.
6. Regenerate the sponsor output from the changed challenge state. Inspect its labels and final values and distinguish it from the untouched baseline export.
7. Write a short operating guide for entering the next record, handling invalid input, checking ranges and exporting a current summary.

What should you see?

13-record cost-corrected challenge and a full coverage/change log. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Close the earlier click instructions. Add T014 on a further disposable copy using the supplied fields, then extend and verify all affected features independently.

2. Use your control table to catch at least one deliberately shortened range. Correct it and explain the cause with cell evidence.
3. Give a partner the operating guide and ask them to find the current source and identify the checks for the next row. Record a handover improvement and leave all scenario copies clearly named.

Evidence to Hand Over

- 13-record cost-corrected challenge and a full coverage/change log.
- Separate 14-record independent test with reconciled totals and repaired range omission.
- Readable current export plus a tested operating guide and preserved baseline.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
The challenge contaminates the next baseline lesson	Keep named scenario copies and compare their population and totals before proceeding.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

You have built, checked and extended a usable register.

Check yourself: Which ranges needed extending after T013 was added?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
Check the handover
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Corrections and extension: T013 ActualCost becomes 150. Further-copy T014: Procurement |
Verify submission checklist | Bala Kumar | week 4 | In Progress | planned 300 | actual 100.
Keep all three states separate: original baseline, 13-record corrected challenge, 14-record
extension.
```

Google Sheets Fundamental and Intermediate

A course review draft with fictional inputs and substantial casework. Captured source controls, separate practice tabs, repaired references, raw text preservation, record integrity, strict entry rules and two saved views. Independent learner validation and later labs remain pending.

Asraf Jaafar Sidik · Microsoft Certified Trainer ·